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# Financial Education - basics

**basics**

**May 30, 2026**



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This book is meant to collect some notes about financial instruments and methods for financial education, and mainly focused asset allocation.

This material is part of the **basics-books project**. It is also available as a .pdf document.

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**Main goal**

The ultimate goal of this material is to develop an understanding of how to manage personal savings efficiently, in line with one's own reasonable objectives.

To achieve this, some intermediate goals include:

- gaining knowledge of the **macroeconomic environment**
- familiarizing with some of the most common **investment tools** (mainly bonds and stocks);
- getting used to some **common-sense** and **investing principles**: minimizing certain costs when conditions are equal, risk/reward, diversification, liquidity, and other constraints/inefficiencies
- learning **what not to do**
- and once the poor choices have been ruled out, evaluating the reasonable options for building and managing an investment portfolio, using mainly *ETFs* as a natural choice of a (usually) liquid asset providing diversification at low cost, even for small capitals.

- 
- Introduction
    - *Summary*
    - *References*
    - *1-Page Draft*
  - Macroeconomic Context for Investing
    - *Actors*
    - *Inflation*
    - *Characteristic times in economy*
    - *Policy*
  - Investing Principles
    - *Introduction to principles of investing*
    - *Il minimo indispensabile*
    - *Principles of investing*
    - *Rebalancing*
    - *Sequence risk*
  - Asset classes
    - *Introduction to asset classes*
    - *Bonds*
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  - Asset allocation and Investing

- *Introduction to investing*
  - *Modern Portfolio Theory and Capital Asset Pricing Model*
  - *Merton's portfolio problem*
  - *Rebalancing*
- **Historical Data**
  - *Historical data*
  - *US Treasuries*
  - *USA equity market*
  - *Gold*
- **Extra**
  - *Extra and Random*
  - *Euristhics and historical correlations*
- **People**
  - *Resources, People and Firms*
  - *The Bull*



## **Part I**

# **Introduction**



## SUMMARY

### Introduction

Financial goals; money; inflation (BC and inflation target);

### Asset classes

### Investing principles

**Goals:** ...

**Fundamentals:** return, risk, risk-adjusted return

**From 1-period to multi-period:** volatility drag, composition-driven skewness,...

**Principles:** diversification, rebalancing,...

Rebalancing keeps the asset allocation (approximately) constant through time, preventing from single asset drift, and aiming at keeping the characteristics of the portfolio constant in time.

Diversification may improve the risk-adjusted return of the portfolio; under certain conditions, diversification may improve the return of the portfolio as well (both in terms of expected multi-period return and dispersion of the results)

**Asset allocation (portfolio building and management).**

- Building: MPT and CAPM, Merton's portfolio theory,...
- Management: rebalancing,...

### Asset allocation



## REFERENCES

Here some references to othere sources, in order to reasonably organize the contents of this book

### **Investment and Portfolio Management - RICE - coursera - A.Ozoguz, J.Foote**

#### **Global Financial Markets**

- Intro and Review of Elementary Finance Tools
- Financial system and financial assets: fixed income, equity and derivatives
- Organization of financial markets and securities trading

#### **Portfolio Selection and Risk Management**

- Intro and R/R: R/R trade-off
- Ptf construction and diversification
- Investor choices: utility functions, mean-variance preferences
- Optimal capital allocation and portfolio choice: mean-variance optimization (Modern Portfolio Theory)
- Equilibrium asset prining models: CAPM, return-beta; multi-factor models (e.g. Fama-French)

#### **Biases and Portfolio Selection**

- Efficient Market Hypotesis (EMH), and anomalies
- Biases and realistic preferences
- Inefficient markets: equity premium, volatility puzzle (?), long-run reversal to the mean, value effect, momentum
- Investor behavior

### Investment Strategies and Portfolio Analysis

- Performance measurement and benchmarking
- Active vs passive investing:  $R^*$  risk-adjusted return measurements: Sharpe, Sortino, Treynor's ratio, Jensen's alpha, ...; comparing the  $R^*$
- Performance evaluation: style analysis and performance attribution

### Capstone: Build a Winning Investment Portfolio

Using software for building ptf and assess its properties

- ...

## 3.1 Inflation and the need for investing

In contemporary economic systems, institutions targets a low non-zero inflation, usually 2%. Thus, the protection fo real valu of money is the bare minimum goal of investing.

Thus, need for investing

- starting from personal goals and constraints (saving rate, risk tolerance,...)
- knowing the available assets and their properties, e.g. 1-period — usually 1-year — return probability distribution
- knowing the fundamentals of composition of returns, and math facts about multi-period performances

First, analysis of 1-asset over 1-period, and eventually multi-asset over many periods of time.

## 3.2 Return, risk, and risk-adjusted return

### 3.2.1 Single asset, single period

The single period return of an asset  $X$  over period  $i$  from time  $i - 1$  and  $i$  is defined as

$$r_t := \frac{X_t - X_{t-1}}{X_{t-1}} = \frac{X_t}{X_{t-1}} - 1 ,$$

being  $X_t$  the value of the asset  $X$  at time  $t$ . A 1-period return of an asset  $X$  can be usually modelled as a random variable with probability distribution  $p_t(r)$ .

### 3.2.2 Single asset, multiple periods

The return compounds over many periods,

$$X_t = (1 + r_t)X_{t-1} = (1 + r_t) \dots (1 + r_1)X_0 = \prod_{i=1}^t (1 + r_i)X_0 = \prod_{i=1}^t \frac{X_i}{X_{i-1}} .$$

The compound average return (usually called *Compound Annual Growth Rate*, if the reference period is a year) can be defined as the constant return  $r_i = r$  for all  $i = 1 : t$  producing the actual result, and thus

$$X_t = (1 + r)^t X_0 ,$$

or

$$r = \left( \frac{X_t}{X_0} \right)^{\frac{1}{t}} - 1 .$$

**Log-return.**

$$\begin{aligned}\log \frac{X_t}{X_0} &= \sum_{i=1}^t \log \frac{X_i}{X_{i-1}} = \sum_i \log(1 + r_i) \\ &= t \log(1 + r),\end{aligned}$$

so that

$$\log(1 + r) = \frac{1}{t} \sum_i \log(1 + r_i).$$

**Probability distribution, expected value and variance**

Skewness of the probability density of the result  $\frac{X_t}{X_0}$ , typically producing *median value lt expected value*. Under some assumptions (...) the expected value of the log-return has expected value that is smaller than the 1-period expected return due to dispersion (**volatility drag**)

$$\mathbb{E}[r] \sim \mathbb{E}[r_i] - \frac{\text{Var}[r_i]}{2},$$

and variance

$$\text{Var}[r] \sim \frac{\text{Var}[r_i]}{t}.$$

**3.2.3 Multiple assets, single period**

Given a portfolio with many assets of value  $X_{t-1}$  at time  $t - 1$ , with weights

$$w_{i,t-1} := \frac{X_{i,t-1}}{X_{t-1}} = \frac{X_{i,t-1}}{\sum_i X_{i,t-1}},$$

and constant composition in time until  $t$ , the value of the final portfolio becomes

$$X_t = \sum_i X_{i,t} = \sum_i (1 + r_{i,t}) X_{i,t-1} = \sum_i r_{i,t} X_{i,t-1} + X_{t-1},$$

and the return

$$r_t := \frac{X_t}{X_{t-1}} - 1 = \frac{\sum_i r_{i,t} X_{i,t-1}}{X_{t-1}} = \sum_i r_{i,t} w_{i,t-1} = \mathbf{w}_{t-1}^T \mathbf{r}_t.$$

The return of the assets  $\mathbf{r}$  can be modelled as a (vector) random variable. The expected value and the variance of the 1-period return of the portfolio with weights  $\mathbf{w}$  read

$$\mu_r := \mathbb{E}[r] = \mathbb{E}[\mathbf{w}^T \mathbf{r}] = \mathbf{w}^T \mathbb{E}[\mathbf{r}] = \mathbf{w}^T \mu_{\mathbf{r}},$$

and

$$\begin{aligned}\sigma_r^2 &:= \text{Var}[r] = \mathbb{E}[(r - \mathbb{E}[r])^2] = \\ &= \mathbb{E}[\mathbf{w}^T (\mathbf{r} - \mu) (\mathbf{r} - \mu)^T \mathbf{w}] = \\ &= \mathbf{w}^T \sigma_{\mathbf{r}}^2 \mathbf{w}.\end{aligned}$$



### 3.2.4 Multiple assets, multiple periods

**Without rebalancing.** No action is taken. Every individual asset of the portfolio evolves independently. At time 0

$$X_0 = \sum_i X_{i,0} = \sum_i w_{i,0} X_0 .$$

At successive times,

$$X_t = \sum_i X_{i,t} = \sum_i X_{i,0} \prod_{\tau=1}^t (1 + r_{i,\tau}) = X_0 \sum_i w_{i,0} \prod_{\tau=1}^t (1 + r_{i,\tau}) .$$

**With rebalancing (assuming no cost...).** At the end of each period, weights of individual assets are set back to the original desired value. Following this strategy, under the assumption of stationarity, the 1-period performance of the portfolio is constant, with 1-period return

$$r_1 = \mathbf{w}^T \mathbf{r} ,$$

and expected value and variance

$$\mu_{(1)} = \mathbf{w}^T \mu_{\mathbf{r}} , \quad \sigma_{(1)}^2 = \mathbf{w}^T \sigma_{\mathbf{r}}^2 \mathbf{w} .$$

Under the assumption of “small” 1-period returns, the geometric average return reads

$$\bar{r} \sim \mathcal{N} \left( \mu_{(1)} - \frac{\sigma_{(1)}^2}{2}, \frac{\sigma_{(1)}^2}{N} \right) .$$

#### Shannon demon and the rebalancing premium: 2-asset portfolio

A 2-asset portfolio is determined by the weights  $\mathbf{w} = (w_1, w_2)$  of the two assets. One-period return of the 2 assets is modeled as a multidimensional random variable  $\mathbf{r}$ , whose expected value and variance read

$$\mu_{(1)} = \mathbb{E}[\mathbf{r}] = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

and

$$\sigma_{(1)}^2 = \mathbb{E}[\Delta \mathbf{r} \Delta \mathbf{r}^T] = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12}^2 \\ \sigma_{12}^2 & \sigma_{22}^2 \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \rho_{12} \sigma_1 \sigma_2 \\ \rho_{12} \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix} .$$

Under certain conditions, the expected value of the multi-period return of the diversified and rebalanced portfolio may exceed the largest expected value of multi-period returns from individual assets, i.e.

$$\mu - \frac{\sigma^2}{2} > \mu_i - \frac{\sigma_i^2}{2} .$$

#### Some algebra and analytic geometry

Let's evaluate the conditions for  $i = 1$

$$\begin{aligned} 0 &< w_1 \mu_1 + w_2 \mu_2 - \frac{w_1^2 \sigma_1^2 + 2 \rho_{12} \sigma_1 \sigma_2 w_1 w_2 + w_2^2 \sigma_2^2}{2} - \mu_1 + \frac{\sigma_1^2}{2} = \\ &= c(w_1, w_2) \end{aligned}$$

Given the properties  $\mu_i, \sigma_i, \rho_{12}$ , the equation  $c(w_1, w_2) = 0$  is the equation of a conic section in the plane  $w_1 - w_2$ . With some [high-school math](#), see [general expression of conic sections](#), the reduced discriminant  $\frac{\Delta}{4} = \frac{B^2}{4} - AC$  determines the nature of the curve. As,

$$\frac{\Delta}{4} = \frac{B^2}{4} - 4AC = \frac{\rho_{12}^2 \sigma_1^2 \sigma_2^2 - \sigma_1^2 \sigma_2^2}{4} = \frac{1}{4} \sigma_1^2 \sigma_2^2 (\rho_{12}^2 - 1) \leq 0 ,$$

the equation  $c(w_1, w_2) = 0$  is the equation of an ellipse for  $\rho_{12} \neq 1$ , and a parabola for  $\rho_{12}^2 = 1$  (perfectly correlated assets).

### Non-perfectly correlated assets, $|\rho_{12}| < 1$

The general expression of the quadratic curve in the  $\mathbf{w}$ -plane reads

$$0 = \frac{1}{2} \mathbf{w}^T \sigma^2 \mathbf{w} - \mu^T \mathbf{w} + \mu_1^{multi}$$

In order to find the **center of the ellipse**, a translation  $\mathbf{w} = \mathbf{w}_C + \mathbf{w}_1$  should produce zero 1-degree terms, i.e.

$$\begin{aligned} 0 &= \frac{1}{2} (\mathbf{w}_1 + \mathbf{w}_C)^T \sigma^2 (\mathbf{w}_C + \mathbf{w}_1) - \mu^T (\mathbf{w}_C + \mathbf{w}_1) + \mu_1^{multi} = \\ &= \frac{1}{2} \mathbf{w}_1^T \sigma^2 \mathbf{w}_1 + (2\mathbf{w}_C^T \sigma^2 - \mu^T) \mathbf{w}_1 + \frac{1}{2} \mathbf{w}_C^T \sigma^2 \mathbf{w}_C - \mu^T \mathbf{w}_C + \mu_1^{multi} , \end{aligned}$$

i.e.

$$\mathbf{w}_C = \frac{1}{2} (\sigma^2)^{-1} \mu .$$

Using coordinates  $\mathbf{x}_1$ , the expression of the ellipse becomes

$$0 = \frac{1}{2} \mathbf{w}_1^T \sigma^2 \mathbf{w}_1 - \mu^T (\sigma^2)^{-1} \mu + \mu_1^{multi} .$$

The angle of the semi-axes of the ellipse w.r.t. the coordinate axes and their lengths appears after applying the rotation transformation,  $\mathbf{w}_1 = \mathbf{R} \mathbf{w}_2$ , that makes the quadratic form diagonal, i.e. as the solution of the spectral decomposition of the covariance matrix,

$$\mathbf{R}^T \sigma^2 \mathbf{R} =: \sigma_d^2 .$$

Using the results obtained for [general expression of conic sections](#), if  $\rho_1 \neq \rho_2$ , the rotation angle  $\theta$  satisfies

$$\tan 2\theta = \frac{2\rho_{12}\sigma_1\sigma_2}{\sigma_1^2 - \sigma_2^2} ,$$

and the length of the semi-axes, given  $\mu^T (\sigma^2)^{-1} \mu - \mu_1^{multi} > 0^1$ , is

$$a_{1,2} = \sqrt{\frac{\mu^T (\sigma^2)^{-1} \mu - \mu_1^{multi}}{s_{1,2}}} ,$$

with the eigenvalue of the covariance matrix

$$s_{1,2} = \frac{\sigma_1^2 + \sigma_2^2}{2} \mp \frac{\sqrt{(\sigma_1^2 - \sigma_2^2)^2 + 4\rho_{12}^2 \sigma_1^2 \sigma_2^2}}{2}$$

<sup>1</sup> Otherwise the general expression doesn't represent any curve in the real plane, as that equation is never satisfied by any values of  $\mathbf{w}$ .

**Perfectly correlated assets,  $|\rho_{12}| = 1$** 

The general expression of the quadratic curve in the  $\mathbf{w}$ -plane reads

$$0 = -c(\mathbf{w}) = \frac{1}{2} (w_1\sigma_1 + w_2\sigma_2)^2 - w_1\mu_1 - w_2\mu_2 + \mu_1 - \frac{\sigma_1^2}{2} .$$

At the end of any period, weights are set back to the desired fractions. At time 0

$$X_0 = \sum_i X_{i,0} = \sum_i w_i X_0 ,$$

At time 1 before rebalancing

$$X_1 = \sum_i X_{i,1}^- = \sum_i X_{i,0}(1 + r_{i,0}) = X_0 \left( 1 + \sum_i w_i r_{i,0} \right) .$$

At time 1 after rebalancing

$$X_1 = \sum_i X_{i,1}^+ = \sum_i w_i X_1$$

At time 2 before rebalancing

$$X_2 = \sum_i X_{i,1}^+(1 + r_{i,1}) = X_1 \sum_i w_i (1 + r_{i,1}) .$$

and thus

$$\frac{X_t}{X_0} = \prod_{\tau=1}^t (1 + \mathbf{w}^T \mathbf{r}_\tau) .$$

The log-return comes from

$$\begin{aligned} \log \frac{X_t}{X_0} &= \sum_{\tau=1}^t \log(1 + \mathbf{w}^T \mathbf{r}_\tau) = \\ &\sim \sum_{\tau=1}^t \left( 1 + \mathbf{w}^T \mathbf{r}_\tau + \frac{1}{2} \mathbf{w}^T \mathbf{r}_\tau \mathbf{r}_\tau^T \mathbf{w} \right) \end{aligned}$$

The expected value and the variance read...



## **Part II**

# **Macroeconomic Context for Investing**



**ACTORS**

**4.1 People**

**4.2 Private companies**

**4.3 Government - public**

**4.4 Central banks**

**4.5 Investment banks**

**4.6 Foreign regions**





## INFLATION

Inflation is the **rate** at which the general level of prices for goods and services changes.

**Contents.** Definition and inflation indices, with examples of indices used in Italy: NIC, FOI, IPCA; components of inflation, with details of IPCA in Italy; correlation with other macroeconomic quantities; who controls inflation; origin of inflation

### 5.1 Inflation Indices (e.g. in Italy)

Overall inflation is the the weighted average of inflation on different classes of goods and services, weighted for their share of expenses.

Everyone perceives its own inflation, depending on its expenses. Different indices are usually used within an economy to track inflation for some “average individual”.

Different indices may differ on values of weights, and other “details” like the effect of discounts and public transfers.

As an example, three indices are used in Italy:

- **NIC** (Prezzi al Consumo per l'intera Collettività Nazionale), usually the general
- **FOI** (Prezzi al Consumo per Famiglie di Operai e Impiegati), usually used for contracts, pension and inflation-linked contracts, ex-tobacco and lotteries.
- **IPCA** (Indice Armonizzato dei Prezzi al Consumo, *HIPC Harmonized Index of Consumer Prices*), used for comparison and statistics in the EU

### 5.2 Weights and Price Indices of Classes of Goods and Services - Italy IPCA

National and International Institutions for Statistics (in Italy, ISTAT) provide open-access databases collecting statistics about society and economics, including data about price.

**ISTAT.** As an example, Italian ISTAT provides data at <https://esploradati.istat.it/databrowser/#/it/dw>

All the data we need here is available under the category “Prezzi” - *Prices*. In order to reach a reasonable stability of the notebook, data have been downloaded, cleaned and stored in a folder on the repository of the project.

## 5.2.1 Inspect Data

Before producing plots, price indices and weights of level-4 categories are visually inspected. Data are usually collected in tables.

### Category Price Indices - Level-4 IPCA

### Category Weights - Level-4 IPCA

## 5.2.2 Plots

### Category weights - Level-2 IPCA

The weights assigned to IPCA (Harmonized Index of Consumer Prices) categories represent the average expenditure share of households on each category of goods and services. These weights reflect how important each category is in the consumption basket.

These weights are revised annually to account for changing consumer behavior, as one can easily realize acting on the slider of the picture below. They are the weights used in computing the overall inflation  $i$  index, as the weighted sum of inflation  $i_c$  of IPCA categories,

$$i = \sum_{c \in \text{Cat}} i_c w_c .$$

### Category Prices - Level-2 IPCA

Some categories in IPCA are subject to strong seasonal effects, meaning prices follow recurring patterns during the year.

As an example:

- Clothing and Footwear: in July–August, retailers apply seasonal discounts (saldi) in Italy and prices in IPCA do include these discounts when they are actually applied in stores, as it's shown by seasonal July/August price drops
- Fresh fruits and vegetables: prone to seasonal availability, leading to fluctuating prices.
- Travel and tourism: prices rise in summer and holidays.

Seasonality can obscure underlying inflation trends: that's why **seasonally adjusted** inflation is evaluated, see below.

```
Index(['[00] Indice generale',
      '[01] -- prodotti alimentari e bevande analcoliche',
      '[02] -- bevande alcoliche e tabacchi',
      '[03] -- abbigliamento e calzature',
      '[04] -- abitazione, acqua, elettricità, gas e altri combustibili',
      '[05] -- mobili, articoli e servizi per la casa',
      '[06] -- servizi sanitari e spese per la salute', '[07] -- trasporti',
      '[08] -- comunicazioni', '[09] -- ricreazione, spettacoli e cultura',
      '[10] -- istruzione', '[11] -- servizi ricettivi e di ristorazione',
      '[12] -- altri beni e servizi'],
      dtype='object', name='Tempo')
```

### Category Price Changes (Inflation) - Level-2 IPCA

The 12-month inflation rate (year-on-year or YoY) compares prices in a given month to the same month the year before. It's already less prone to seasonal effects than the month-to-month rate.

However, even YoY rates can exhibit seasonal patterns, especially in volatile components like food, energy, and clothing. In order to reduce volatility of inflation indices, it's possible to use:

- **Core inflation**, as a measure of inflation that excludes the most volatile items (e.g., unprocessed food, energy), in order to provide a smoothed measure of inflation trends.
- Statistical filtering, and moving averages

---

#### Energy post-2022

Since 2022, prices in the energy and utility sectors have shown exceptional volatility. Different causes may have contributed, like geopolitical tensions (notably, the war in Ukraine), “liberalized” electricity/gas markets in Italy where price caps were adjusted or removed. Inflation in energy and electricity was also influenced by a *base effect* (e.g., very low prices in 2020–2021 followed by spikes in 2022).

Policy interventions like tax reductions and bonuses - that are not “free” -, which may or may not be reflected in consumer prices, depending on implementation.

The use of *core inflation* in 2022–2023 was arguable, as energy prices didn't just spiked and reverted, but was/is quite a long-term shock (war, sanctions, market and supply restructuring,...); as energy price influences many other sectors, food price rose as well, due to input cost shocks /fertilizers, transports,...) not as a result of seasonality only. Using core inflation and excluding energy and food components masked the true **cost-of living** impact on households.

---

### Category contributions to overall inflation - Level-2 IPCA

## 5.3 Correlations in macroeconomics with inflation

Some correlations exist<sup>1</sup> between inflation and other macroeconomics quantities.

- **Phillips Curve**: inverse relation between inflation and unemployment (in the short-run)
- **Money supply** in the long-run “*Inflation is a monetary phenomenon*”, M.Friedman.

## 5.4 Control of Inflation

Control of inflation is one of the goals of **central banks**, like the FED and the ECB.

*Central banks* aims at controlling inflation, matching target inflation (usually set as 2%) by means of **monetary policy**:

- interest rates (cost of money)
- non-conventional actions, like quantitative easing (QE)/tightening (QT)

---

<sup>1</sup> ...

A government may indirectly influence inflation with **fiscal policy**, as taxation and government spending can influence demand.

**Credibility** of targets, and actors through their actions and forward guidance may influence inflation as well: expectations influences inflation.

## 5.5 Origin of inflation

Origin of inflation?

- *short-run, medium-run*: cost-push, demand-pull, built-in (triangle model)
  - *long-run*: “inflation is always and everywhere a monetary phenomenon” M.Friedman
-

## CHARACTERISTIC TIMES IN ECONOMY

**6.1 The short run**

**6.2 The medium run**

**6.3 The long run**



**POLICY**

|               | Monetary Policy               | Fiscal Policy                   |
|---------------|-------------------------------|---------------------------------|
| Controlled by | CB                            | Government                      |
| Main tools    | IR, Money supply              | Taxes, Spending, Transfers      |
| Speed         | Usually faster                | Politically slower, debated     |
| Focus         | Inflation, lliquidity, credit | Employment, Income distribution |
| Independence  |                               |                                 |

**7.1 Monetary policy**

**7.2 Fiscal policy**





# **Part III**

## **Investing Principles**



## INTRODUCTION TO PRINCIPLES OF INVESTING

Investing is a core part of personal financial management—it's how individuals navigate uncertainty to meet their financial goals under real-world constraints. The most basic objective is to preserve the real value of wealth, protecting it against inflation; more ambitious goals include growing capital to fund retirement, education, or other life plans.

Sound investing requires understanding *return* and *risk* of available assets, and the fundamental *R/R trade off*. It also demands attention to **constraints** such as *liquidity* needs, *time horizon*, *acceptable volatility*, and *risk tolerance*. One of the main principle is *diversification* - which can reduce risk and, in some cases, enhance returns.

This section introduces the core concepts needed to build a robust investment strategy: how compound returns shape long-term growth, how *volatility drag* reduces expected performance, and how a clear, principle-based approaches - like *rebalancing* - may improve performance under uncertainties.

Given its set of constraints, an informed and intelligent agent, see *Portfolio construction* would take actions that try to maximise return for a given accepted risk, or minimize risk for a given desired return: this behavior can be summarized in choosing actions on a *Pareto front*, i.e. within the set of all Pareto efficient solutions.

### Sections

| Section                            | Key Concepts                    |
|------------------------------------|---------------------------------|
| 0. <i>Il minimo indispensabile</i> |                                 |
| 1. <i>Return</i>                   |                                 |
| 2. <i>Risk</i>                     |                                 |
| 3. <i>Risk-Return Trade-Off</i>    |                                 |
| 4. <i>Diversification</i>          |                                 |
| 5. <i>Portfolio Construction</i>   |                                 |
| 6. <i>Time and Compounding</i>     | Compounding and volatility drag |
| 7. <i>Disciplined Investing</i>    | PIC/PAC, rebalancing,...        |



## IL MINIMO INDISPENSABILE

Questa pagina viene scritta in lingua italiana, per alcuni semplici motivi: l'Italia ha la popolazione con il minor tasso di alfabetizzazione finanziaria tra i paesi OCSE<sup>1</sup>, e il grado di alfabetizzazione finanziaria è legato al livello di istruzione (con correlazione positiva: in media, meno hai studiato, meno sai rispondere a domande su concetti base come inflazione, interesse composto, diversificazione del rischio, e più alta è la probabilità che tu faccia vaccate; meglio scrivere in italiano quindi se l'obiettivo è raggiungere l'italiano medio).

Prima di affrontare argomenti più complessi, è quindi meglio stabilire una linea di galleggiamento e fissare un minimo di decenza e igiene dal quale partire.

### Necessità dell'investimento.

- Si vive con il valore reale dei soldi, non con quello reale
- Nel sistema attuale, gli obiettivi di inflazione sono attorno al 2%<sup>2</sup> — misurata con un adeguato indice dei prezzi nell'area monetaria di riferimento
- L'inflazione erode il valore reale dei soldi
- Questo rende necessario cercare dei modi per difendere il valore del risparmio/patrimonio, attraverso investimento.

### Come investire: i principi

- rendimento, rischio, R/R: composizione rendimenti, ruolo dell'incertezza (e della volatilità), minimizzazione costi certi (a parità di altre condizioni)
- **diversificazione**
- Strumenti per investire

**Come investire: costruzione di un portafogli, e gestione nel tempo.** Una volta definiti gli obiettivi personali, compatibilmente con la propria tolleranza a rischi e volatilità:

- Costruzione portafogli con strumenti finanziari, secondo i principi di investimento
- Gestione su un orizzonte temporale medio-lungo (**ribilanciamento**,...)

<sup>1</sup> Paola Tamma, *Why Europeans need to learn more about money*, FT, e riferimenti contenuti; Mario Seminerio, tweet 2025.12.12; OECD 2022 survey.

<sup>2</sup> Pagine di alcune banche centrali riguardo il valore obiettivo della *politica monetaria* e le motivazioni: Banca Centrale Europea, BCE o ECB; Federal Reserve, US Fed; Bank of England, BoE; Bank of Japan; Bloomberg, *China may struggle to hit inflation target even after cutting it*, 2025.03.05 riguardo l'adeguamento del target della Bank of China, dal 3% al 2%. Sono numeri arbitrari? Entro una certa tolleranza probabilmente sì, anche se un numero o un intervallo va scelto, e la scelta è motivata nelle pagine delle Banche Centrali. Più importante del numero in sé, è la credibilità e la capacità delle BC di riuscire a ottenere un valore di inflazione vicino all'obiettivo, tramite la politica monetaria — anche se questa può avere dei limiti, soprattutto se non accompagnata da una politica fiscale coerente da parte dei governi: per chi fa scelte economiche risulta fondamentale che gli obiettivi di inflazione siano rispettati, e che gli attori preposti siano credibili. In un mondo normale, attese di inflazione e incertezza futura sugli obiettivi di inflazione vengono scontate dagli attori del sistema economico. Ad esempio, è lecito attendersi che un investitore che acquista un'obbligazione — statale di uno stato affidabile per escludere spread rispetto a ciò che può essere considerato "risk-free"; nella stessa valuta in cui vive per escludere rischio-cambio — a scadenza 10 anni richieda almeno un rendimento reale positivo su tale orizzonte temporale. In una situazione in cui l'investitore ha la certezza (beato lui che ha certezze) che il target di inflazione sia rispettato, può chiedere un interesse nominale netto del 2.5% se si accontenta di un rendimento reale dello 0.5% annuo. Lo stesso investitore chiederebbe lo stesso interesse se si attendesse un'inflazione superiore al target, o se non considerasse attendibile l'impegno della BC di rispettare il target, o se temesse qualche "fiammata inflazionistica"? Vedi *rischio inflazione per obbligazioni*.

## 9.1 Inflazione

Breve nota sull'inflazione italiana negli ultimi anni. Se necessario, usate il traduttore automatico del browser.

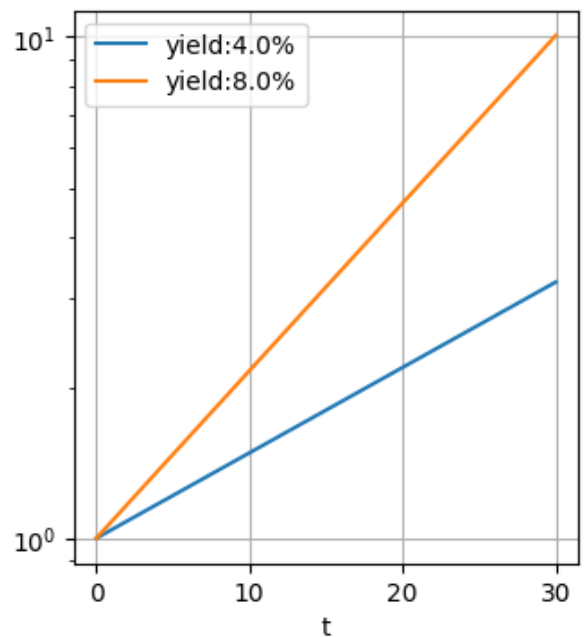
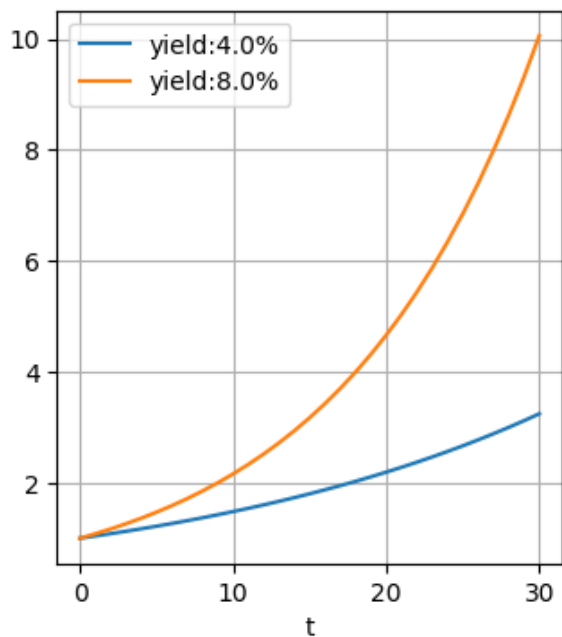
## 9.2 Composizione dei rendimenti/dell'interesse

### 9.2.1 Rendimenti costanti

**Composizione del rendimento: “se migliori l'1% ogni giorno, alla fine dell'anno sei un supereroe”**

- Il rendimento (o l'interesse, con reinvestimento se necessario) di un investimento su più intervalli temporale si compone con il prodotto. Una piccola differenza di rendimento sul singolo intervallo può avere effetti enormi su orizzonti temporali sufficientemente lunghi.
- Su orizzonti temporali lunghi, è misura minima di decenza e igiene usare la **scala logaritmica** nei grafici.

```
Final value after 30 periods:  
1-period yield: 0.04; wealth: 3.243  
1-period yield: 0.08; wealth: 10.063
```



## 9.2.2 Rendimenti non costanti - volatilità

- Concetti di base su composizione di rendimenti non costanti
- Volatility drag: la volatilità riduce il rendimento composto atteso

---

**Se perdi il 33.33%, successivamente devi guadagnare il 50% per tornare in pari**

- Se perdi il 33.33%, successivamente devi guadagnare il 50% per tornare in pari.

$$\begin{aligned}X_1 &= X_0(1 - .\bar{3}) = .\bar{6} X_0 \\X_2 &= X_1(1 + .5) = X_0\end{aligned}$$

- Se perdi il 50%, successivamente devi guadagnare il 100%

$$\begin{aligned}X_1 &= X_0(1 - .5) = .5 X_0 \\X_2 &= X_1(1 + 1.) = X_0\end{aligned}$$

- Se perdi il  $\ell\%$  prima e guadagni il  $g\%$  poi o se guadagni il  $g\%$  prima e perdi il  $\ell\%$  poi, alla fine il risultato è lo stesso

$$\begin{aligned}X_2^a &= X_1^a \left(1 + \frac{g}{100}\right) = X_0 \left(1 - \frac{\ell}{100}\right) \left(1 + \frac{g}{100}\right) \\X_2^b &= X_1^b \left(1 - \frac{\ell}{100}\right) = X_0 \left(1 + \frac{g}{100}\right) \left(1 - \frac{\ell}{100}\right) .\end{aligned}$$

- ...
- 

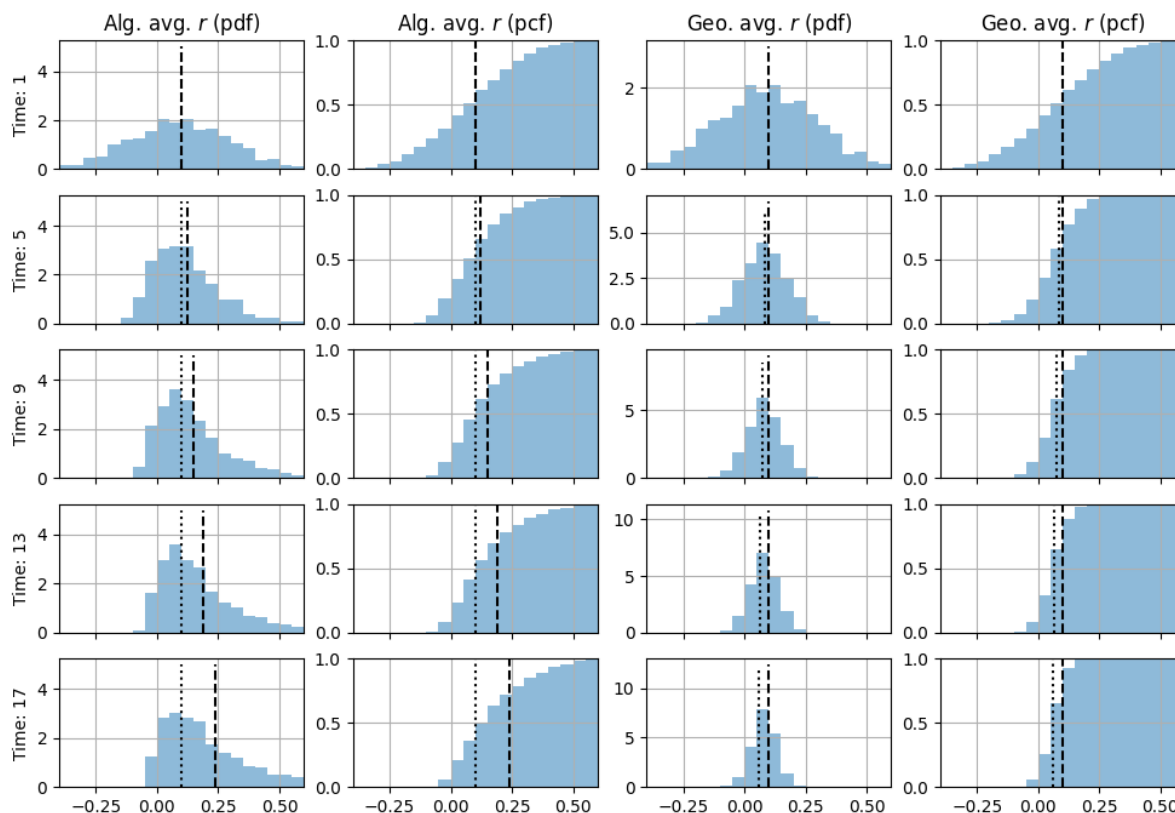
### Il rendimento composto produce un risultato asimmetrico

Anche un processo con rendimento di 1-periodo con probabilità simmetrica attorno a un valore medio, produce un **rendimento algebrico** medio con distribuzione di probabilità asimmetrica: in particolare si verifica la maggior parte degli eventi con risultati inferiori alla media, che risulta superiore alla mediana a causa di una minoranza di eventi estremamente positivi.

---

```
0.100000000000000009
0.122102000000000011
0.150883076777777799
```

```
0.18863624726100797
0.23849825205840852
```



L'esempio considera un investimento con densità di probabilità di 1-periodo con distribuzione gaussiana con valore atteso  $\mu_1 = .1$  e deviazione standard  $\sigma_1 = .2$

Le due righe mostrano i rendimenti algebrici (:) e geometrici (–) medi corrispondenti al rendimento atteso di un 1-periodo. La disuguaglianza  $(1 + r)^N > 1 + rN$  assicura l'**effetto compounding**, che la composizione di un rendimento  $r$  su  $N$  periodi produce un risultato migliore di rendimento algebrico  $rN$ .

Così, un rendimento algebrico medio del 10% corrisponde a un rendimento composto medio decrescente nel tempo

| Anni | CAGR |
|------|------|
| 1    | 10%  |
| 5    | 8.4% |
| 9    | 7.4% |
| 13   | 6.6% |
| 17   | 6.0% |

Viceversa un rendimento composto medio del 10% corrisponde a un rendimento algebrico medio crescente nel tempo (effetto compounding)

| Anni | Rendimento algebrico medio |
|------|----------------------------|
| 1    | 10%                        |
| 5    | 12.2%                      |
| 9    | 15.1%                      |
| 13   | 18.9%                      |
| 17   | 23.8%                      |

La prima colonna dei grafici mostra come la distribuzione di probabilità del rendimento algebrico medio diventa asim-



metrica.

La *terza colonna* mostra la distribuzione del rendimento geometrico: sotto opportune ipotesi, il rendimento geometrico tende a una distribuzione gaussiana con valore atteso  $\mu - \frac{\sigma^2}{2}$  (minore del rendimento di 1-periodo) e varianza  $\sim \frac{\sigma}{T}$  (che quindi tende a diminuire con il tempo, come mostrato dalla distribuzione più compatta, con dispersione minore al crescere del tempo) come discusso con qualche dettaglio matematico nella sezione a proposito del *rendimento composto*.

La *seconda* e la *quarta colonna* mostrano la funzione di probabilità cumulativa del rendimento algebrico e geometrico, rispettivamente. Questi grafici mostrano come il rendimento mediano sia inferiore al rendimento atteso della distribuzione di 1-periodo: per  $T = 5$ , circa il 60% delle realizzazioni ha rendimento negativo, per  $T = 17$ , oltre il 70% delle realizzazioni ha un rendimento negativo.

```
#> Realizations
# fig, ax = plt.subplots(2,1, figsize=(5,4))
# for (i_ret, cum_ret) in enumerate(cum_rets):
#     ax[0].plot(tv, cum_ret, color=plt.cm.tab10.colors[i_ret], lw=.2)
#     ax[1].semilogy(tv, cum_ret, color=plt.cm.tab10.colors[i_ret], lw=.2)
# ax[0].grid()
# ax[1].grid()
```

### Volatility-drag (I)

Una successione di eventi che comportano la perdita o il guadagno del 30%, nel tempo portano il patrimonio atteso a zero. Se la possibile perdita è del 50%, il patrimonio atteso tende a zero più velocemente.

```
#> Method 1. Successive independent Bernoulli = binomial distribution
from scipy.stats import binom

percs = np.array([.0, 1./30., .1, 1./3., .5, 2./3., .9, 1.-1./30., 1.])
n_percs = len(percs)
perc_pairs = [ [ i, n_percs-1-i ] for i in np.arange(int(np.floor((n_percs-1)/2))) ]

l1, b1 = binom.support(nt, p_win_1)
x = np.arange(l1, b1+1)
p = binom.cdf(x, nt, p_win_1)
y = ( 1. + gain_1 )**x * ( 1 + loss_1 )**(nt-x)

tv = np.arange(0, nt+1)
y_percs_m = []

for it in tv:
    #> Support, domain, and cdf of the Bernoulli distribution
    l1, u1 = binom.support(it, p_win_1) # lower, upper bound
    x = np.arange(l1, u1+1) # domain
    p = binom.cdf(x, it, p_win_1) # cumulative prob
    y_1 = ( 1. + gain_1 )**x * ( 1 + loss_1 )**(u1-x) # outcomes

    #>
    idx = np.searchsorted(p, percs, side='left')
    y_percs = y_1[idx]

    y_percs_m += [ y_percs ]

y_percs_m = np.array(y_percs_m)

fig, ax = plt.subplots(1,1, figsize=(5,3))
```

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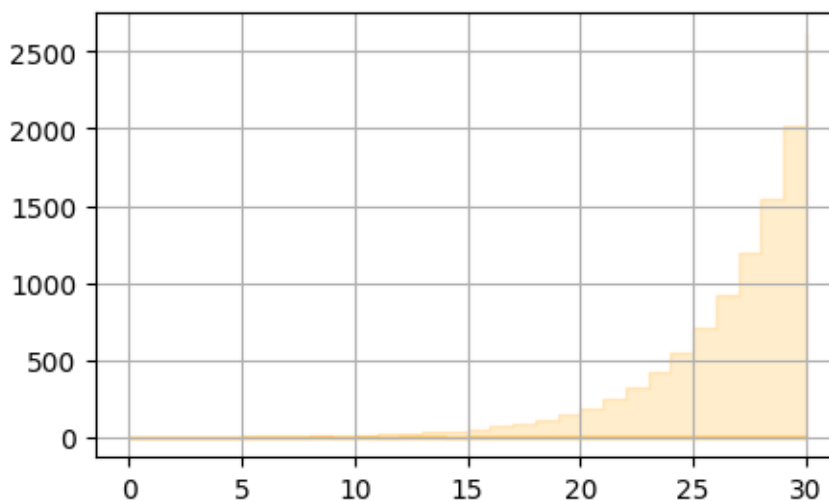
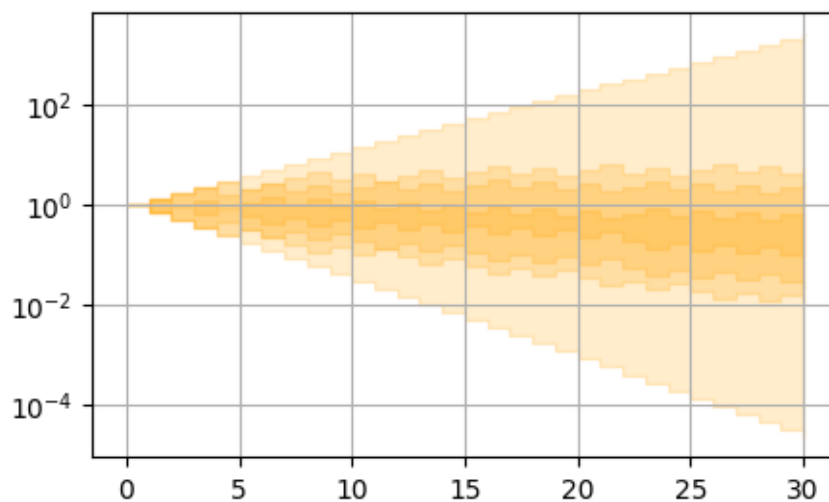
```

# ax.step(tv, y_percs_m, where='post', color='black', lw=.5)
for perc_pair in perc_pairs:
    ax.fill_between(tv, y_percs_m[:,perc_pair[0]], y_percs_m[:,perc_pair[1]], step=
        ↪ 'post', alpha=.2, color='orange')

ax.set_yscale('log')
ax.grid()

fig, ax = plt.subplots(1,1, figsize=(5,3))
# ax.step(tv, y_percs_m, where='post', color='black', lw=.5)
for perc_pair in perc_pairs:
    ax.fill_between(tv, y_percs_m[:,perc_pair[0]], y_percs_m[:,perc_pair[1]], step=
        ↪ 'post', alpha=.2, color='orange')
ax.grid()

```



```

# #> Method 2. Realizations of the stochastic processes
# rng = np.random.default_rng().choice
# rng_params_1 = { 'a': [gain_1, loss_1], 'p': [p_win_1, p_loss_1], 'size': nt }

```

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```
# rng_params_2 = { 'a': [gain_2, loss_2], 'p': [p_win_2, p_loss_2], 'size': nt }

# w1, w2 = [], []

# for ireal in np.arange(nreals):
#     w1 += [ np.array([ 1. ] + list(np.cumprod(1+rng(**rng_params_1)))) ]
#     w2 += [ np.array([ 1. ] + list(np.cumprod(1+rng(**rng_params_2)))) ]

# w1 = np.array(w1).T
# w2 = np.array(w2).T
```

### Volatility-drag (II)

A pari ritorno atteso di due investimenti, l'investimento con minore dispersione ha un rendimento composto maggiore.

Come discusso in dettaglio *più avanti*, il rendimento composto di un investimento con rendimento atteso  $\mu$  e deviazione standard dei rendimenti  $\sigma$  vale  $\mu - \frac{\sigma^2}{2}$ .

### La composizione di rendimenti simmetrici è asimmetrica

La composizione del rendimento di eventi simmetrici sul singolo intervallo temporale (es. probabilità uguale di guadagno o perdita di una stessa quantità) porta a un rendimento asimmetrico, *con skewness*, di solito con valore mediano inferiore al valore medio: tipicamente, il valore medio risulta superiore al valore mediano a causa di eventi estremamente positivi ma estremamente rari.

## 9.3 Diversificazione

### “Non mettere tutte le tue uova nello stesso paniere”

Citazione?

#### 9.3.1 Portafogli con 2 asset

Dati due asset con rendimento atteso  $\mu$ , e covarianza  $\sigma^2$ ,

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

$$\sigma^2 = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12}^2 \\ \sigma_{12}^2 & \sigma_{22}^2 \end{bmatrix} = \begin{bmatrix} \sigma_{11}^2 & \rho_{12}\sigma_1\sigma_2 \\ \rho_{12}\sigma_1\sigma_2 & \sigma_{22}^2 \end{bmatrix}$$

un portafoglio completamente investito, senza leva o short-selling, è identificato da i due pesi degli asset in portafogli

$$w_1, w_2 = 1 - w_1 \in [0, 1].$$

Il rendimento del portafoglio è la variabile casuale

$$r = w_1 r_1 + w_2 r_2 = w_1 r_1 + (1 - w_1) r_2,$$

e quindi il rendimento atteso è

$$\mu := \mathbb{E}[r] = \mathbb{E}[\dots] = w_1\mu_1 + (1 - w_1)\mu_2 ,$$

e la varianza è

$$\begin{aligned}\sigma^2 &= \mathbb{E}[(r - \mu)^T(r - \mu)] = \mathbf{w}^T \mathbb{E}[(\mathbf{r} - \mu)(\mathbf{r} - \mu)^T] \mathbf{w} = \\ &= \mathbf{w}^T \sigma^2 \mathbf{w} = \\ &= w_1^2 \sigma_{11}^2 + 2\rho_{12}\sigma_1\sigma_2 w_1 w_2 + w_2^2 \sigma_{22}^2 = \\ &= w_1^2 \sigma_{11}^2 + 2\rho_{12}\sigma_1\sigma_2 w_1(1 - w_1) + (1 - w_1)^2 \sigma_{22}^2 ,\end{aligned}$$

---

## PRINCIPLES OF INVESTING

### 10.1 Return

Return is the reward for investing. It can come from **capital gain** (price increase of assets bought), or **periodic cashflows**, like interest (from bonds), or dividends (from stocks). Some assets produce predictable return (either nominal, or real), other assets have less predictable returns. Any asset has some level of uncertainty, or *risk*<sup>1</sup>.

Most returns are quoted on a **per-period** basis - usually annually - and expressed as the percentage of the reward over the initial amount of the investment.

For a many-year investment, single-period returns **compound** over time.

#### 10.1.1 Costs

While return are uncertain, at least to a certain level, usually costs - fees, expenses, taxes - or part of them, are certain. With equal other conditions, the intelligent investor should reduce costs (known), as higher costs reduce returns w/o changing the level of risk.

### 10.2 Risk

Risk measures uncertainty and its effects, combining probability of events and consequences of specific events. *All the assets have some systematic and some specific risks*.

Key measures (*should give info about magnitude, frequency/probability, and duration*) include:

- standard deviation or **volatility**: how much returns may deviate from their expected value),
- max loss (usually 100% can't be neglected for catastrophic although rare events), value at risk (VaR, max loss with a given probability), drawdown (maximum peak-to-trough loss)
- time-to-recover (time to recover drawdowns, in a temporal perspective)

Usually, risk metrics measure uncertainty, without discerning from positive and negative events: these metrics perceive a higher-than-expected return as a risk as well. Some metrics instead, see *Sortino ratio* in *risk-return* section, aims at quantifying only negative events as risk.

---

<sup>1</sup> Even the most safe assets could undergo some (really) **rare**, but usually (really) **catastrophic events**. Just as an example, it's hard to imagine what could happen even to bonds issued by the most (perceived and priced) safe government or institution, in case of its participation in a war.

## 10.3 Risk-Return Trade Off

---

### “There’s no free lunch”

Higher expected returns usually come with higher risk.

---

---

### ...but high risk doesn’t imply high expected return

Very stupid actions usually implies poor return with high risk. Just as an example, playing Russian roulette for fun implies an expected return worse than an alternative “do-nothing and have an ice-cream instead” scenario (at least, if your goal is not to kill yourself, and your return function does not positively weight this outcome) with higher uncertainty on the final status of your health.

Sometimes the same could happen if one plays doing trading with some random meme-stocks or shit-coins.

---

**Risk-adjusted return** provides an indication of the expected return per unit of risk. Common metrics are:

- **Sharpe ratio**, comparing excess return and volatility compared with a “risk-free” asset - or a benchmark

$$S := \frac{\mathbb{E}[R - R_b]}{\sqrt{\text{var}[R - R_b]}}$$

- **Sortino ratio**

$$So := \frac{\mathbb{E}[R] - T}{\text{DR}},$$

with  $T$  target return, and DR the downside deviation, i.e. the deviation w.r.t the target return evaluated only for returns  $r$  lower than the target return  $T$

$$\text{DR}^2 = \int_{r=-\infty}^T (T - r)^2 f(r) dr,$$

being  $f(r)$  the probability density function of the (continuous) random variable  $R$  representing return

## 10.4 Diversification

Diversification spreads risk across different investments so no single event can ruin your portfolio. Diversification works well with assets that are not - or at least they’re loosely - correlated: in this case, diversification could increase return per unit of risk.

## 10.5 Portfolio Construction

## 10.6 Time

### 10.6.1 Compound Return

Return of multi-period investment compounds and not simply sums. Return  $r_n$  in the  $n^{th}$  period from  $n$  to  $n + 1$ , is defined as

$$x_{n+1} = x_n(1 + r_n).$$

Over  $N$  periods, the final wealth  $x_N$  can be written as a function of the initial wealth and the 1-year returns  $r_n$ ,

$$x_N = x_0 \prod_{n=1}^N (1 + r_n) .$$

**Geometric mean** provides by definition the value of the average — constant — return,  $\bar{r}$

$$x_N = x_0 \prod_{n=1}^N (1 + \bar{r}) = x_0 (1 + \bar{r})^N ,$$

and thus

$$\bar{r} = \left( \frac{x_N}{x_0} \right)^{\frac{1}{N}} - 1 = \left( \prod_{n=1}^N (1 + r_n) \right)^{\frac{1}{N}} - 1 .$$

**Logarithmic return.** By the very definition of the exponential function

$$\begin{aligned} \ln(\bar{r} + 1) &= \ln \left\{ \left( \prod_{n=1}^N (1 + r_n) \right)^{\frac{1}{N}} \right\} = \\ &= \frac{1}{N} \sum_{n=1}^N \ln(1 + r_n) \end{aligned}$$

While  $\ln(1 + r_n)$  is not Gaussian even if  $r_n$  is Gaussian, under the assumptions of the [central limit theorem](#),

$$\ln(\bar{r} + 1) \rightarrow \mathcal{N} \left( \mu_{\ln}, \frac{\sigma_{\ln}^2}{N} \right) ,$$

being  $\mu_{\ln}$  and  $\sigma_{\ln}^2$  the expected value and the variance of the [i.i.d. random variables](#)  $\ln(1 + r_n)$ . Given the probability density  $p(r)$  of the 1-period return  $r$ , these values read

$$\begin{aligned} \mu_{\ln} &= \int_r p(r) \ln(1 + r) dr \\ \sigma_{\ln}^2 &= \int_r p(r) (\ln(1 + r) - \mu_{\ln})^2 dr . \end{aligned}$$

For “sufficiently small” values of  $\bar{r}$ , the first-order expansion  $x \sim \ln(1 + x)$ , for  $x \rightarrow 0$ , provides a good approximation relating the logarithmic return and the geometric return, that thus have approximately Gaussian distribution for  $N$  “sufficiently large”,

$$\bar{r} \sim \ln(1 + \bar{r}) \rightarrow \mathcal{N} \left( \mu_{\ln}, \frac{\sigma_{\ln}^2}{N} \right) .$$

Is it possible to provide an approximate value of  $\mu_{\ln}$ ,  $\sigma_{\ln}$ , at least for *compact pdf* (even if not identically zero outside). Does this approximation match [GBM](#), [Geometric Brownian motion](#)? See the following dropdown.

### Approximated value for compact probability density

Let's broadly defined a compact probability density  $p(r)$  a probability density that is *very close to 0* outside a certain range, so small that a second-degree polynomial expansion of the function  $\ln(1 + r)$ ,

$$\ln(1 + r) \sim \ln(1 + \mu) + \frac{1}{1 + \mu} (r - \mu) - \frac{1}{2(1 + \mu)^2} (r - \mu)^2 + o((r - \mu)^3)$$

is a good approximation of the function.

**Expected value.** Taking expectation of the Taylor expansion,

$$\begin{aligned}\mu_{\ln} &:= \mathbb{E}[\ln(1+r)] = \\ &\sim \mathbb{E}\left[\ln(1+\mu) + \frac{1}{1+\mu}(r-\mu) - \frac{1}{2(1+\mu)^2}(r-\mu)^2\right] = \\ &= \ln(1+\mu) - \frac{\sigma^2}{2(1+\mu)^2}.\end{aligned}$$

**Variance.** Taking the expectation of the square of the deviation from the expected value,

$$\begin{aligned}\Delta \ln(1+r) &:= \ln(1+r) - \mathbb{E}[\ln(1+r)] = \\ &\sim \ln(1+r) - \ln(1+\mu) + \frac{\sigma^2}{2(1+\mu)^2} = \\ &\sim \frac{1}{1+\mu}(r-\mu) - \frac{1}{2(1+\mu)^2}(r-\mu)^2\end{aligned}$$

gives

$$\sigma_{\ln}^2 := \mathbb{E}[(\Delta \ln(1+r))^2] \sim \frac{\sigma^2}{(1+\mu)^2},$$

having neglected terms beyond the second degree.

**Approximation for small values of  $\mu$ .** Exploiting Taylor expansion

$$f(x) \sim f(x_0) + f'(x_0)x + f''(x_0)\frac{x^2}{2} + \dots,$$

for  $x_0 = 0$ ,

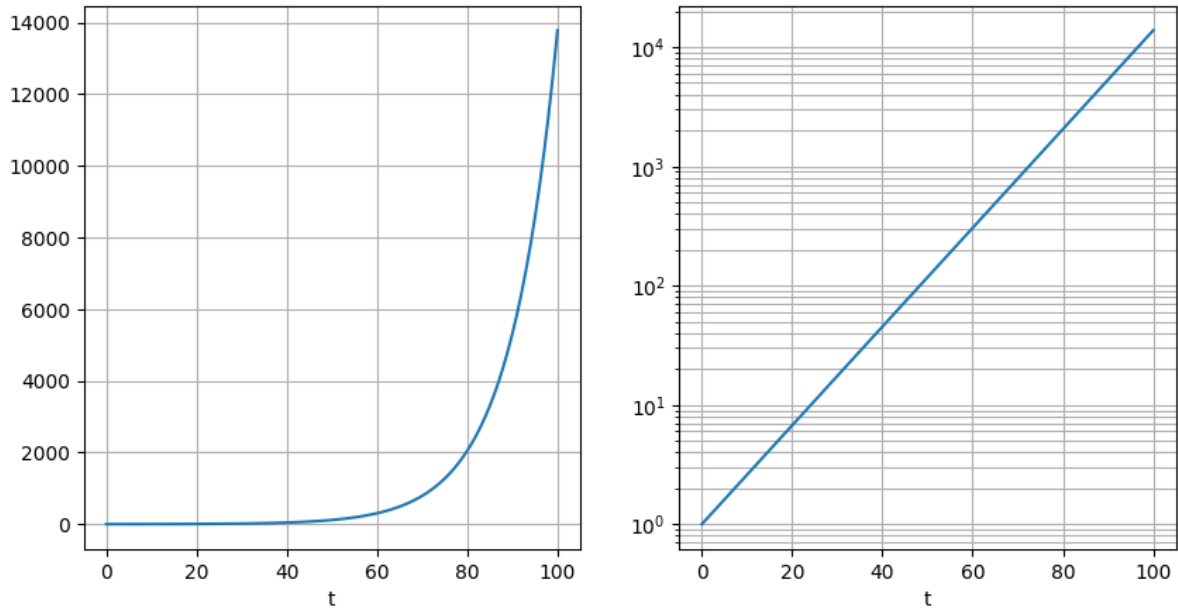
$$\begin{aligned}\ln(1+x) &\sim \left[\ln(1+x_0) + \frac{1}{1+x_0}x - \frac{1}{(1+x_0)^2}\frac{x^2}{2}\right]\Bigg|_{x_0=0} = 0 + x - \frac{x^2}{2} \\ \frac{1}{(1+x)^2} &\sim \left[\frac{1}{(1+x_0)^2} - \frac{2x_0}{(1+x_0)^3}x + \frac{-2+x_0}{(1+x_0)^4}\frac{x^2}{2}\right]\Bigg|_{x_0=0} = 1 - 0 \cdot x - x^2\end{aligned}$$

the low-degree approximation of the expected value and the variance of the logarithmic return reads

$$\begin{aligned}\mu_{\ln} &\sim \mu - \frac{\mu^2}{2} - \frac{\sigma^2}{2}(1-\mu^2) = \mu - \frac{\sigma^2}{2} - \frac{\mu^2}{2}(1-\sigma^2) \\ \sigma_{\ln}^2 &\sim \sigma^2 - \sigma^2\mu^2\end{aligned}$$

```
(Text(0.5, 0, 't'), None)
```

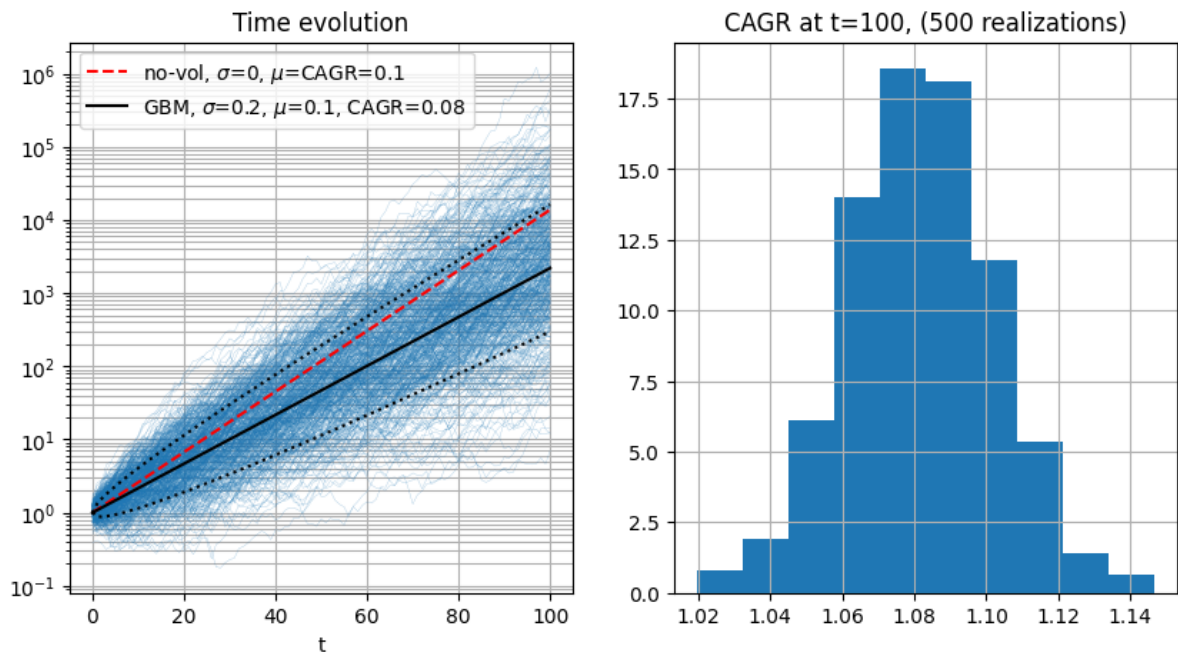




### Volatility Drag

Under the assumption of normal distribution of 1-period return, the price of an asset with constant expected return  $\mu$  and variance of returns  $\sigma$  can be modelled as a **geometric Brownian motion**, whose compound 1-period (usually annual, and thus the acronym *CAGR*, *compound annual growth rate*) growth rate has expected value

$$CAGR = \mu - \frac{\sigma^2}{2}.$$



todo

- "Time and risk?" Listen to The Logic of Risk

## 10.7 Disciplined Investing

### 10.7.1 Rebalancing

Colab Notebook, [rebalancing.ipynb](#)

#### Rebalancing premium

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## REBALANCING

While investing should not be confused with gambling, some games, gambling and betting strategies could provide useful toy-problems and introduction to investing principles. This is the case of gambling on a coin flip game that is discussed here and used to introduce the concept of **rebalancing**.

Rebalancing:

- reduces the dispersion of the composite return
- improves risk-adjusted return
- may also improve the expected value of the composite return as well (**rebalancing premium**, or **Shannon's demon**), for processes that meet some particular conditions

### Contents

*Example: coin flip game*

*Resources*

## 11.1 Example: coin flip game

In this section, the effect of rebalancing is discussed in a coin flip game. This game can be interpreted as a very simple model of a 2-asset portfolio, with 1 risky asset with only two outcomes (win or loss), and 1 safe asset (no *real* return).

---

### Example 11.1.1 (Shannon demon in a coin flip game)

Starting with 100€, and a fair coin with 50% probability of for each outcome, either H: head or T: tail. If outcome is H you gain 50%, if the outcome is T you lose 33.3%.

Let's evaluate two different strategies:

1. play with all the money you have
2. at every toss, bet 50% of the amount you have

What's the expected amount at the end of the game? What's the amount distribution? ...

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Here the problem is investigated for a set of different strategies uniquely determined by different values of the betting fraction  $f$ .

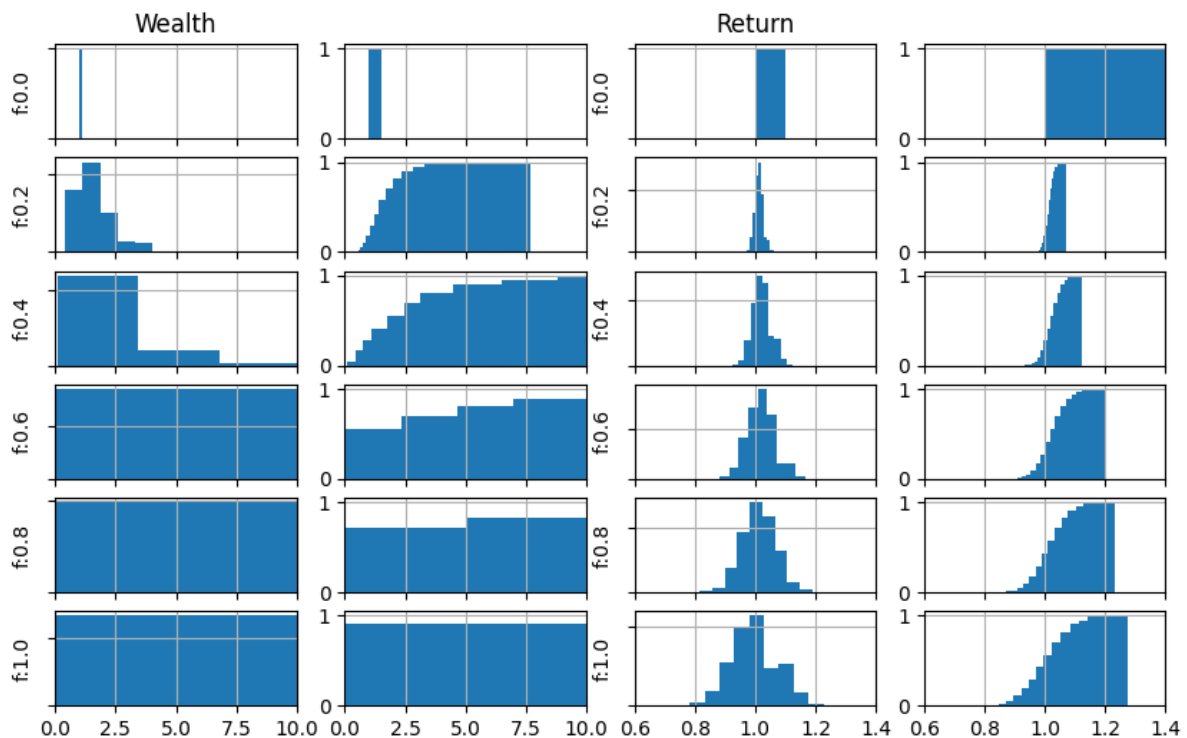
As discussed below, the *optimal fraction* (optimal in the sense of maximum expected geometric return; but does this definition of optimum meet your taste — e.g. your risk tolerance to negative results and dispersion of the results?) is

$$f^* = \frac{p}{l} - \frac{1-p}{g} = \frac{\frac{1}{2}}{\frac{1}{3}} - \frac{\frac{1}{2}}{\frac{1}{2}} = \frac{1}{2} = .5.$$

```
fraction, avg wealth, avg geo return, n.negative outcomes
f:0.0, w_avg: 1.000, r_avg: 1.000, n_neg: 0
f:0.2, w_avg: 1.640, r_avg: 1.013, n_neg: 356
```

```
f:0.4, w_avg: 2.715, r_avg: 1.021, n_neg: 571
f:0.6, w_avg: 4.588, r_avg: 1.022, n_neg: 551
```

```
f:0.8, w_avg: 6.172, r_avg: 1.014, n_neg: 871
f:1.0, w_avg: 9.417, r_avg: 1.002, n_neg: 857
```



#### todo

- Improve plot quality: very poor hist outcome
- Discuss:
  - difference between algebraic and geometric return (compounding)
  - probability of negative outcomes and dispersion of the results
  - probability of intermediate negative outcomes and dispersion of results (time effect and volatility)
  - comparison with no-rebalancing strategy

**Example 11.1.2 (Interludio: is the coin fair?)**

After tossing a coin 10 times and getting 10 Heads in a row, would you bet on Head or Tail?

Although every toss is a statistically independent event, a *very* unlikely series of outcomes under the assumption of a fair coin may induce some doubts about the truthfulness of this assumption.

See [here: Inferential statistics - Hypotesis testing \(Fisher\) - Is the coin fair?](#) as an example. If you want to play a bit with the notebook, just open it in Colab: the script is originally meant for  $n = 30$  coin flips; just 1) set `n_flips = 10`; 2) run it again; 3) get the value of the extreme event probabability (either 10 Heads or 10 Tails) to address the question of the interludio,

i.e. the probability of getting 10 consecutive outcomes in the first 10 flips with a fair coin is less than 0.1%. A pretty low probability that the null hypotesis “the coin is fair” holds...

[Here](#) for the application of Fisher criterion for hypotesis testing for two coins with  $(p_H, p_T)^1 = (0.5, 0.5)$  and  $(p_H, p_T)^2 = (0.45, 0.55)$ .

**Example 11.1.3 (Kelly criterion for the coin flip game)**

Let  $p$  the probability of a win,  $q = 1 - p$  the probability of a loss,  $g$  is the net fraction gained in a win,  $l$  is the net fraction lost in a loss. The strategy is simple: always bet the fraction  $f$  of the amount you have. Is there an optimal value  $f^*$  that maximizes the expected return?

**todo** Clean this section containing mathematical details about Kelly's criterion.

**Amount after  $n$  coin tosses**

$$x_n = \dots$$

**Geometric return**

$$\ln \frac{x_n}{x_0} = \dots$$

**Optimization of the expected value of the geometric return**

$$\frac{1}{n} \frac{d}{df} \mathbb{E} \left[ \ln \frac{x_n}{x_0} \right] = \dots$$

- free or constrained optimization,...

Starting with the amount  $x_0$ , the expected amount after 1 toss reads

$$\mathbb{E}[x_1] = x_0(1 - f) + fx_0(1 + g) + fx_0(1 - l),$$

while the actual amount,

the amount after 1 coin toss is a random variable depending on the result of the toss, represented by a  $X_1 \in \{0, 1\}$  of the coin toss, 0: loss, 1: win, reads

$$x_1 = x_0 (1 + fg)^{X_1} (1 - fl)^{1-X_1}$$

After the second toss,

$$x_2 = x_1 (1 + fg)^{X_2} (1 - fl)^{1-X_2}$$

and after the  $n^{th}$  toss

$$x_n = x_0 \prod_{i=1}^n (1 + fg)^{X_i} (1 - fl)^{1-X_i}$$

The maximization of the  $\ln \frac{x_n}{x_0}$  (equivalent to the maximization of  $\frac{x_n}{x_0}$ ),

$$\ln \frac{x_n}{x_0} = \sum_{i=1}^n \{X_i \ln(1 + fg) + (1 - X_i) \ln(1 - fl)\}$$

w.r.t.  $f$  gives

$$\frac{d}{df} \ln \frac{x_n}{x_0} = \sum_{i=1}^n \left\{ \frac{X_i g}{1 + fg} - \frac{(1 - X_i)l}{1 - fl} \right\}$$

The **expected value** of the logarithm of the ratio reads

$$\mathbb{E} \left[ \ln \frac{x_n}{x_0} \right] = n \{p \ln(1 + fg) + (1 - p) \ln(1 - fl)\} ,$$

and its derivative w.r.t.  $f$  reads

$$\frac{d}{df} \mathbb{E} \left[ \ln \frac{x_n}{x_0} \right] = n \left\{ \frac{pg}{1 + fg} - \frac{(1 - p)l}{1 - fl} \right\}$$

and becomes zero (so that  $\mathbb{E}[x_n] = x_0$  and return is zero) when

$$\begin{aligned} 0 &= pg(1 - f^*l) - (1 + f^*g)(1 - p)l = \\ &= pg - pgf^*l - l - f^*gl + pl + f^*gpl = \\ &= pg - l - f^*gl + pl , \end{aligned}$$

and

$$f^* = \frac{pg - l + pl}{gl} = \frac{p}{l} - \frac{1 - p}{g} ,$$

**Is this an maximum?** Compute the second-order derivative  $\frac{d^2}{df^2} \mathbb{E} \left[ \ln \frac{x_n}{x_0} \right]$ , evaluate it in  $f^*$  and check if it's  $< 0$ . By direct computation, the second order derivative

$$\frac{1}{n} \frac{d^2}{df^2} \mathbb{E} \left[ \ln \frac{x_n}{x_0} \right] = -\frac{pg^2}{(1 + fg)^2} - \frac{(1 - p)l^2}{(1 - fl)^2} ,$$

is always negative for  $p \in [0, 1]$ , i.e. for all the possible values of the win probability of the probability distribution. Thus, the extreme point found above is a maximum.

**First order derivative for  $f = 0$ .**

$$\frac{1}{n} \frac{d}{df} \mathbb{E} \left[ \ln \frac{x_n}{x_0} \right] \Bigg|_{f=0} = pg - (1 - p)l = p(g + l) - l$$

The condition for the first derivative in  $f = 0$  to be positive is equivalent to the condition of  $f^* > 0$ , i.e. optimal fraction is feasible,  $f^* \in [0, 1]$ .

**Is this maximum feasible?** If *no short* (betting against,  $f \geq 0$ ) and *no leverage* (betting more than you have,  $f \leq 1$ ) betting is possible, the fraction  $f$  must lie in range  $[0, 1]$ : the maximization problem is not a free optimization problem, but a simple constrained optimization problem. In these problems — under the assumptions of sufficient regularity of the function — extreme points may occur where first order derivative is zero or at the boundary of the domain.

Thus:

- either the maximum is at  $f^*$ , if  $f^* \in [0, 1]$ ,  $0 \leq p(l + g) - l \leq lg$
- or the maximum is at  $f^* = 0$ : the probability is against you, so you'd better do nothing
- or the maximum is at  $f^* = 1$ : the probability is with you, so you'd better bet (if your **risk tolerance** accepts the **dispersion of the results**)

If *short* is allowed, there's no constraint  $f \geq 0$ ; if *leverage* betting is allowed, there's no constraint  $f \leq 1$ .

**What's the optimal expected value?** Evaluate  $\mathbb{E} \left[ \ln \frac{x_n}{x_0} \right] (f^*)$

$$\begin{aligned} \frac{1}{n} \mathbb{E} \left[ \ln \frac{x_n}{x_0} \right] (f^*) &= p \ln \left( 1 + \frac{pg}{l} - 1 + p \right) + (1 - p) \ln \left( 1 - p + \frac{(1 - p)l}{g} \right) = \\ &= p \ln \left( p \left( 1 + \frac{g}{l} \right) \right) + (1 - p) \ln \left[ (1 - p) \left( 1 + \frac{l}{g} \right) \right] = \\ &= \ln \left\{ \left[ p \left( 1 + \frac{g}{l} \right) \right]^p \left[ q \left( 1 + \frac{l}{g} \right) \right]^q \right\} \end{aligned}$$

## 11.2 Resources

- *The Bull* podcast. Puntate:
  - **217.** Il modo migliore per Ribilanciare il portafoglio
  - **152.** La magia del ribilanciamento e il demone di Shannon
  - **117.** Come ribilanciare il portafoglio (e previsioni per i prossimi 10 anni)
- *R.Arnett*, over-rebalancing
- *Market sentiment, Shannon's demon*





## SEQUENCE RISK

### 12.1 Introduction

Sequence risk occurs when investment or withdrawal is distributed in time. These two scenarios may be representative of:

- Dollar Cost Averaging (**DCA**, or **PAC** in Italian for “Piano di Accumulo di Capitale”)
- **Withdrawal** during old age

Sequence in time of 1-period returns may strongly influence the composite return of a portfolio.

#### 12.1.1 Mathematical model

In a continuous-time model, sequence risk of constant-amount DCA or withdrawal can be modeled with a Geometric Brownian Motion with “drift”,

$$dX_t = \mu X_t dt + \sigma X_t dW_t + C dt ,$$

being  $C_t$  the rate of investment ( $> 0$ ) or withdrawal ( $< 0$ ),  $\mu, \sigma$  the expected value and variance of the rate of return. A discrete-time counterpart may be

$$\Delta X_{n,n+1} = (\mu_{n,n+1} + \sigma_{n,n+1} W_{n,n+1}) X_n + C_{n,n+1} ,$$

with  $\mu_{n,n+1}, \sigma_{n,n+1}$  the expected value and the variance of the 1-period return,  $W_{n,n+1}$  a unit-variance random variable representing the distribution of the returns from  $n$  to  $n + 1$ , and  $C_{n,n+1}$  the investment or withdrawal from  $n$  to  $n + 1$ .

#### 12.1.2 Constant investment or withdrawal rate: analytical solution

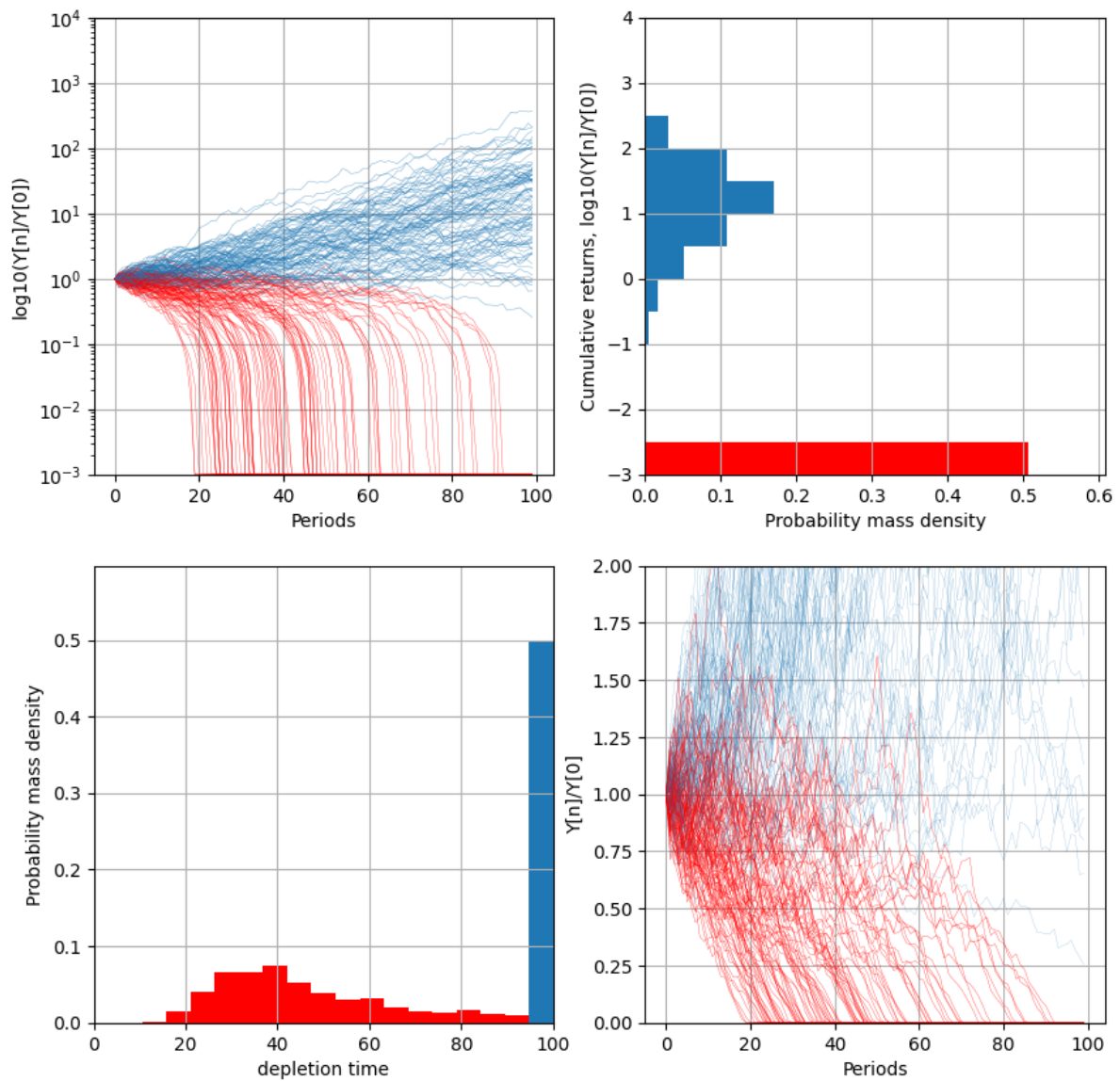
The solution of the continuous-time equation with reads

$$X_t = X_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t} + C \int_{s=0}^t e^{\left(\mu - \frac{\sigma^2}{2}\right)(t-s) + \sigma(W_t - W_s)} ds$$

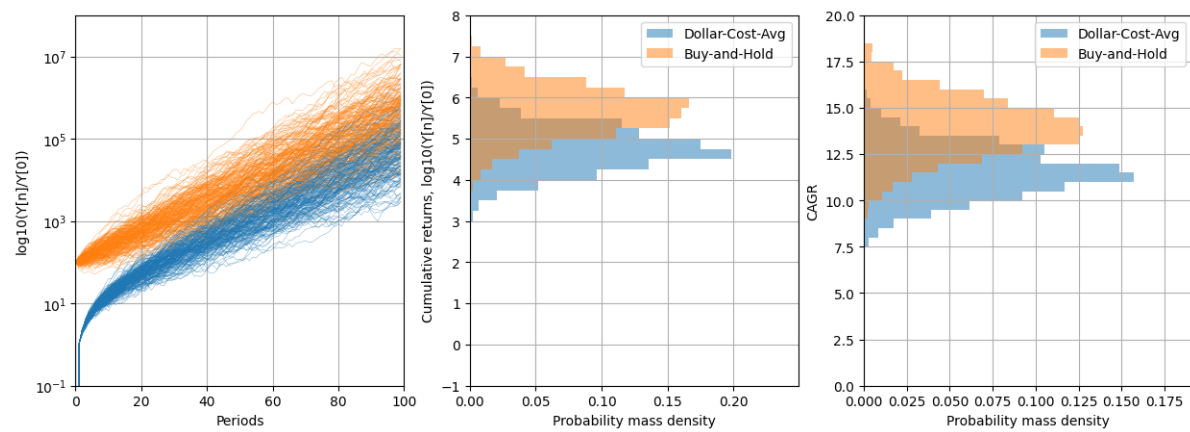
## 12.2 Realizations

### 12.2.1 Libraries and parameters

### 12.2.2 Constant Withdrawals



### 12.2.3 Dollar Cost Averaging (DCA)





## **Part IV**

# **Asset classes**



## INTRODUCTION TO ASSET CLASSES





## BONDS

**What is a bond? ...**

**Characteristics of a bond. ...**

*Return*

*Risks*

### 14.1 Return

Here the most general expression for nominal and real **yield** are derived as a function of prices, face value of coupon, taxation and year to maturity, both in case of coupon reinvestment or not (reinvestment not always possible); a closed form solution is then derived under some assumptions, like constant (or average) rates; the effect on price and yield of credit rating and rating change, coupon, year to maturity are discussed on both examples and real-world cases.

Extra:

- definition of duration
- risks: inflation; reinvestment (at lower rates) for bonds with same maturity and different coupons
- inflation linked

#### 14.1.1 Constant coupon bonds

**W/o reinvestment**

At time  $t_0$  the unit price of a bond is  $p_0$ ; investing  $Y_0$  allows to buy  $N_0 = \frac{Y_0}{p_0}$  titles; each title has the right of receiving net coupon  $C(1-t)$ , with  $t$  taxation rate, per period (here assumed 1-year coupon range).

$$N_0 = \frac{Y_0}{p_0} = \frac{Y_0}{p_{in}} \frac{p_{in}}{p_0}$$

W/o reinvestment, the number of titles hold is constant and equal to  $N_0$ . As capital  $Y_i$  can be written as the product of unit price and number of bond in portfolio, the DCF of a bond w/o coupon reinvestment reads

$$\begin{aligned} \widetilde{DCF} &= -Y_0 + Y_N \prod_{k=1}^N (1+r_k)^{-1} + \sum_{k=1}^N N_0 C(1-t) \prod_{j=1}^k (1+r_j)^{-1} \\ &= N_0 \left[ -p_0 + p_N \prod_{k=1}^N (1+r_k)^{-1} + C(1-t) \sum_{k=1}^N \prod_{j=1}^k (1+r_j)^{-1} \right], \end{aligned}$$

This DCF must be corrected a CF at time  $t_N$  corresponding to tax on capital gain if  $p_N > p_0$ , discounted as

$$-N_0(p_N - p_0)t \prod_{k=1}^N (1 + r_k)^{-1} \quad (\text{only if } p_N > p_0)$$

The cumulative real return (if the discount ratio is inflation) is the ratio between the  $DCF$  and the actual value of the investment  $Y_0$ ,

$$\frac{\widetilde{DCF}}{Y_0} = -1 + \frac{p_N}{p_0} \prod_{k=1}^N (1 + r_k)^{-1} + \frac{C}{p_0} (1 - t) \sum_{k=1}^N \prod_{j=1}^k (1 + r_j)^{-1}$$

If the discount rate is constant, or the average (which average) discount rate is used, the expression of the cumulative return reads

$$\frac{\widetilde{DCF}}{Y_0} = -1 + \frac{p_N}{p_0} (1 + r)^{-N} + \frac{C}{p_0} (1 - t) \sum_{k=1}^N (1 + r)^{-k}$$

### W/ reinvestment

| Time    | Cashflows             | $\Delta$ Quantity                              | Quantity                           | DF                                 |
|---------|-----------------------|------------------------------------------------|------------------------------------|------------------------------------|
| 0       | $-Y_0$                | $N_0 = \frac{Y_0}{p_0}$                        | $N_0 = \frac{Y_0}{p_0}$            | 1                                  |
| 1       | $+N_0 C(1 - t)$       |                                                |                                    | $(1 + r_1)^{-1}$                   |
| 1       | $-N_0 C(1 - t)$       | $N_1 = \frac{N_0 C(1 - t)}{p_1}$               | $N_{0:1} = N_0 + N_1$              | $(1 + r_1)^{-1}$                   |
| 2       | $+N_{0:1} C(1 - t)$   |                                                |                                    | $(1 + r_1)^{-1} (1 + r_2)^{-1}$    |
| 2       | $-N_{0:1} C(1 - t)$   | $N_2 = \frac{N_{0:1} C(1 - t)}{p_2}$           | $N_{0:2} = N_0 + N_1 + N_2$        | $(1 + r_1)^{-1} (1 + r_2)^{-1}$    |
| ...     |                       |                                                |                                    |                                    |
| $T - 1$ | $+N_{0:T-2} C(1 - t)$ |                                                |                                    | $\prod_{k=1}^{T-1} (1 + r_k)^{-1}$ |
| $T - 1$ | $-N_{0:T-2} C(1 - t)$ | $N_{T-1} = \frac{N_{0:T-2} C(1 - t)}{p_{T-1}}$ | $N_{0:T-1} = \sum_{k=0}^{T-1} N_k$ | $\prod_{k=1}^{T-1} (1 + r_k)^{-1}$ |
| $T$     | $+N_{0:T-1} C(1 - t)$ |                                                |                                    | $\prod_{k=1}^T (1 + r_k)^{-1}$     |
| $T$     | $+N_{0:T-1} p_T$      |                                                |                                    | $\prod_{k=1}^T (1 + r_k)^{-1}$     |

All the cashflows from coupons are immediately reinvested so the DCF is

$$\begin{aligned}
 DCF &= -Y_0 + \underbrace{N_{0:T-1} (p_T + C(1 - t))}_{Y_T} \underbrace{\prod_{k=1}^T (1 + r_k)^{-1}}_{DF_T} = \\
 &= -Y_0 + Y_T DF_T,
 \end{aligned}$$

with

$$\begin{aligned}
 N_{0:T-1} &= N_{0:T-2} + N_{T-1} = N_{0:T-2} + N_{0:T-2} \frac{C(1-t)}{p_{T-1}} = N_{0:T-2} \left[ 1 + \frac{C(1-t)}{p_{T-1}} \right] = \\
 &= N_{0:T-3} \left[ 1 + \frac{C(1-t)}{p_{T-2}} \right] \left[ 1 + \frac{C(1-t)}{p_{T-1}} \right] = \\
 &= \dots = \\
 &= N_{0:1} \prod_{k=2}^{T-1} \left[ 1 + \frac{C(1-t)}{p_k} \right] = \\
 &= N_0 \prod_{k=1}^{T-1} \left[ 1 + \frac{C(1-t)}{p_k} \right].
 \end{aligned}$$

Cumulative discounted return reads

$$\begin{aligned}
 \frac{DCF}{Y_0} &= -1 + \frac{Y_T}{Y_0} DF_T = \\
 &= -1 + \frac{N_0}{N_0 p_0} \prod_{k=1}^{T-1} \left( 1 + \frac{C(1-t)}{p_k} \right) (p_T + C(t-1)) DF_T \\
 &= -1 + \frac{p_T}{p_0} \prod_{k=1}^T \left( 1 + \frac{C(1-t)}{p_k} \right) DF_T \\
 &= -1 + \frac{p_T}{p_0} \prod_{k=1}^T \left( \frac{1 + \frac{C(1-t)}{p_k}}{1 + r_k} \right).
 \end{aligned}$$

Composite discounted return is obtained, after writing the diiscounted cashflow as the difference between discounted cashflow at time  $t_T$  and  $t_0$ ,  $DCF = Y_T DF_T - T_0$ ,

$$\begin{aligned}
 (1 + DCAGR)^T &= \frac{Y_T DF_T}{Y_0} = \frac{DCF}{Y_0} + 1 = \frac{p_T}{p_0} \prod_{k=1}^T \left( \frac{1 + \frac{C(1-t)}{p_k}}{1 + r_k} \right) \\
 DCAGR &= \left( \frac{p_T}{p_0} \prod_{k=1}^T \frac{1 + \frac{C(1-t)}{p_k}}{1 + r_k} \right)^{\frac{1}{T}} - 1
 \end{aligned}$$

If<sup>1</sup> price of the bond is constant throughout its whole life,  $p_k = 1$ ,  $\forall k = 0 : T$ , and discount rate  $r$  is constant, the number of held bonds at time  $T - 1$  is

$$N_{0:T-1} = N_0 (1 + C(1-t))^{T-1},$$

the discounted cashflow is

$$\begin{aligned}
 DCF &= -N_0 + N_0 (1 + C(1-t))^{T-1} (1 + C(1-t)) (1+r)^{-T} = \\
 &= N_0 \left[ -1 + \left( \frac{1 + C(1-t)}{1+r} \right)^T \right],
 \end{aligned}$$

cumulative discounted return

$$\frac{DCF}{Y_0} = -1 + \left( \frac{1 + C(1-t)}{1+r} \right)^T$$

<sup>1</sup> It's a big if. Even if credit rating and inflation are constant throughout the life of the bond, years to maturity decreases and thus - usually - the required rate decreases as well.

and the composite discounted return reads

$$DCAGR = \frac{1 + C(1 - t)}{1 + r} - 1 .$$

---

## 14.2 Risks

- Inflation risk
- Interest rate fluctuations
- Reinvestment risk
- Rating, credit and default risks
- ...

### 14.2.1 Inflation risk

Independently from any other mechanism, a change in inflation alters the real return of constant nominal yield bonds. Just as an example, if you buy a 3% nominal rate bond with expected inflation at 2%, you're aiming at 1% real return. If average inflation grows and remains constant at 4%, you get a negative -1% return.

---

#### Example 14.2.1 (Zero-coupon bond)

10-year  $r = 3\%$  nominal net yield bond, with expected inflation at target inflation  $i = i^* = 2\%$ .

- What's the probability of getting at least the expected (and desired if you buy it?) real yield?
  - What's the probability of getting negative real yield?
  - What's the nominal net yield required to halve the probability of getting a real return lower than an expected 1% CAGR?
- 

```
Final price: 100.000  
Initial price to get compound return r=0.03: 74.409
```

## 14.3

## EQUITY

### What's equity?

### Contents

*Valuation methods.* Comparison and intrinsic value methods.

*Financial statements.* Introduction to financial statements of a company.

## 15.1 Equity Valuation

### Detailed introduction

Equity valuation blends common sense, mathematics, expectations, estimation—and a bit of art. Buying shares in a company, whether directly or through a fund, means owning a (tiny) stake in a real business that produces goods and/or services and has the potential to generate earnings or free cash flows. As a shareholder, you are not just investing in market prices—you're becoming a part-owner of the enterprise. This ownership entitles you to a share of the company's profits through dividends or capital appreciation. It also comes with certain rights and responsibilities, especially during difficult periods.

When companies face financial stress or pursue growth opportunities, they may issue new shares to raise capital. This can lead to dilution, reducing the percentage ownership of existing shareholders. However, shareholders often have preemptive rights, allowing them to participate in new issuances to maintain their ownership stake. Moreover, owning equity means having a claim on the residual value of the company—what's left after all debts are paid—in both prosperous and challenging times. Understanding these dynamics is crucial to valuing equity: you're not just buying into today's performance, but into a stream of future cash flows and the complex, evolving structure of ownership.

**Sensitivity analysis** could provide an estimate of the effects of different parameters/assumptions on the final result.

**Different valuation methods** exist, and can be broadly classified in

- **comparison** approach: P/E, EV/EBITDA, or other indices used to compare companies of the same sector, marked, dimension,...<sup>1</sup>
- **intrinsic value** approach, based on **DCF**
- ...other methods for general firms (cost approach,...); valuation of financials;...

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<sup>1</sup> It's not always possible to find "equivalent" companies for the comparison...; P/E, EV/EBITDA,... would be projected into the future to keep into account future in the value of a firm.

### 15.1.1 Comparison

### 15.1.2 Intrinsic value

- Future cash flows are estimated,
- CFs are discounted, usually for the *WACC* (Weighted Average Cost of Capital) to find the *NPV* (net present value) of the **enterprise value** *EV*
- Cash and equivalents are added to the *NPV* to find the **equity value**

---

*WACC*

$$WACC = \frac{E}{V}R_e + \frac{D}{V}R_d(1 - t)$$

being  $R_e$  the **cost of equity** and  $R_d$  the **cost of debt** (maybe the easiest part to estimated accurately, since the debt structure is usually known/programmed). The factor  $(1 - t)$  usually appears as interest payments are tax-deductible.

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#### Equity Risk Premium $R_e$ - Sharpe

Following W.Sharpe, equity risk premium can be estimated as

$$R_e = R_f + (R_m + R_f)\beta,$$

being  $R_f$  the risk-free rate (usuallly 10Y US Treasuries), and  $R_m$  the annual return of the market/sector of the investment,  $\beta$  is a measure of risk or stock volatility of returns of the investment relative to that of the market/sector.

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## 15.2 Three Financial Statements

Financial statements are written records that illustrates the business activities and the financial performance of a company. In most cases they are audited to ensure accuracy for tax, financing, or investing purposes.

**Uses.** *Management* uses them for decision-making, budgeting and performance evaluation. *Investors* use them to asses profitability, financial health, future performance, and creditworthiness (especially *lenders*).

- **Income statement:** company performance (profit and loss) over a period. Broadly speaking:

$$\text{net earnings} = (\text{revenues} - \text{total expenses}) \times (1 - \text{tax rate}),$$

with total expenses = operative (labor + non-labor + DA) + Interest (due to debt holders), and “partial earnings” EBITDA, EBIT, EBT with trivial definition (Earnings Before: I:interest, DA: depreciation and amortization, T: tax)

- **Balance sheet:** financial position at a specific point in time, in terms of:
  - assets: cash and equivalent + acc.receive. + inventory + PPE (Plant property and equipment, subject to CapEx and depreciation,  $PPE(n) = PPE(n - 1) + \text{CapEx}(n) + \text{DA}(n)$ )
  - liabilities: debt + acc.pay.

- equity:

$$\text{retained earnings}(n) = \text{retained earnings}(n-1) + \text{net earnings}(n) - \text{dividends}(n)$$

$$\text{shareholder equity}(n) = \text{equity capital}(n) + \text{retained earnings}(n),$$

being retained earnings( $n$ ) the **cumulative** retained earnings not distributed to shareholders.

The 2 contributions shareholder equity, total liabilities shows how the company's asset are financed: either through capital raised or retained earnings (equity), or through debt (liabilities). The **identity**

$$\text{total liabilities} + \text{shareholders equity} = \text{total asset}$$

must hold in a proper filled balance.

- **Cash flow statement** tracks the flows of cash in and out of the business over a period. Cashflows over a period modifies cash,

$$\text{closing cash}(n) = \text{opening cash}(n) + \text{total cashflow}(n)$$

$$\text{opening cash}(n) = \text{closing cash}(n-1)$$

Cashflows are usually classified as:

- operating CF (DA is added back to net income, since it's not a cashflow going anywhere; it lowers income, but it's not a cashflow)
- investing CF
- financing CF

$$\text{Op.CF}(n) = \text{net earnings}(n) + \text{DA}(n) - \Delta \text{WC}(n)$$

$$\text{Inv.CF}(n) = \text{investment in PPE}(n)$$

$$\text{Fin.CF}(n) = \text{issuance of debt}(n) + \text{issuance of equity}(n) - \text{dividends}(n)$$

being  $\text{WC}(n) = \text{acc.rec}(n) + \text{inventory}(n) - \text{acc.pay}(n)$  the **working capital**.





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CHAPTER  
**SIXTEEN**

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**ETFS**



## **Part V**

# **Asset allocation and Investing**



**INTRODUCTION TO INVESTING**



## MODERN PORTFOLIO THEORY AND CAPITAL ASSET PRICING MODEL

Modern Portfolio Theory (MPT), introduced by Harry Markowitz<sup>1</sup>, operates under a simple premise: a rational investor wants to either minimize risk (variance) for a given level of expected return, or maximize expected return for a given level of risk.

MPT treats risk as the volatility - the variance - of the portfolio return, for a given desired expected value of the return. MPT shows how assets can interact with each others, and shows the effect of **diversification** in a portfolio with non-correlated assets.

### 18.1 Modern Portfolio Theory

**Assets.** There are  $N$  available assets, whose  $\mathbf{r} = (r_i)$  return over the period of interest has

- expected value  $\mu = (\mu_i) = \mathbb{E}(r_i)$ ,
- covariance matrix  $\sigma^2 = (\sigma_{ij}^2) = \mathbb{E}[(r_i - \mu_i)(r_j - \mu_j)]$ .

Given a set of  $N$  assets with expected return  $\mu_i = \{\mu\}_i$ ,  $i = 1 : N$ , and covariance (matrix)  $\sigma^2$ , a portfolio is defined by the weights (percentage)  $w_i = \{\mathbf{w}\}_i$  of the individual assets in the portfolio.

**Return of the portfolio.** The return of an amount  $x_0$  invested in a portfolio with asset allocation  $w_i$  is found evaluating the amount  $x_1$  after the period for which the return is defined,

$$x_1 = \sum_{i=1}^N x_0 w_i (1 + r_i) .$$

Not-fully invested portfolio can be represented with an asset 0 (cash with zero return). The return is defined as

$$r = \frac{x_1 - x_0}{x_0} = \frac{x_1}{x_0} - 1 = \sum_{i=1}^N w_i (1 + r_i) - 1 .$$

If (or as? Even with not-fully invested portfolio, or portfolio w/leverage)  $\sum_{i=1}^N w_i = 1$ , the return of the portfolio reads

$$r = \sum_{i=1}^N w_i r_i = \mathbf{w}^T \mathbf{r} .$$

**Expected value and variance of the return of the portfolio.** The expected value and the variance of the return thus read

$$\mu = \mathbb{E}[r] = \mathbb{E}[\mathbf{w}^T \mathbf{r}] = \mathbf{w}^T \boldsymbol{\mu} . \tag{18.1}$$

---

<sup>1</sup> Markowitz, H.M. "Portfolio Selection", *The Journal of Finance*, 1952.

and

$$\begin{aligned}
 \sigma^2 &= \mathbb{E}[(r - \mu)^2] = \\
 &= \mathbb{E} [((\mathbf{r} - \mu)^T \mathbf{w})^T ((\mathbf{r} - \mu)^T \mathbf{w})] = \\
 &= \mathbf{w}^T \mathbb{E} [(\mathbf{r} - \mu)(\mathbf{r} - \mu)^T] \mathbf{w} = \mathbf{w}^T \sigma^2 \mathbf{w} .
 \end{aligned} \tag{18.2}$$

**Optimal portfolio.** The evaluation of weights  $\mathbf{w}^*$  of optimal asset allocation for MPT is recast as the optimization problem of *finding asset allocation with minimum volatility (variance) for the given expected return  $\bar{\mu}$ , under some constraints about asset allocation*

$$\begin{aligned}
 \mathbf{w}^* &= \operatorname{argmin}_{\mathbf{w}} \sigma^2 \quad \text{s.t.} \quad \mathbf{w}^T \boldsymbol{\mu} = \bar{\mu} \\
 &\quad \text{other constraints ,}
 \end{aligned}$$

where other constraints could be:

- fully invested portfolio

$$\sum_k w_k = 1 ,$$

- no leverage on asset  $k$

$$w_k \leq 1 ,$$

- no short-selling on asset  $k$

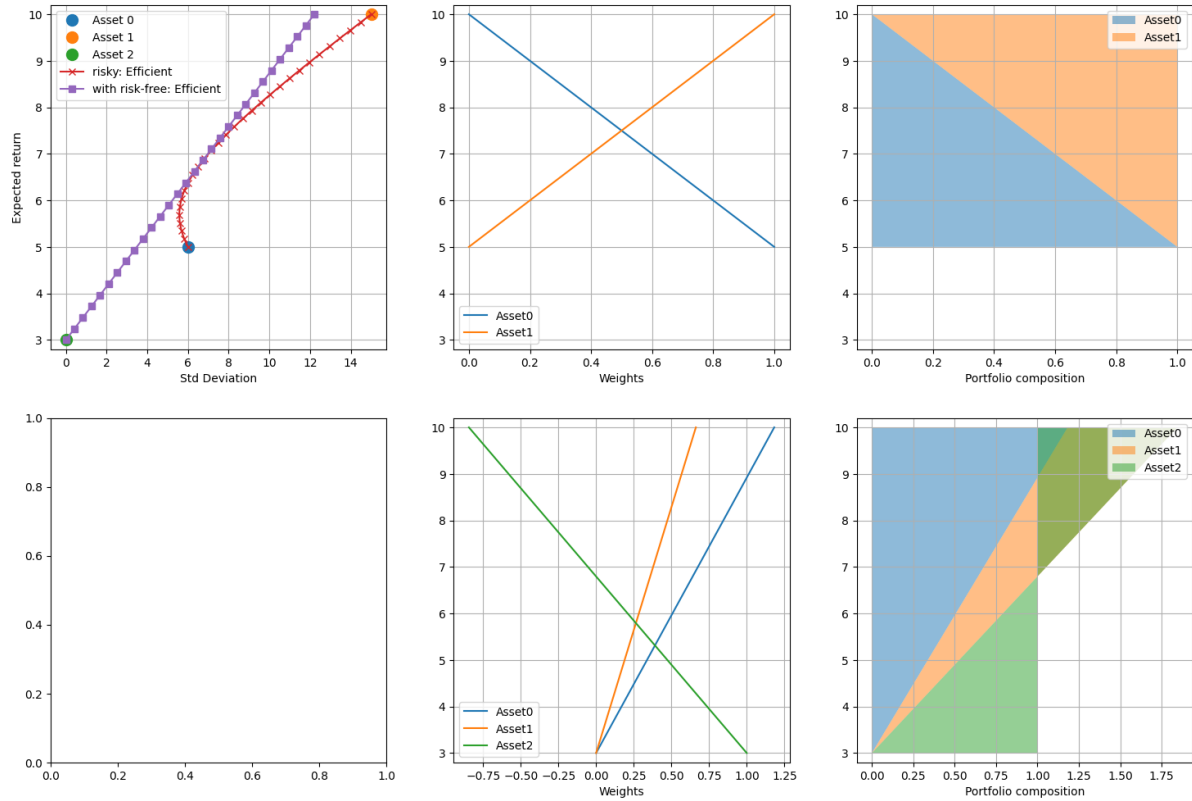
$$w_k \geq 0 .$$

**Useful arrays and functions.** Functions to be used in the optimization are defined here. The optimization process aims at finding the asset allocation with minimum variance of the return, given the expected value of the return.

## 18.2 Efficient frontier and CAPM with possibility of leverage and short-selling

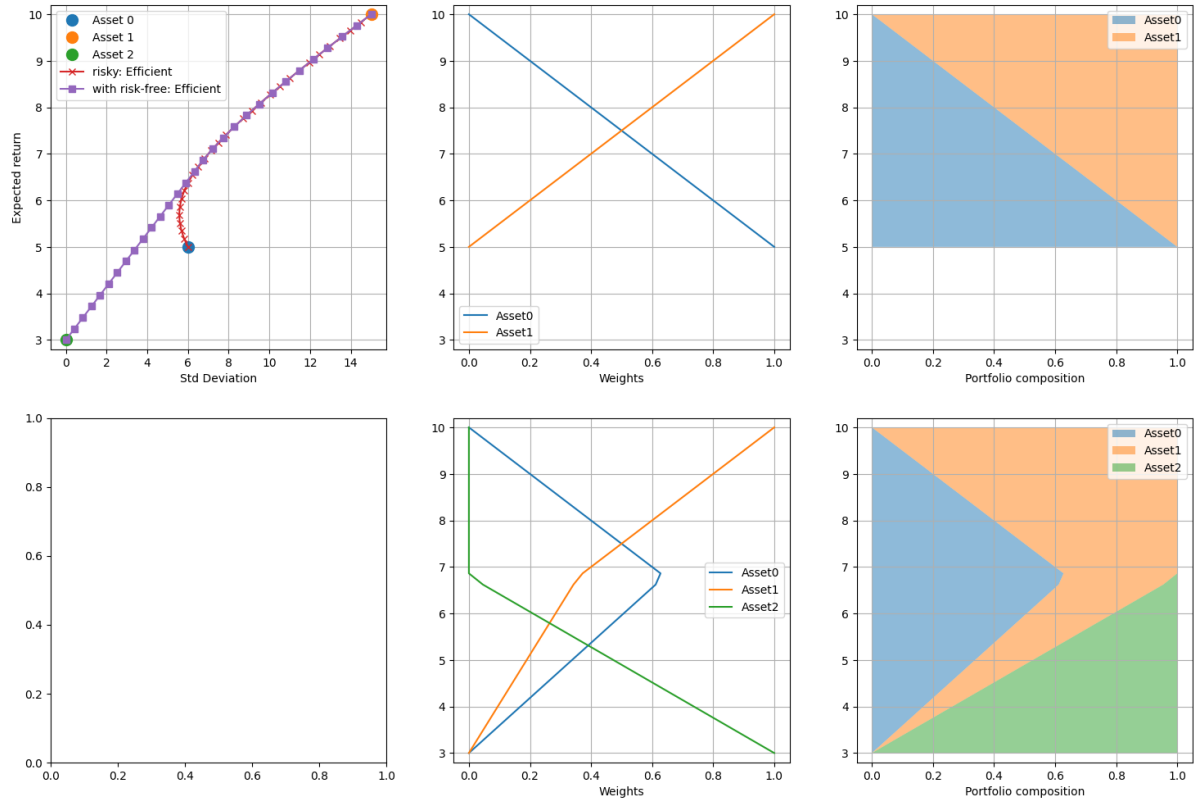
Plots





## 18.3 Efficient frontier and CAPM without leverage and short-selling

Plots.



## 18.4 MPT and CAPM: analytical solution

This section details the closed-form matrix derivations for Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM).

### 18.4.1 MPT: fully invested portfolio with no risk-free asset

When short-selling and leverage are allowed without limits, the optimization problem can be solved exactly using the method of **Lagrange Multipliers**.

**Problem setup.** MPT can be formulated as a constrained optimization problem, to find the vector  $\mathbf{w}$  describing the asset allocation of the portfolio that produces the minimum variance portfolio, for a given expected return of the portfolio and under the constraint of fully invested portfolio,

$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmin}} \sigma^2 \quad \text{s.t.} \quad \begin{cases} \mathbf{w}^T \boldsymbol{\mu} = \bar{\mu} & \text{(Portfolio expected return)} \\ \sum_{i=1}^N w_i = 1 & \text{(Fully invested portfolio)} \end{cases}$$

The expression of portfolio expected return  $\bar{\mu}$  and variance  $\sigma^2$  are given by (18.1) and (18.2) as functions of the expected value  $\mu$  and variance matrix  $\sigma^2$  of the assets and portfolio weights  $\mathbf{w}$ .

**Constrained optimization, with Lagrange multipliers.** Using Lagrange multiplier method, and adding a factor  $\frac{1}{2}$  to the variance of the portfolio (to avoid a factor 2 later), the augmented objective function reads

$$\tilde{\mathcal{J}}(\mathbf{w}; a, b) = \frac{1}{2} \mathbf{w}^T \sigma^2 \mathbf{w} - a(\mathbf{w}^T \boldsymbol{\mu} - \bar{\mu}) - b(\mathbf{w}^T \mathbf{1} - 1). \quad (18.3)$$

If the constraints hold, then  $\sigma^2 = 2\tilde{J}$ .

**Solution of the optimization problem.** Setting to zero the gradient of the augmented objective function<sup>1</sup> w.r.t. asset weights  $\mathbf{w}$ , and Lagrange multipliers  $a, b$ , the optimal solution is found as the solution of the following linear system,

$$\begin{cases} \mathbf{0} = \nabla_{\mathbf{w}} \tilde{J} = \sigma^2 \mathbf{w}^* - a\mu - b\mathbf{1} \\ 0 = \nabla_a \tilde{J} = \mu^T \mathbf{w}^* - \mu \\ 0 = \nabla_b \tilde{J} = \mathbf{1}^T \mathbf{w}^* - 1 \end{cases}, \quad \begin{bmatrix} \sigma^2 & -\mu & -\mathbf{1} \\ -\mu^T & 0 & 0 \\ -\mathbf{1}^T & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{w}^* \\ a \\ b \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ -\mu \\ -1 \end{bmatrix} \quad (18.4)$$

### Solution of the linear system - Details

Without any risk-free asset, the covariance matrix is non-singular, and thus invertible. (Formally) solving the first equation for  $\mathbf{w}$ ,

$$\mathbf{w} = \sigma^{-2} \mu a + \sigma^{-2} \mathbf{1} b,$$

a system of 2 equations in 2 unknowns  $a, b$  reads

$$\begin{bmatrix} \mu^T \sigma^{-2} \mu & \mu^T \sigma^{-2} \mathbf{1} \\ \mathbf{1}^T \sigma^{-2} \mu & \mathbf{1}^T \sigma^{-2} \mathbf{1} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \mu \\ 1 \end{bmatrix}.$$

and thus

$$\begin{aligned} \begin{bmatrix} a \\ b \end{bmatrix} &= \frac{1}{A_{11}A_{22} - 2A_{12}} \begin{bmatrix} A_{22} & -A_{12} \\ -A_{12} & A_{11} \end{bmatrix} \begin{bmatrix} \mu \\ 1 \end{bmatrix} \\ &= \frac{1}{\Delta} \left( \begin{bmatrix} A_{22} \\ -A_{12} \end{bmatrix} \mu + \begin{bmatrix} -A_{12} \\ A_{11} \end{bmatrix} \right). \end{aligned}$$

Thus, the optimal asset allocation is a 1-degree function of  $\mu$ ,

$$\mathbf{w}^* = \mathbf{w}_0 + \mathbf{w}_{/\mu} \mu.$$

and its variance is a 2-degree function of  $\mu$ ,

$$\begin{aligned} \sigma^{2*} &= \mathbf{w}^{*T} \sigma^2 \mathbf{w}^* = \sigma_0^2 + \sigma_1^2 \mu + \sigma_2^2 \mu^2 = \\ &= (a\mu^T + b\mathbf{1}^T) \underbrace{\sigma^2 \sigma^{-2}}_{=\mathbf{I}} (\mu a + \mathbf{1} b) = \\ &= a^2 \mu^T \sigma^{-2} \mu + 2ab \mathbf{1}^T \sigma^2 \mu + b^2 \mathbf{1}^T \sigma^{-2} \mathbf{1} = \\ &= \left( \frac{1}{\Delta} (A_{22}\mu - A_{12}) \right)^2 A_{11} + 2 \frac{1}{\Delta^2} (A_{22}\mu - A_{12}) \frac{1}{\Delta} (-A_{12}\mu + A_{11}) A_{12} + \left( \frac{1}{\Delta} (-A_{12}\mu + A_{11}) \right)^2 A_{22} = \\ &= \frac{1}{\Delta^2} (\mu^2 (A_{22}^2 A_{11} - 2A_{22}A_{12}^2 + A_{12}^2 A_{22}) + \mu \cdot 2(-A_{11}A_{22}A_{12} + A_{22}A_{11}A_{12} + A_{12}^3 - A_{11}A_{22}A_{12}) + (A_{12}^2 A_{11} - 2A_{12}^2 A_{11} - \\ &= \frac{1}{\Delta^2} [\mu^2 A_{22} (A_{11}A_{22} - A_{12}) - 2\mu A_{12} (A_{11}A_{22} - A_{12}^2) + A_{11} (A_{11}A_{22} - A_{12}^2)] = \\ &= \frac{1}{\Delta} (\mu^2 A_{22} - 2\mu A_{12} + A_{11}) = \\ &= [\mu \quad 1] \mathbf{A}^{-1} \begin{bmatrix} \mu \\ 1 \end{bmatrix}. \end{aligned}$$

As the matrix  $\mathbf{A}$  is definite positive (its inverse is definite positive as well?), it follows that  $\sigma^2 > 0$  for any value of  $\mu$ , as expected for the value of a variance.

<sup>1</sup> The augmented objective function is a quadratic function  $\mathbf{w}$ , and thus it has only one extreme point.

**Some analytic geometry.** The function

$$\begin{aligned}\text{Var}[r](\mu) &= \frac{1}{\Delta} (A_{22}\mu^2 - 2A_{12}\mu + A_{11}) = \\ &= \frac{A_{22}}{\Delta} \left( \mu - \frac{A_{12}}{A_{22}} \right)^2 + \frac{A_{11}}{\Delta} - \frac{A_{12}^2}{A_{22}\Delta} = \\ &= \frac{A_{22}}{\Delta} \left( \mu - \frac{A_{12}}{A_{22}} \right)^2 + \frac{1}{A_{22}} .\end{aligned}$$

is the function of a parabola, with vertex in

$$\begin{aligned}\mu_v &= \frac{A_{12}}{A_{22}} \\ \text{Var}[r]_v &= \text{Var}[r](\mu_v) = \frac{1}{\Delta} \left( \frac{-A_{12}^2 + A_{11}A_{22}}{A_{22}} \right) = \frac{1}{A_{22}} .\end{aligned}$$

Using  $\sigma$  as an independent coordinate (and not  $\text{Var}[r] = \sigma^2$ )...

### Properties of matrix $\mathbf{A}$

Is it positive definite? Covariance matrix is positive matrix, so for  $\forall \mathbf{v}$

$$0 < \mathbf{v}\sigma^2\mathbf{v} ,$$

and choosing  $\mathbf{v} = [\mu \quad \mathbf{1}] \begin{bmatrix} a \\ b \end{bmatrix}$ , for  $\forall a, b$ , it immediately follows

$$\begin{aligned}0 < \mathbf{v}\sigma^2\mathbf{v} &= \\ &= [a \quad b] \begin{bmatrix} \mu^T \\ \mathbf{1}^T \end{bmatrix} \sigma^2 [\mu \quad \mathbf{1}] \begin{bmatrix} a \\ b \end{bmatrix} = \\ &= [a \quad b] \begin{bmatrix} \mu^T \sigma^2 \mu & \mu^T \sigma^2 \mathbf{1} \\ \mathbf{1}^T \sigma^2 \mu & \mathbf{1}^T \sigma^2 \mathbf{1} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} ,\end{aligned}$$

and thus matrix  $\mathbf{A}$  is definite positive.

### Sensitivity of the MPT to data uncertainty

**Sensitivity of the solution to data uncertainty.** Once a solution is found, the sensitivity of this solution w.r.t. variation in the expected return and the variance of the assets are evaluated as the gradient of the augmented objective function (18.3) w.r.t. to  $\mu$ , and  $\sigma^2$  respectively,

$$\begin{aligned}\nabla_{\mu}\sigma^2 &= 2\nabla_{\mu}\tilde{\mathcal{J}} = -2a\mathbf{w} \\ \nabla_{\sigma^2}\sigma^2 &= 2\nabla_{\sigma^2}\tilde{\mathcal{J}} = \mathbf{w} \otimes \mathbf{w} ,\end{aligned}$$

and, since  $\nabla\sigma^2 = 2\sigma\nabla\sigma$ ,

$$\nabla_{\mu}\sigma = \frac{\nabla_{\mu}\tilde{\mathcal{J}}}{\sigma} = -\frac{a\mathbf{w}}{\sigma} .$$

Some example is provided is discussed in the Jupyter notebook [Uncertainty in MPT and CAPM Models](#).

## Sensitivity and Taylor expansion

Let the objective function

$$\tilde{J}(\mathbf{w}; a, b; \mu, \sigma^2) = \frac{1}{2} \mathbf{w}^T \sigma^2 \mathbf{w} - a(\mathbf{w}^T \mu - \mu) - b(\mathbf{w}^T \mathbf{1} - 1),$$

having explicitly written the dependence from the parameters (input data) of the model,  $\mu, \sigma^2$ . If the expected value and the variance of the returns become  $\mu + \Delta\mu, \sigma^2 + \Delta\sigma^2$ , the objective function becomes

$$\tilde{J}(\mathbf{w}; a, b; \mu + \Delta\mu, \sigma^2 + \Delta\sigma^2) = \tilde{J}(\mathbf{w}; a, b; \mu, \sigma^2) + \frac{1}{2} \mathbf{w}^T \Delta\sigma^2 \mathbf{w} - a \mathbf{w}^T \Delta\mu + o(|\Delta\mu|, |\Delta\sigma^2|).$$

**Sensitivity of the asset allocation.** The optimal solution comes from the solution of the linear system (18.4), that can be formally written as

$$\mathbf{A}(\mu, \sigma^2) \mathbf{x} = \mathbf{b}.$$

The change in the asset allocation  $\Delta\mathbf{w}$  due to a change of the expected return of the asset  $\Delta\mu$ , follows from a linear expansion around the optimal solution,

$$\begin{aligned} \mathbf{b} &= \mathbf{A}(\mu + \Delta\mu) (\mathbf{x} + \Delta\mathbf{x}) = \\ &= (\mathbf{A}(\mu) + \Delta\mathbf{A}) (\mathbf{x} + \Delta\mathbf{x}) = \\ &= \mathbf{A}\mathbf{x} + \Delta\mathbf{A}\mathbf{x} + \mathbf{A}\Delta\mathbf{x} + o(|\Delta|), \end{aligned}$$

and thus

$$\mathbf{A}(\mu) \Delta\mathbf{x} = -\Delta\mathbf{A}\mathbf{x}.$$

## Details - Rearranging terms

As

$$\Delta\mathbf{A} = \begin{bmatrix} \cdot & -\Delta\mu & \cdot \\ -\Delta\mu^T & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

it's possible to rearrange the linear system for the variation of asset allocation as

$$\Delta\mathbf{w} = -\mathbf{A}^{-1} \begin{bmatrix} -a\Delta\mu \\ -\Delta\mu^T \mathbf{w} \\ 0 \end{bmatrix} = -\mathbf{A}^{-1} \begin{bmatrix} -a\mathbf{I}_{n,\text{assets}} \\ -\mathbf{w}^T \\ 0 \end{bmatrix} \Delta\mu.$$

Some example is provided is discussed in the Jupyter notebook [Uncertainty in MPT and CAPM Models](#), where the changes of asset allocation due to unitary variation of the expected return of individual assets are discussed.

## 18.4.2 CAPM

Analytical solution of the MPT-CAPM optimization problem, with a risk-free asset. If a risk-free asset exists, the covariance matrix is singular. However, the risk-free asset can be partitioned from the risky assets, so that the covariance matrix of the return of the risky asset is non-singular. The problem becomes

$$(\mathbf{w}, w_0)^* = \operatorname{argmin}_0 \sigma^2 \quad \text{s.t.} \quad \dots$$

...

$$\begin{bmatrix} \sigma^2 & \mathbf{0} & -\mu & -\mathbf{1} \\ \mathbf{0}^T & 0 & -\mu_0 & -1 \\ -\mu^T & -\mu_0 & 0 & 0 \\ -\mathbf{1}^T & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{w}^* \\ w_0^* \\ a \\ b \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ 0 \\ -\mu \\ -1 \end{bmatrix}$$

From the second and the fourth equation,

$$\begin{aligned} b &= -\mu_0 a \\ w_0^* &= -\mathbf{1}^T \mathbf{w}^* + 1 \end{aligned}$$

and thus

$$\begin{bmatrix} \sigma^2 & -\mu + \mu_0 \mathbf{1} \\ -\mu^T + \mu_0 \mathbf{1}^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{w}^* \\ a \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ -(\mu - \mu_0) \end{bmatrix}$$

whose solution reads

$$\begin{aligned} a &= \frac{\mu_e}{\mu_e^T \sigma^{-2} \mu_e} \\ \mathbf{w}^* &= \frac{\mu_e \sigma^{-2} \mu_e}{\mu_e^T \sigma^{-2} \mu_e} \end{aligned}$$

and the relationship between the variance and the expected value of the optimal portfolios,

$$\sigma^2 = \frac{\mu_e^2}{\mu_e^T \sigma^{-2} \mu_e}$$

or the linear relation between the standard deviation of the portfolio  $\sigma$  and the excess return  $\mu_e$  of the portfolio w.r.t. the risk-free asset,

$$\sigma = \frac{\mu_e}{\sqrt{\mu_e^T \sigma^{-2} \mu_e}}.$$

### Solution of the linear system - Details

$$\begin{aligned} \mathbf{w}^* &= \sigma^{-2} (\mu - \mu_0 \mathbf{1}) a, \\ \mu - \mu_0 &= (\mu - \mu_0 \mathbf{1})^T \mathbf{w}^* = \\ &= (\mu - \mu_0 \mathbf{1})^T \sigma^{-2} (\mu - \mu_0 \mathbf{1}) a, \\ a &= \frac{\mu_e}{\mu_e^T \sigma^{-2} \mu_e}, \end{aligned}$$

and thus

$$\mathbf{w}^* = \frac{\mu_e \sigma^{-2} \mu_e}{\mu_e^T \sigma^{-2} \mu_e}.$$

Eventually, the variance of the portfolio reads

$$\begin{aligned} \sigma^2 &= \mathbf{w}^{*T} \sigma^2 \mathbf{w} = \\ &= \mu_e^T \frac{\mu_e^T \sigma^{-2} \sigma^2 \sigma^{-2} \mu_e}{(\mu_e^T \sigma^{-2} \mu_e)^2} = \\ &= \frac{\mu_e^2}{\mu_e^T \sigma^{-2} \mu_e}, \end{aligned}$$

with  $\mu_e := \mu - \mu_0$  the excess desired return of the portfolio w.r.t. the risk-free asset, and  $\mu_e := \mu - \mu_0 \mathbf{1}$  the vector of the excess returns of each risky asset w.r.t. the risk-free asset. Taking the square root of the last relation, a 1-degree function relates the standard deviation and the return of the portfolio,

$$\sigma = \frac{\mu_e}{\sqrt{\mu_e^T \sigma^{-2} \mu_e}}.$$

**Tangency condition** as a maximization of a measure of *risk-adjusted return*, namely **Sharpe ratio** comparing the excess return and the variance of the portfolio w.r.t. a risk-free (zero variance) asset  $(\cdot)_0$  used as a benchmark  $(\cdot)_b$

$$S := \frac{\mathbb{E}[r - r_b]}{\sqrt{\text{Var}[r - r_b]}} = \frac{\mathbf{w}^T \mu - \mu_0}{\sqrt{\mathbf{w}^T \sigma^2 \mathbf{w}}}$$

as the variance reads

$$\text{Var}[r - r_0] = \dots$$

### Tangency condition between optimal portfolio lines w/ and w/o risk-free asset

W/o risk-free asset

$$\sigma_r(\mu) = \sqrt{\frac{A_{22}\mu^2 - 2\mu A_{12} + A_{11}}{\Delta}},$$

with  $A_{22} = \mathbf{1}^T \sigma^{-2} \mathbf{1}$ ,  $A_{11} = \mu^T \sigma^{-2} \mu$ ,  $A_{12} = \mu^T \sigma^{-2} \mathbf{1}$ , and  $\Delta = A_{11}A_{22} - A_{12}^2$

W/ risk-free asset

$$\sigma_f(\mu) = \frac{\mu - \mu_0}{\sqrt{(\mu - \mu_0 \mathbf{1})^T \sigma^{-2} (\mu - \mu_0 \mathbf{1})}}.$$

Tangency condition

$$\begin{aligned}\sigma_r(\bar{\mu}) &= \sigma_f(\bar{\mu}) \\ \sigma'_r(\bar{\mu}) &= \sigma'_f(\bar{\mu})\end{aligned}$$

or with the variance,

...

$$0 = \mu^2 \left( \frac{A_{22}}{\Delta} - \frac{1}{\mu_e^T \sigma^{-2} \mu_e} \right) - 2\mu \left( \frac{A_{12}}{\Delta} - \frac{\mu_0}{\mu_e^T \sigma^{-2} \mu_e} \right) + \left( \frac{A_{11}}{\Delta} - \frac{\mu_0^2}{\mu_e^T \sigma^{-2} \mu_e} \right)$$

## 18.5 Modern Portfolio Theory and Capital Asset Pricing Model

Modern Portfolio Theory (MPT), introduced by Harry Markowitz<sup>1</sup>, operates under a simple premise: a rational investor wants to either minimize risk (variance) for a given level of expected return, or maximize expected return for a given level of risk.

MPT treats risk as the volatility - the variance - of the portfolio return, for a given desired expected value of the return. MPT shows how assets can interact with each others, and shows the effect of **diversification** in a portfolio with non-correlated assets.

<sup>1</sup> Markowitz, H.M. "Portfolio Selection", *The Journal of Finance*, 1952.

### 18.5.1 Modern Portfolio Theory

**Assets.** There are  $N$  available assets, whose  $\mathbf{r} = (r_i)$  return over the period of interest has

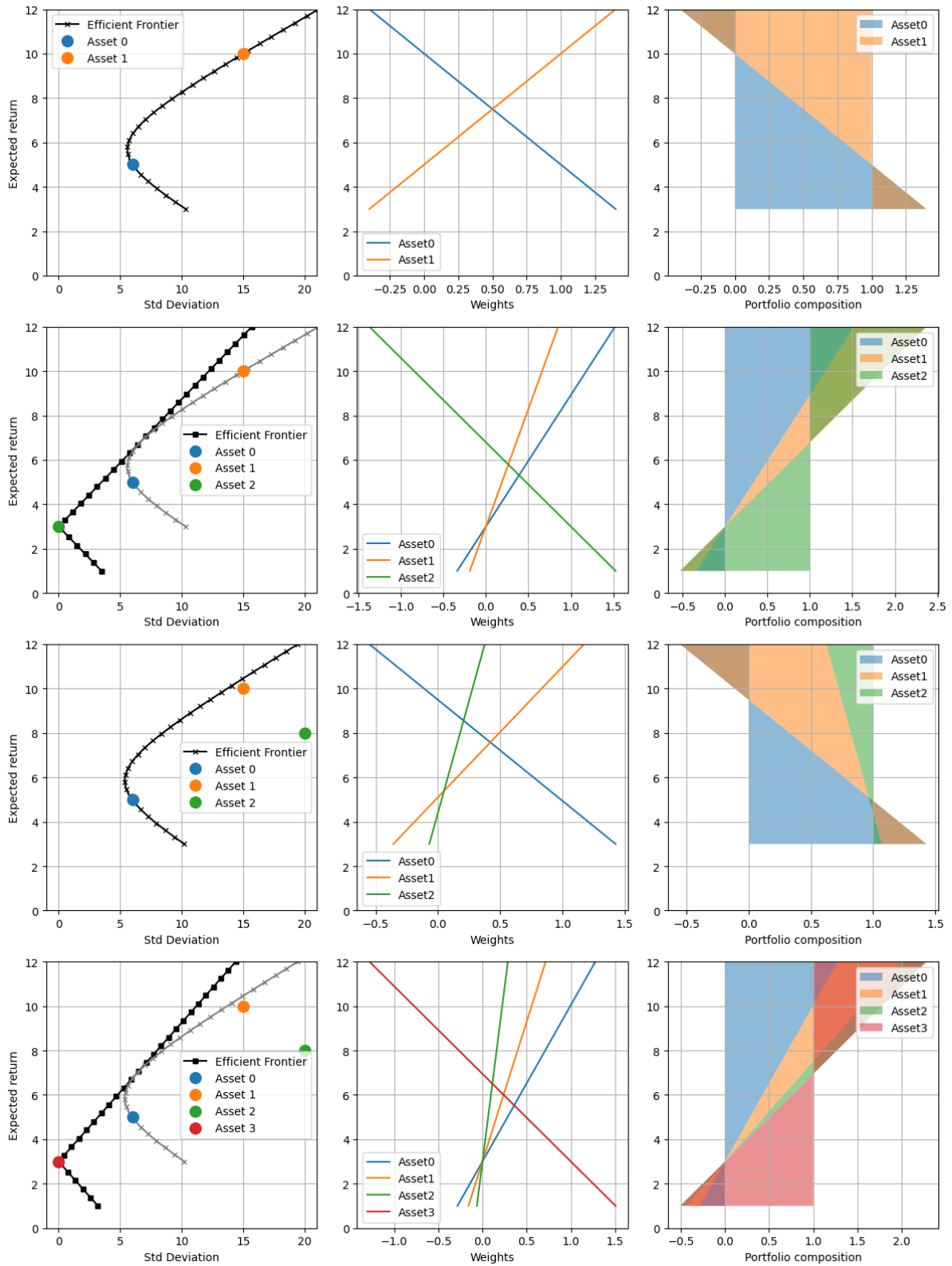
- expected value  $\mu = (\mu_i) = \mathbb{E}(r_i)$ ,
- covariance matrix  $\sigma^2 = (\sigma_{ij}^2) = \mathbb{E}[(r_i - \mu_i)(r_j - \mu_j)]$ .

**Useful arrays and functions.** Functions to be used in the optimization are defined here. The optimization process aims at finding the asset allocation with minimum variance of the return, given the expected value of the return.

### 18.5.2 Efficient frontier and CAPM with possibility of leverage and short-selling

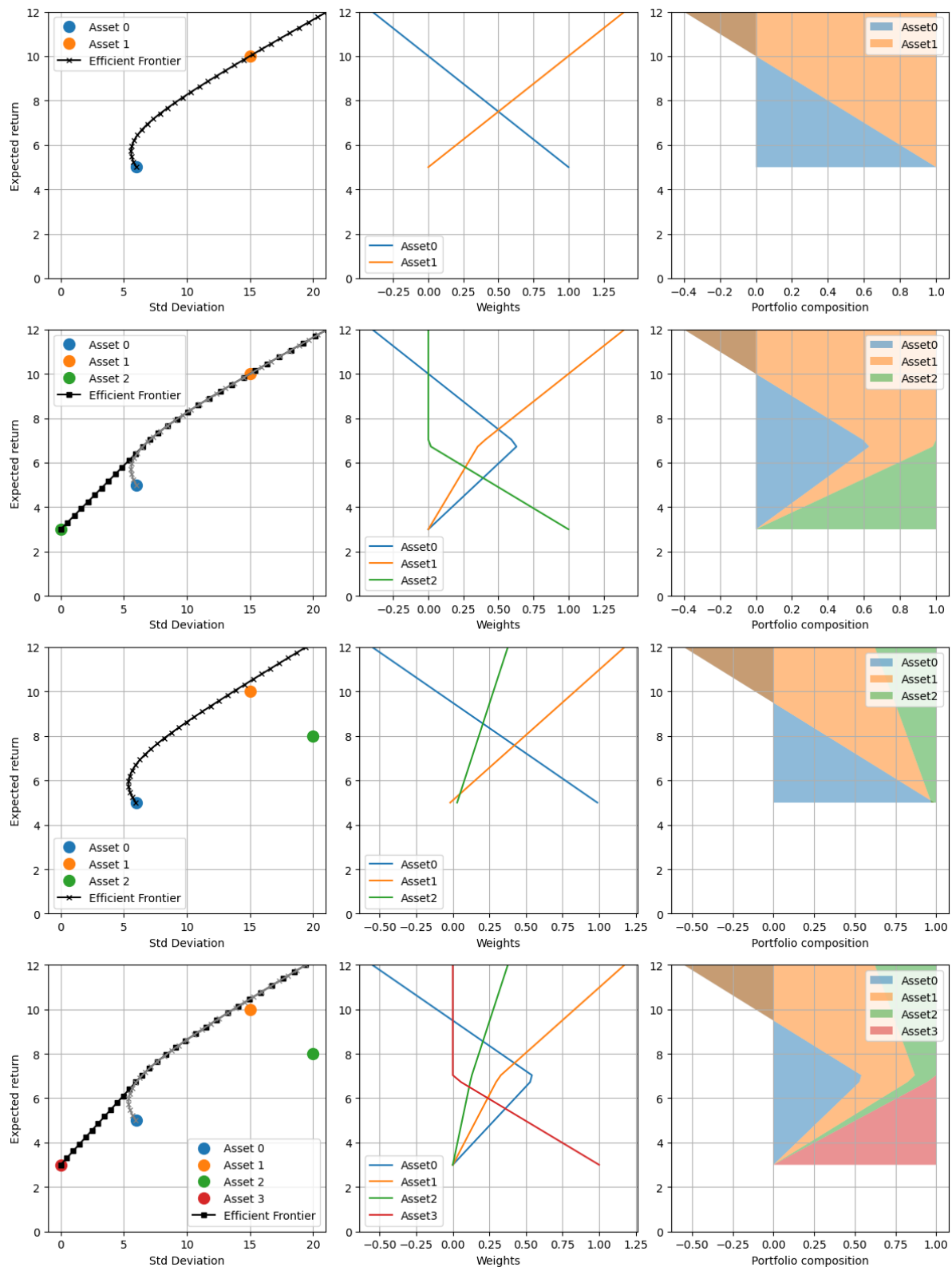
**Plots**





### 18.5.3 Efficient frontier and CAPM without leverage and short-selling

Plots.



## 18.6 Uncertainty in MPT and CAPM Models

In this section, the sensitivity of the results of MPT and CAPM models w.r.t. variation of expected return and variance of the individual assets is discussed.

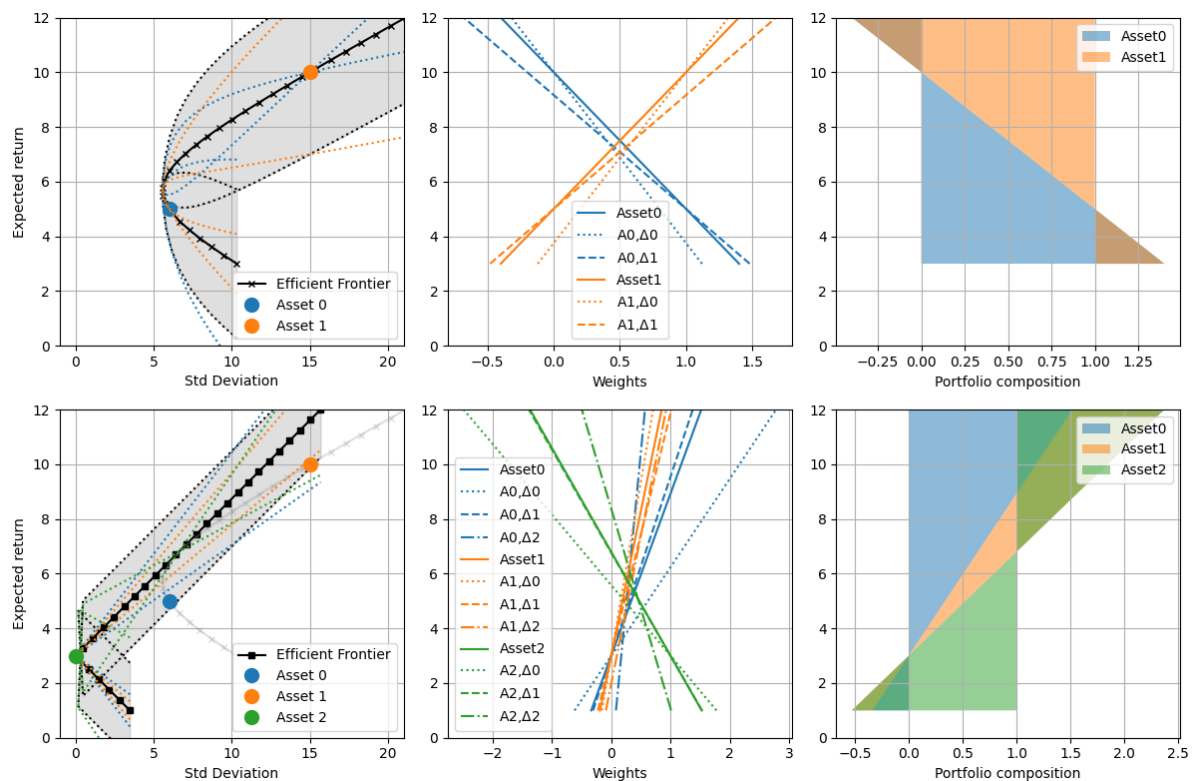
...

**Useful arrays and functions.** Functions to be used in the optimization are defined here. The optimization process aims at finding the asset allocation with minimum variance of the return, given the expected value of the return.

### 18.6.1 Efficient frontier and CAPM with possibility of leverage and short-selling, and fully invested portfolio

This problem has analytical solution, and sensitivity immediately follows from constrained optimization of the augmented objective function  $\tilde{J}$ , as discussed in *Sensitivity of the MPT to data uncertainty.s*

#### Plots



## 18.7 Correlation between Assets in MPT and CAPM Models

In this section, the effects of correlation between assets are discussed in a few-asset portfolio.

```

"""
Modern Portfolio Theory
Effect of the correlation in a 2-asset portfolio

"""

import numpy as np
from scipy.optimize import minimize
from funtools import partial

import matplotlib.pyplot as plt

#> Dictionary collecting different assets for building portfolio
ptfs = {}

#> 1. Two-asset portfolio
# Here, un-correlated assets, r12 = 0
#> Expected returns and covariance
mu1, mu2 = 5., 10.
si1, si2 = 6., 15.
r12 = .0

mu = np.array([mu1, mu2])

si_fun = lambda r12: np.array([
    [ si1**2, si1*si2*r12],
    [ si1*si2*r12, si2**2]
])

ptfs['risky'] = { 'mu': mu, 'si_fun': si_fun, 'marker': 'x' }
ptfs['risky']['w/o risk-free'] = None

#> 2. Three-asset portfolio
# The same as the two-asset portfolio above, with an extra risk-free asset
# (risk-free = zero variance, si3 = .0)
#> Expected returns and covariance
mu1, mu2, mu3 = 5., 10., 3.
si1, si2, si3 = 6., 15., 0.
r12, r13, r23 = .0, .0, .0

mu = np.array([mu1, mu2, mu3])
si_fun = lambda r12: np.array([
    [ si1**2, si1*si2*r12, .0 ],
    [ .0, si2**2, .0 ],
    [ .0, .0, si3**2 ]
])

ptfs['with risk-free'] = { 'mu': mu, 'si_fun': si_fun, 'marker': 's' }
ptfs['with risk-free']['w/o risk-free'] = 'risky'

```

**Useful arrays and functions.** Functions to be used in the optimization are defined here. The optimization process aims

at finding the asset allocation with minimum variance of the return, given the expected value of the return.

```
# Find min, max returns for plots
mu_v = np.concatenate( [ i['mu'] for k,i in ptfs.items() ] )

min_mu = np.min(mu_v)
max_mu = np.max(mu_v) + 2.0

#> MPT
# An optimization problem is solved for a set of desired return to find the efficient
# frontier of a portfolio, under the assumptions of MPT

#> Array of desired returns, between min and max return of single assets
# des_ret_v = np.linspace(min_mu, max_mu, 30)

#> Constraints
def eq_desired_return(x, mu, desired_ret):
    return np.sum( x * mu ) - desired_ret

def eq_weight_sum(x):
    return np.sum(x) - 1

#> Objective function
def ptf_var(x, sigma):
    return x.T @ sigma @ x
```

```
n_des_return = 30
n_corr = 9

for kptf, ptf in ptfs.items():

    n_assets = len(ptf['mu'])

    ptf['wmat'] = []
    ptf['min_sig'] = []
    ptf['des_return'] = np.linspace(np.min(ptf['mu']) - 2. , np.max(ptf['mu']) + 2.,
    ↪n_des_return)
    ptf['corr'] = np.linspace(-1., 1., n_corr)
    ptf['des_ret'] = []

    for corr in ptf['corr']:
        for desired_ret in ptf['des_return']:

            #> Constraints (just comment if you don't want some)
            eq_cons = [
                {'type': 'eq', 'fun': partial(eq_desired_return, mu=ptf['mu'],
    ↪desired_ret=desired_ret)},
                {'type': 'eq', 'fun': eq_weight_sum,      # sum(w) = 1 (fully
    ↪invested)
            ]
            # eq_cons += [ {'type': 'ineq', 'fun': lambda x: x[i]} for i in np.
    ↪arange(n_assets) ]
            # eq_cons += [ {'type': 'ineq', 'fun': lambda x: 1-x[i]} for i in np.
    ↪arange(n_assets) ]
```

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```

cost_fun = partial(ptf_var, sigma=ptf['si_fun'](corr))

x0 = np.zeros(n_assets); x0[0] = 1 # np.ones(3) / 3.
res = minimize(cost_fun, x0, constraints=eq_cons,)

# print(f"Desired Return: {desired_ret}, res: {res.x}")

ptf_si = np.sqrt(res.x @ ptf['si_fun'](corr) @ res.x)
ptf_mu = np.sum(res.x * ptf['mu'])

#> Store weights and min variance, for the desired expected return
ptf['wmat'] += [ res.x ]
ptf['min_sig'] += [ ptf_si ]

ptf['des_ret'] += [ desired_ret ]

# print(f"des_ret, cor: {desired_ret}, {corr}")
# print(res.x)

ptf['wmat'] = np.array(ptf['wmat'])
ptf['min_sig'] = np.array(ptf['min_sig'])

```

```

#> Initialize plot
fig, ax = plt.subplots(2, 2, figsize=(8, 8))

cmap = plt.get_cmap('tab10')

nptf = 0
for kptf, ptf in ptfs.items():

    #>
    # si = ptf["si"]
    mu = ptf["mu"]

    #> Weights
    wmat = ptf['wmat']

    # for kptf, ptf in ptfs.items():
    # ax[nptf,0].plot(ptf['min_sig'], ptf['des_ret'], ptf['marker']+'-', label=
    ↪ 'Efficient Frontier', color='black', markersize=5,)
    for i_corr in np.arange(n_corr):
        ax[nptf,0].plot(
            ptf['min_sig'][i_corr*n_des_return:(i_corr+1)*n_des_return],
            ptf['des_ret'][i_corr*n_des_return:(i_corr+1)*n_des_return],
            ptf['marker']+'-', markersize=0,
            label=f"$\\rho=${ptf['corr'][i_corr]:.2f}",
            color = f"{0.+0.8*(i_corr+1)/n_corr}"
            # color='black', alpha = .0 + 1. * i_corr/n_corr
        )

    for iass in np.arange(len(mu)):
        ax[nptf,0].plot(si_fun(.0)[iass,iass]**.5, mu[iass], 'o', markersize=10, ↪
        ↪ label=f'Asset {iass}')

```

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```

"""
if ( not ptf['w/o risk-free'] == None ):
    ptf_rf = ptf['w/o risk-free']
    # ax[nptf,0].plot(ptf_rf['min_sig'], ptf_rf['des_ret'], ptf_rf['marker']+'-',
    color='grey', markersize=5,)
    for i_des_return in np.arange(n_des_return):
        ax[nptf,0].plot(
            ptf_rf['min_sig'][i_des_return::n_corr],
            ptf_rf['des_ret'][i_des_return::n_corr],
            ptf_rf['marker']+'-', color='grey', markersize=0,
        )
"""

ax[nptf,0].set_xlim([-1, 21])
ax[nptf,0].set_ylim([ 0, 12])
# ax[nptf,0].set_ylim([min_mu-.2, max_mu+.2])
ax[nptf,0].set_xlabel("Std Deviation")
ax[nptf,0].set_ylabel("Expected return")
ax[nptf,0].grid()
ax[nptf,0].legend()

#>
for i_corr in np.arange(n_corr):

    # for i_des_return in np.arange(n_des_return):

        for i in np.arange(np.shape(wmat)[1]):
            if i_corr == n_corr-1:
                ax[nptf,1].plot(wmat[i_corr*n_des_return:(i_corr+1)*n_des_return,i],
                ptf['des_ret'][i_corr*n_des_return:(i_corr+1)*n_des_return], alpha=(i_corr+1)/n_
                corr, color=cmap(i%10), label=f"Asset{i}")
            else:
                ax[nptf,1].plot(wmat[i_corr*n_des_return:(i_corr+1)*n_des_return,i],
                ptf['des_ret'][i_corr*n_des_return:(i_corr+1)*n_des_return], alpha=(i_corr+1)/n_
                corr, color=cmap(i%10),)

ax[nptf,1].set_xlabel("Weights")
ax[nptf,1].legend(); ax[nptf,1].grid()
# ax[nptf,1].set_ylim([min_mu-.2, max_mu+.2])
ax[nptf,1].set_ylim([ 0, 12])

"""
#> Stacked - with expected return on y
cum = np.zeros((np.shape(wmat)[0], np.shape(wmat)[1]+1))

for i in np.arange(1, np.shape(cum)[1]):
    cum[:,i] = cum[:,i-1] + wmat[:,i-1]

for i in np.arange(1, np.shape(cum)[1]):
    ax[nptf,2].fill_betweenx(ptf['des_ret'], cum[:,i-1], cum[:,i], label=f"Asset
    {i-1}", alpha = .5)

ax[nptf,2].legend(); ax[nptf,2].grid() # loc='upper right'
ax[nptf,2].set_xlabel('Portfolio composition')

```

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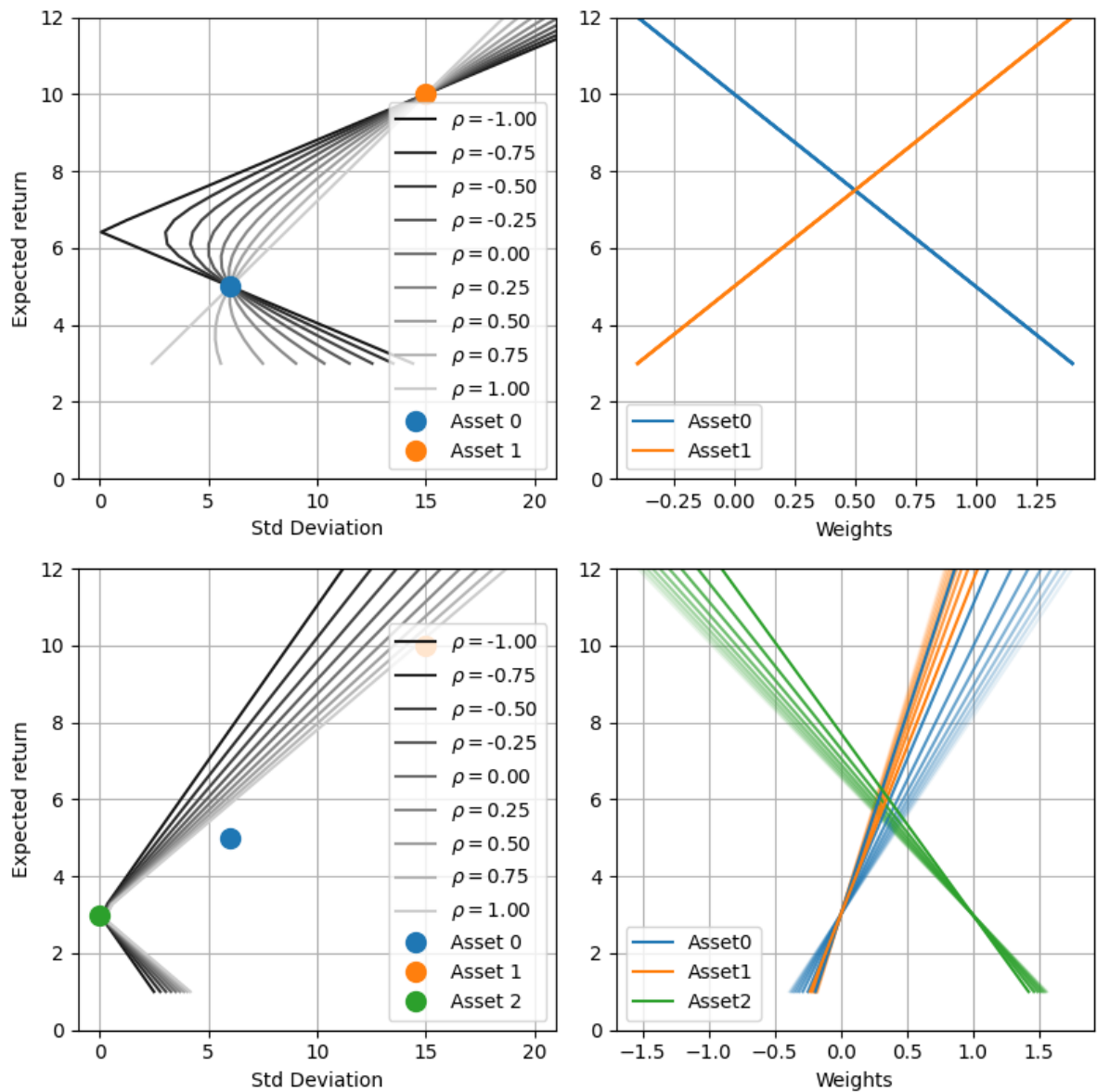
```

# ax[nptf,2].set_ylim([min_mu-.2, max_mu+.2])
ax[nptf,2].set_ylim([ 0, 12])
"""

nptf += 1

fig.tight_layout()
plt.show()

```





## MERTON'S PORTFOLIO PROBLEM

Merton's portfolio problem is an **optimization problem** dealing with the choice of optimal fraction of investment in a risky asset  $\pi_t$  and the consumption  $c_t$ , making a value function maximum. Merton's assume that the portfolio is build with a risk-free and a risky asset only. In this problem, there's no minimum spending level (subsistence level, what you need for survival), or transaction costs. Under some assumptions, the optimal fraction of the investment in the risky asset  $\pi_t^*$  is constant (need for - free, as costs are neglected - rebalancing) and equal to the ratio of the excess return of the risky asset  $\mu - r$ , scaled by its volatility  $\sigma^2$  and a personal risk aversion  $\gamma$ ,

$$\pi^* = \frac{\mu - r}{\gamma \sigma^2} ,$$

and the consumption is proportional to the wealth by a factor depending on time  $t$ ,

$$c_t^* = \frac{X_t}{f(t)}$$

### Contents

#### Sections.

- *Wealth dynamics*
- *Value function*
- *Solution of the problem*

#### Limits and extensions.

- J.J.Choi: "Why you probably don't invest enough in stocks", [The Bull podcast 306](#).

## 19.1 Wealth dynamics

Assuming that the wealth of a family or an individual can is invested with a fraction  $\pi_t$  in a risky part of the portfolio with expected return  $\mu$  and standard variation  $\sigma_t$ , and with a fraction  $1 - \pi_t$  in a risk-free asset with expected return  $r_t$  and zero standard deviation, the wealth  $x_t$  of a family or an individual evolves with the SDE

$$dX_t = \underbrace{(1 - \pi_t)r_t X_t dt}_{\text{risk-free asset}} + \underbrace{\pi_t (\mu_t X_t dt + \sigma_t X_t dW_t)}_{\text{risky asset}} - \underbrace{c_t dt}_{\text{consumption}} ,$$

or, rearranging for  $dt$ , and  $dW_t$ ,

$$dX_t = \underbrace{[r_t + (\mu_t - r_t) \pi_t] X_t dt}_{\text{expected return}} - \underbrace{c_t dt}_{\text{consumption}} + \underbrace{\pi_t \sigma_t X_t dW_t}_{\text{volatility of the return}} ,$$

i.e. the equation of a [geometric Brownian motion with drift](#), with

- 1-period (or percentage) expected value  $(1 - \pi_t)r_t + \pi_t\mu_t = r_t + \pi_t(\mu_t - r_t)$
- 1-period (or percentage) std-deviation  $\pi_t\sigma_t$
- 1-period (or percentage) drift  $-c_t$

This is the same equation as the one that can be used to discuss *sequence risk* in investment, especially dealing with withdrawal.

## 19.2 Value function

The goal of the problem is the maximization of a objective function representing the life-time cumulative discounted reward, using *continuous-time reinforcement learning* vocabulary, see as an example [Math:Introduction to RL](#), or - roughly - cumulative happiness. The objective function can be written as a value function,

$$V(x, t) = \mathbb{E} \left[ \int_{s=t}^T e^{-\rho(s-t)} u(c_s) ds + e^{-\rho(T-t)} B(T) u(X_T) \middle| X_t = x \right],$$

with

- $u(c)$  the **utility function**, the “joy from spending”
- $\rho$  the **discount rate**, i.e. “the **personal** weight of the present”, or a “measure of impatience”
- $B(T)$ , the **bequest weight**, i.e. “the importance of leaving money behind”

Optimization problem can be solved using **continuous-time reinforcement learning**, see as an example [Math:Introduction to RL](#), and [Statistics:RL \(todo\)](#). Optimal solution  $\pi_t^*$ ,  $c_t^*$  is the fraction invested in the risky asset and consumption that maximises the **value function**,  $V(x, t)$ .

---

### Arguments of the value function

Value function may be written as a function of several arguments, that can be treated as

- independent variables, like:
  - the initial time  $t$
  - the initial value of the wealth  $x$
- parameters, like
  - the final time  $T$
  - the discount rate  $\rho$ ,
  - the parameters in wealth dynamics, e.g.  $r, \mu, \sigma$
- functions and its parameters, like
  - the wealth fraction invested in the risky asset  $\pi_t$
  - the utility function  $u_t(c; \gamma)$ , and the risk aversion therein  $\gamma$
  - ...

For brevity, the value function is usually written as a function of the independent variables  $x, t$  only, but it could be written with the explicit dependence from parameters and functions,

$$V(x, t; \mathbf{p}, \mathbf{f}_t).$$


---

For constant expected return and volatility of the assets, under the assumption of **constant relative risk aversion (CCRA)**, i.e. utility function  $u(x) = \frac{x^{1-\gamma}}{1-\gamma}$ , it's possible to find an analytical solution with optimal fraction invested in the risky asset

$$\pi^* = \frac{\mu - r}{\gamma \sigma^2},$$

and consumption that's proportional to the wealth with a function of time  $f(t)$

$$c_t = \frac{X_t}{f(t)}.$$

## 19.3 Solution of the optimization problem

The solution of the optimization problem in the framework of **reinforcement learning** (or stochastic control?) may exploit the **principle of optimality**, getting the global optimization of the value function over time range  $[t, T]$  from the local optimization over all the elementary time-steps  $dt$  in the time range of interest.

### Recursive relation and Hamilton-Jacobi-Bellman relation

Writing the value function  $V(X_{t+dt}, t + dt)$  and expanding as a Taylor series up to the first order in  $dt$ , it's possible to find a **Hamilton-Jacobi-Bellman** equation from a recursive relation. Let the value function be

$$V(x, t) = \mathbb{E}_W \left[ \int_{s=t}^T e^{-\rho(s-t)} u(c_s) ds + e^{-\rho(T-t)} B(T) u(X_T) \middle| X_t = x \right],$$

**subject to** *wealth dynamics*, with initial condition  $X_t = x$ . This constraint allows to move the conditioned probability from  $X_{t+dt}$  to  $X_t = x$ , in  $V(X_{t+dt}, t + dt)$ ,

$$\begin{aligned} V(X_{t+dt}, t + dt) &= \mathbb{E}_W \left[ \int_{s=t+dt}^T e^{-\rho(s-t-dt)} u(c_s) ds + e^{-\rho(T-t-dt)} B(T) u(X_T) \middle| X_{t+dt} \right] = \\ &= \mathbb{E}_W \left[ \int_{s=t+dt}^T e^{-\rho(s-t-dt)} u(c_s) ds + e^{-\rho(T-t-dt)} B(T) u(X_T) \middle| X_t = x \right] = \end{aligned}$$

**todo** Be careful about arguments of  $V$  function.

Now, let's evaluate  $e^{-\rho t} V(x, t)$  for  $t$  and  $t + dt$ ,

$$\begin{aligned} e^{-\rho t} V(X_t, t) &= \\ &= \mathbb{E}_W \left[ \int_{s=t}^T e^{-\rho s} u(c_s) ds + e^{-\rho T} B(T) u(X_T) \middle| X_t \right] \\ e^{-\rho(t+dt)} V(X_t + dX_t, t + dt) &= \\ &= \mathbb{E}_W \left[ \int_{s=t+dt}^T e^{-\rho s} u(c_s) ds + e^{-\rho T} B(T) u(X_T) \middle| X_t = x \right] = \\ &= \mathbb{E}_W \left[ \int_{s=t}^T e^{-\rho s} u(c_s) ds + e^{-\rho T} B(T) u(X_T) \middle| X_t = x \right] - \mathbb{E}_W \left[ \int_{s=t}^{t+dt} e^{-\rho s} u(c_s) ds \middle| X_t = x \right]. \end{aligned}$$

and approximating the latter integral with the average value theorem,

$$e^{-\rho(t+dt)} V(X_t + dX_t, t + dt) = e^{-\rho t} V(X_t, t) - e^{-\rho t} u(c_t) dt,$$

collecting the increment,

$$d(e^{-\rho t} V(X_t, t)) = -e^{-\rho t} u(c_t) dt$$

and using the rule of product of the differentials,

$$\rho V dt = dV + u(c_t) dt . \quad (19.1)$$

### Taylor expansion

In this section, the expression of  $dV$  as a function of  $dt$  is found, and later introduced in equation (19.1) to find a relation not involving  $dt$ . Taylor expansion of value function  $V(t, x)$  subject to *wealth dynamics* - thus  $dV(t, X_t)$  - providing the expression of  $dX_t$ , and up to  $dt$  (and thus, to  $dX_t^2 \sim dt$ ), valued with  $x = X_t$ , reads

**todo** Add expectation value properly

$$\begin{aligned} dV &= \left[ \partial_t V dt + \partial_x V dx + \frac{1}{2} \partial_{xx} V dx^2 + o(dt) \right] \Big|_{x=X_t} = \\ &= \partial_t V dt + \partial_x V dX_t + \frac{1}{2} \partial_{xx} V dX_t^2 + o(dt) = \\ &= dt (\partial_t V + \partial_x V (r_t + (\mu_t - r_t) \pi_t) X_t - \partial_x V c_t) + \mathbb{E}[\partial_x V \pi_t \sigma_t X_t dW_t] + \frac{1}{2} \partial_{xx} V (\pi_t \sigma_t X_t)^2 dt + o(dt) , \\ &= dt (\partial_t V + \partial_x V (r_t + (\mu_t - r_t) \pi_t) X_t - \partial_x V c_t) + \frac{1}{2} \partial_{xx} V (\pi_t \sigma_t X_t)^2 dt + o(dt) , \end{aligned} \quad (19.2)$$

having used the wealth dynamics for  $dX_t$ ,  $\mathbb{E}[dW_t] = 0$ , and the expression at the first order of  $dX_t^2 = (\pi_t \sigma_t X_t)^2 dW_t^2 + o(dt) = (\pi_t \sigma_t X_t)^2 dt + o(dt)$ .

### Recursive relation for optimal solution

Introducing the expression (19.2) into the relation (19.1) the following relation holds

$$\rho V = (\partial_t V + \partial_x V (r_t + (\mu_t - r_t) \pi_t) X_t - \partial_x V c_t) + \partial_x V \pi_t \sigma_t X_t + \frac{1}{2} \partial_{xx} V (\pi_t \sigma_t X_t)^2 + u(c_t) , \quad (19.3)$$

at first order in  $dt$ . This condition holds for every  $\pi_t, u_t$ . **Optimality condition** requires that the value function is maximum, i.e.

$$\rho V^* = \max_{\pi_t, c_t} \left\{ \partial_t V^* + \partial_x V^* [(\pi_t (\mu - r) + r) X_t - c_t] + \frac{1}{2} \partial_{xx} V^* (\pi \sigma X_t)^2 + u(c_t) \right\} = \max_{\pi_t, c_t} \Phi$$

Explicitly writing independent variables and parameters, and keeping all the parameters constant except for the risky asset function  $\pi_t$  and the consumption function  $c_t$

$$\rho V^*(X_t, t; \mathbf{p}) := \rho V(X_t, t; \mathbf{p}; \pi_t^*, c_t^*) = \Phi(\pi_t = \pi_t^*, c_t = c_t^*; X_t, t, \mathbf{p}) ,$$

for a given set of parameters  $\mathbf{p}$ , initial time  $t$ , and wealth dynamics  $X_t$ , in  $[t, T]$  subject to a given initial condition  $X_t = x$ .

### Optimality condition

With  $\pi_t, c_t$  as the only free parameters, zero gradient w.r.t. these variables (function evaluated at a time  $t$  can be treated as a variable) is a necessary condition for local extreme points,

$$\begin{cases} 0 = \partial_{\pi_t} \Phi(\pi_t^*, c_t^*) = (\mu - r)X_t \partial_x V^* + \partial_{xx} V^* \pi_t^* (\sigma_t X_t)^2 \\ 0 = \partial_{c_t} \Phi(\pi_t^*, c_t^*) = -\partial_x V^* + \partial_{c_t} u(c_t^*) \end{cases}$$

and thus

$$\begin{cases} \pi_t^* = -\frac{\partial_x V^*}{\partial_{xx} V^*} \frac{\mu - r}{X_t \sigma^2} \\ 0 = -\partial_x V^* + \partial_{c_t} u(c_t^*) \end{cases}$$

**Remark.** Optimal value function  $V^*$  is not a function of  $\pi_t, c_t$ , as it's the value function evaluated in the optimal fraction and consumption by definition.

### Necessary conditions on $u()$ for the Hessian to be definite negative

todo

**CCRA assumption: utility function**  $u(x) = \frac{x^{1-\gamma}}{1-\gamma}$

With the expression  $u(x) = \frac{x^{1-\gamma}}{1-\gamma}$ ,  $\partial_{c_t} u(c_t) = c_t^{-\gamma}$ , and thus

$$\begin{cases} \pi_t^* = -\frac{\partial_x V^*}{\partial_{xx} V^*} \frac{\mu - r}{X_t \sigma^2} \\ c_t^* = (\partial_x V^*)^{-\frac{1}{\gamma}} \end{cases}$$

**CCRA assumption: value function**  $V^*(t, x) = f^\gamma(t) \frac{x^{1-\gamma}}{1-\gamma}$

With value function  $V^*(x, t) = f^\gamma(t) \frac{x^{1-\gamma}}{1-\gamma}$  and bequest function  $B(T) = \varepsilon^T$ , first and second order derivatives read

$$\begin{aligned} \partial_x V^* &= f^\gamma x^{-\gamma} \\ \partial_{xx} V^* &= -\gamma f^\gamma x^{-\gamma-1} \end{aligned}$$

so that the ratio  $\frac{\partial_x V^*(X_t, t)}{\partial_{xx} V^*(X_t, t) X_t}$  reads

$$\frac{\partial_x V^*(X_t, t)}{\partial_{xx} V^*(X_t, t) X_t} = -\frac{1}{\gamma}$$

and the optimal ratio invested in the risky asset is

$$\pi_t^* = \frac{\mu - r}{\gamma \sigma^2},$$

and the optimal consumption is

$$c_t^* = (f^\gamma(t) X_t^{-\gamma})^{-\frac{1}{\gamma}} = \frac{X_t}{f(t)}.$$

### Differential function for $f(t)$

Plugging the expressions of the optimal fraction of the risky asset and the optimal consumption into the recursive Bellman relation (19.3), for the optimal value function  $V^*$

$$\rho V^* = (\partial_t V^* + \partial_x V^* (r_t + (\mu_t - r_t) \pi_t^*) X_t - \partial_x V^* c_t^*) + \frac{1}{2} \partial_{xx} V^* (\pi_t \sigma X_t)^2 + u(c_t^*), \quad (19.4)$$

$$\begin{aligned} 0 &= -\rho f^\gamma \frac{X_t^{1-\gamma}}{1-\gamma} + \gamma \dot{f} f^{\gamma-1} \frac{X_t^{1-\gamma}}{1-\gamma} + f^\gamma X_t^{-\gamma} X_t \left( r_t + (\mu_t - r_t) \frac{\mu - r}{\gamma \sigma^2} - \frac{1}{f(t)} \right) + \frac{1}{2} \sigma^2 X_t^2 \frac{(\mu - r)^2}{\gamma^2 \sigma^4} (-\gamma f^\gamma X_t^{-\gamma-1}) + \frac{1}{1-\gamma} \left( \frac{X_t}{f(t)} \right) \\ &= \left[ -\frac{\rho}{1-\gamma} + \left( r + \frac{(\mu - r)^2}{\gamma \sigma^2} - \frac{1}{f(t)} \right) - \frac{1}{2} \frac{(\mu - r)^2}{\gamma \sigma^2} \right] f + \dot{f} \frac{\gamma}{1-\gamma} + \frac{1}{1-\gamma} - 1 = \\ &= \left[ -\frac{\rho}{1-\gamma} + r + \frac{(\mu - r)^2}{2\gamma \sigma^2} \right] f + \dot{f} \frac{\gamma}{1-\gamma} + \frac{\gamma}{1-\gamma}, \end{aligned}$$

i.e.

$$0 = \frac{1}{\gamma} \left[ -\rho + (1-\gamma) \left( r + \frac{(\mu - r)^2}{2\gamma \sigma^2} \right) \right] f(t) + \dot{f}(t) - 1.$$

Thus, function  $f(t)$  satisfies the ODE

$$\dot{f}(t) = \nu f(t) - 1$$

with final condition  $f(T) = \varepsilon$ , and

$$\nu = \frac{1}{\gamma} \left\{ \rho - (1-\gamma) \left( \frac{(\mu - r)^2}{2\sigma^2 \gamma} + r \right) \right\}.$$

The solution of the ODE reads

$$f(t) = \begin{cases} \frac{1}{\nu} + \left( \varepsilon - \frac{1}{\nu} \right) e^{-\nu(T-t)} & , \quad \text{if } \nu \neq 0 \\ T - t + \varepsilon & , \quad \text{if } \nu = 0 \end{cases}$$

## 19.4 Merton's portfolio problem - Example

Merton's portfolio problem aims at finding the optimal value of the fraction the risky asset  $\pi_t$ , in a 2-asset portfolio: the risky asset and a risk-free asset. The optimal solution is the fraction  $\pi_t^*$  that maximizes the value function

$$V(x, t) = \mathbb{E} \left[ \int_{s=t}^T e^{-\rho(s-t)} u(c_s) ds + e^{-\rho(T-t)} B(T) u(X_T) \middle| X_t = x \right],$$

i.e. a discounted return of a utility function  $u(c_s)$ , depending on the consumption  $c_s$  at time  $s \in [t, T]$  with a weight of utility function evaluated for the final value of the wealth  $X_T$ , modeling the bequest, weighted with  $B(T)$ , and subject to the dynamics of wealth  $X_t$  governed by the SDE

$$dX_t = \underbrace{[r_t + (\mu_t - r_t) \pi_t] X_t dt}_{\text{expected return}} - \underbrace{c_t dt}_{\text{consumption}} + \underbrace{\pi_t \sigma_t X_t dW_t}_{\text{volatility of the return}}.$$

Under the assumption of CCRA, i.e. with the expression of the utility function

$$u(x) = \frac{x^{1-\gamma}}{1-\gamma},$$

being  $\gamma$  the **personal risk-adversion factor**, the optimal solution provides:

- a value of optimal fraction invested in the risky asset  $\pi^*$  **constant in time** (did you say *rebalancing?*),

$$\pi_t^* = \frac{\mu - r}{\gamma \sigma^2} ,$$

with  $\mu, \sigma^2$  the expected value and the variance of the return of the risky asset, and  $r$  the risk-free return

- the expression of the consumption  $c_t$ , that's proportional to the wealth  $X_t$ ,

$$c_t = \frac{X_t}{f(t)} ,$$

through a function of time  $f(t)$ .

Function  $f(t)$  is

$$f(t) = \begin{cases} \frac{1}{\nu} + \left( \varepsilon - \frac{1}{\nu} \right) e^{-\nu(T-t)} & , \quad \text{if } \nu \neq 0 \\ T - t + \varepsilon & , \quad \text{if } \nu = 0 \end{cases}$$

with

$$\nu = \frac{1}{\gamma} \left\{ \rho - (1 - \gamma) \left( \frac{(\mu - r)^2}{2\sigma^2\gamma} + r \right) \right\} .$$





## REBALANCING

In this Notebook, two strategies on a 2-asset portfolio are discussed and compared:

- **rebalanced portfolio**, after each period
- **buy-and-hold portfolio**, without rebalancing

Effects of rebalancing and conditions for **rebalancing premium** are discussed: sometimes the expected log-return of the rebalanced portfolio may exceed the expected log-return of each single asset.

### 20.1 Libraries, parameters and useful functions

Libraries are imported and useful functions to treat conics below are defined here

#### 20.1.1 Libraries

#### 20.1.2 Parameters

#### 20.1.3 Functions for conics

### 20.2 Rebalanced portfolio

Let the **1-period return** of the assets be normal (**todo** is this necessary? Can't one rely on central limit theorem? How long the summation must be for convergence to normal distribution, in presence of **heavy-tails** distribution? If one can't rely on central limit theorem, let use numerical methods to investigate the effect of heavy tails distributions),

$$\mathbf{r} \sim \mathcal{N}(\mu, \sigma^2) .$$

Compound return of the portfolio has **expected value**

$$\mu_p^c = \mathbb{E}[r_p^c] = \mathbf{w}^T \mu - \frac{1}{2} \mathbf{w}^T \sigma^2 \mathbf{w}$$

and **variance**

$$\sigma_{r_p^c}^2 = \mathbb{E}[(r_p^c - \mu_p^c)^2] = \dots = \mathbf{w}^T \sigma \mathbf{w}$$

### 20.2.1 Shannon demon

Sometimes the expected value of the compound return of the rebalanced portfolio can be larger than the expected return of each asset class.

$$\mu_k^c = \mathbb{E}[r_k^c] = \mu_k - \frac{\sigma_k^2}{2}$$

**Example: 2-asset portfolio.** As an example, the expected value of the compound return of a 2-asset rebalanced portfolio,

$$\mathbb{E}[r_p^c] = w_1\mu_1 + w_2\mu_2 - \frac{1}{2}(w_1^2\sigma_1^2 + 2w_1w_2\rho\sigma_1\sigma_2 + w_2^2\sigma_2^2)$$

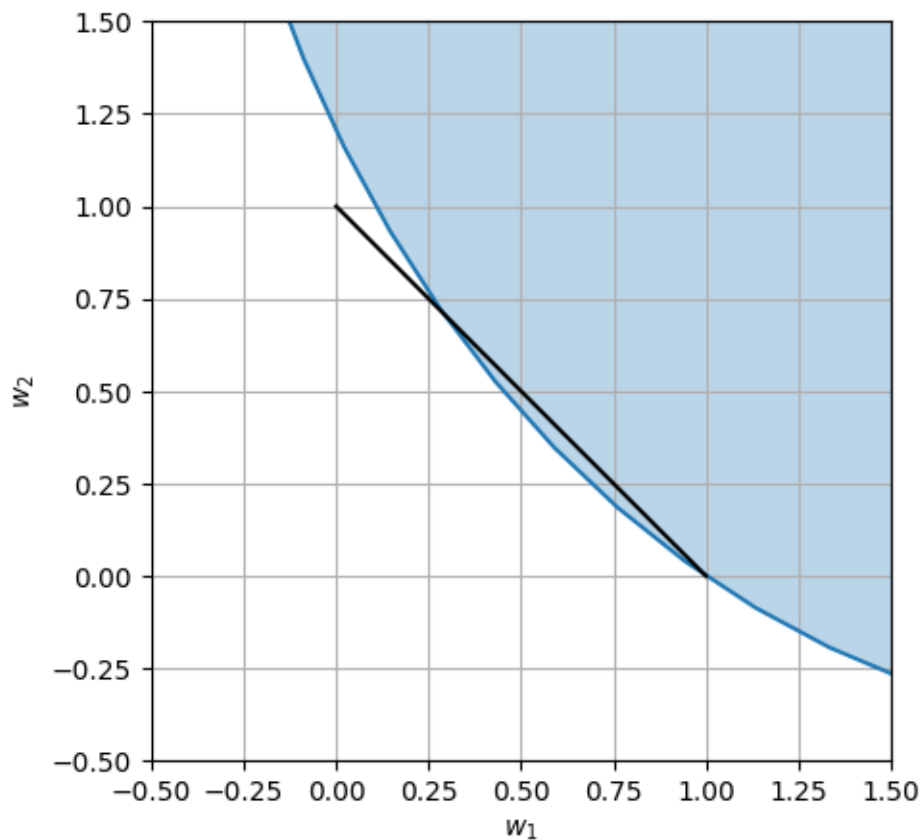
Using  $w_1, w_2$  as independent variables, for any value of the expected return, the expression of the return itself can be represented as a **conic section** in the  $w_1, w_2$ -plane. In particular,

- for  $\rho \neq 1$ , it's an **ellipse** ( $\Delta = B^2 - 4AC < 0$ ),
- for  $\rho = 1$ , it's a **parabola** ( $\Delta = 0$ )

**Portfolio allocation.** Some constraints may hold on portfolio allocation:

- fully-invested:  $w_1 + w_2 = 1$
- no short-selling:  $w_1, w_2 \geq 0$
- no leverage:  $w_1, w_2 \leq 1$

```
interactive(children=(FloatSlider(value=-0.25, description='rho', max=1.0, min=-1.0, step=0.01), FloatSlider(v...
```



This plot represents in **blue** asset allocations of the balanced 2-asset portfolio with expected value of the compound return larger than the compound return of any individual asset. **Black line** represents all the possible allocations of a fully invested portfolio,  $w_1 + w_2 = 1$  with no leverage  $w_k \leq 1$  and not short selling  $w_k \geq 0$ .

## 20.3 Comparison of portfolios: realizations of stochastic processes

In this section, rebalanced portfolio and buy-and-hold portfolio are compared. Different realizations of these two portfolio strategies are built, and used to build statistics, and discuss their properties in terms of **compound return**, **drawdowns**,...

**Note.** Here, 1-period returns are modelled as *normal random variable* so far. Anyways, it's possible (and suggested) to implement the most suited random process for modelling the return of the desired assets.

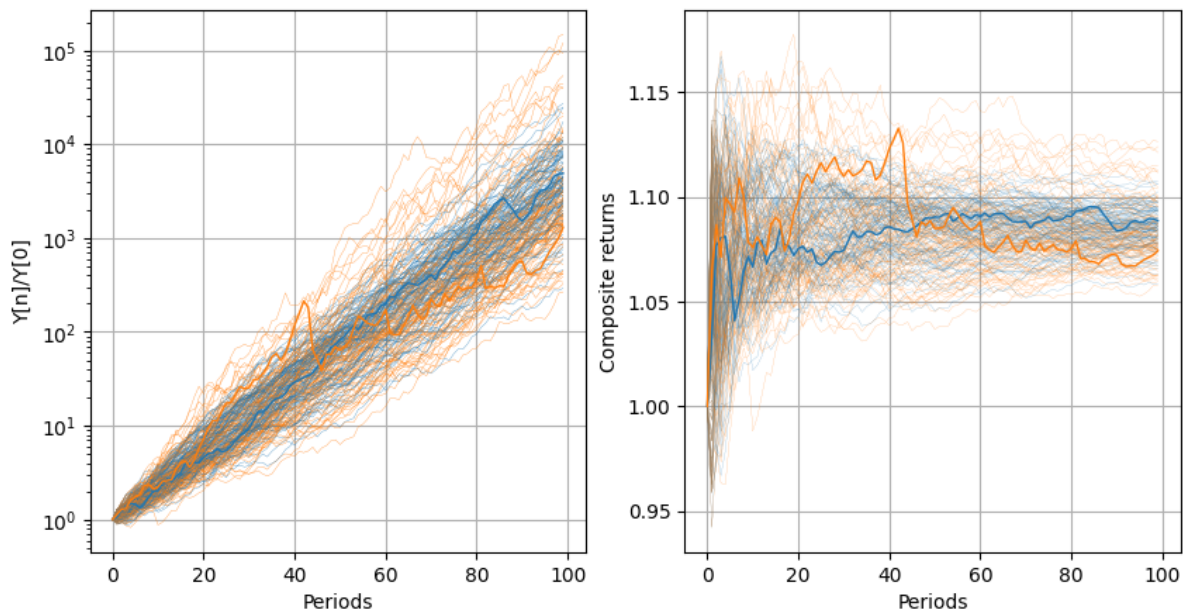
### 20.3.1 Useful functions

A useful function is introduced here to build correlated random variables with the desired expected values and covariance.

Main Colab notebook can be found here: [https://colab.research.google.com/drive/1n5py0Zf8i3\\_jrTTk0AR7Noq2kBYwpaqx?authuser=1#scrollTo=gmbjbprjCHto](https://colab.research.google.com/drive/1n5py0Zf8i3_jrTTk0AR7Noq2kBYwpaqx?authuser=1#scrollTo=gmbjbprjCHto)

### 20.3.2 Realizations

```
[<matplotlib.lines.Line2D at 0x7f825f3f6fa0>]
```

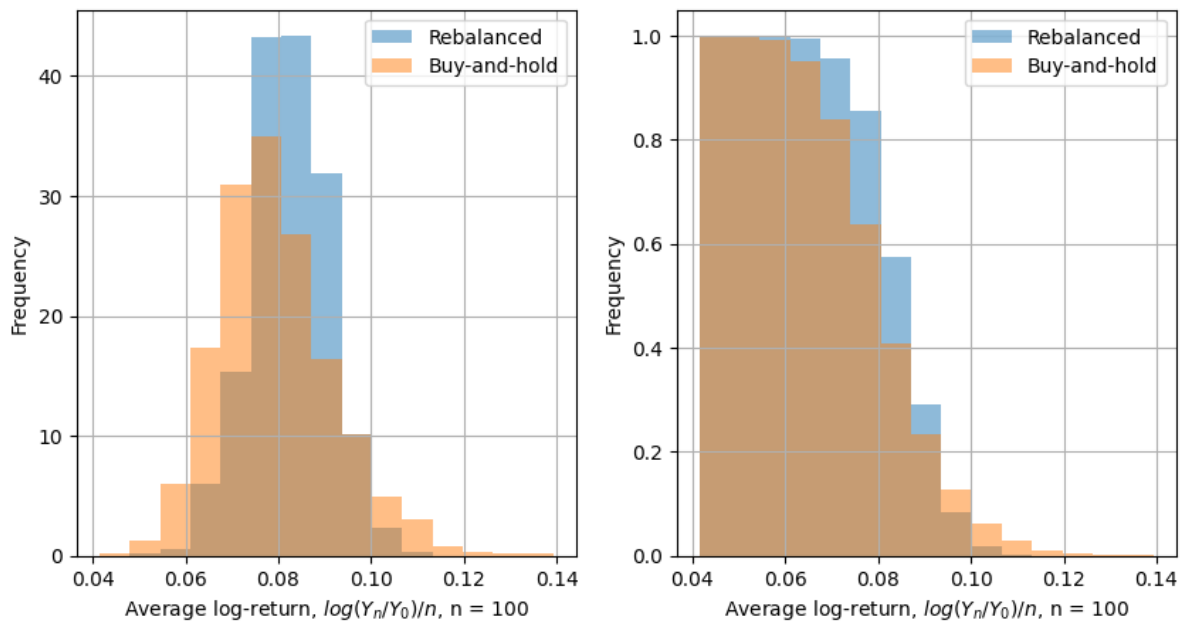


### 20.3.3 Composite return

```

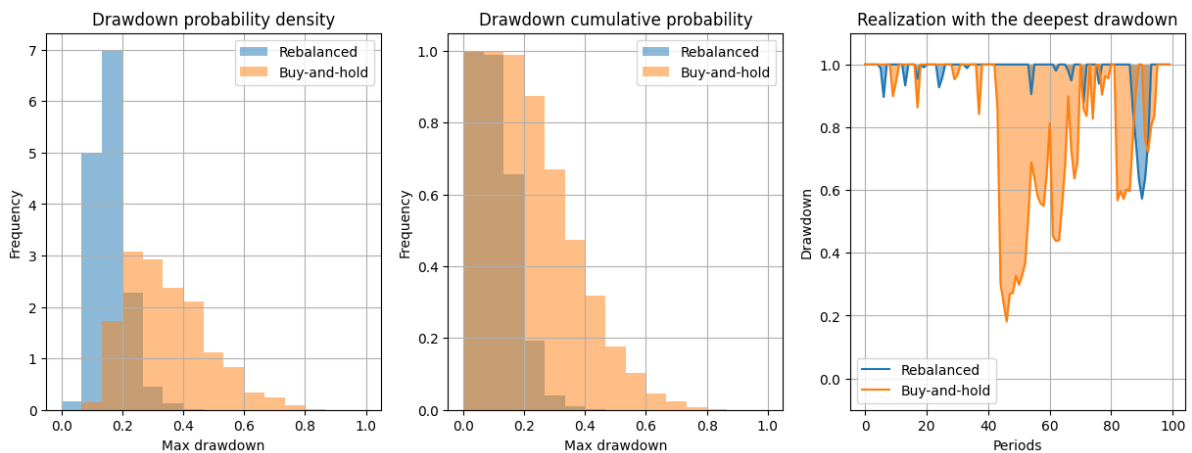
Rebalanced portfolio. Compsite return
Exp. value: 0.086
Std. dev. : 0.009
Buy-and-Hold portfolio. Compsite return
Exp. value: 0.083
Std. dev. : 0.014

```



### 20.3.4 Maximum drawdown

```
Text(0.5, 1.0, 'Realization with the deepest drawdown')
```



# **Part VI**

## **Historical Data**



**HISTORICAL DATA**





## US TREASURIES

...

```
#> Libraries
import numpy as np

import pandas as pd
# from IPython.display import display # import not required, maybe

import plotly.express as px
import plotly.graph_objects as go
import plotly.io as pio
from plotly.subplots import make_subplots

import yfinance as yf
```

```
#> Mount GDrive if required - for Colab editing
from pathlib import Path

# Detect Colab and mount Drive if needed
try:
    from google.colab import drive
    IN_COLAB = True
    pio.renderers.default = "colab"
except ImportError:
    IN_COLAB = False
    pio.renderers.default = "iframe"

if IN_COLAB:
    drive.mount('/content/drive')
    %cd /content/drive/MyDrive/basics-books/repos/financial-edu/bbooks-financial-edu/
    ↪ch/history

FILE_ROOT = Path.cwd()

print("Repo root:", FILE_ROOT)
```

```
Repo root: /home/davide/Documents/basics-books/books/bbooks-financial-edu/ch/
↪history
```

```
#> Import Shiller data (.xls)
shiller_file = str(FILE_ROOT) + "../../../code/data/shiller/ie_data.xlsx"
```

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```
df_shiller = pd.read_excel(shiller_file, sheet_name="Data", skiprows=7,) # engine=
↳ "xlrd")
```

```
#> Clean data
#> Shiller
#> Remove non-data last row
df_shiller = df_shiller.iloc[:-1].copy()

#> Convert Data to proper datetime type
df_shiller["Date"] = df_shiller["Date"].map(lambda x: f"{x:.2f}")
df_shiller["Date"] = pd.to_datetime(df_shiller["Date"], format="%Y.%m")
```

```
#> Shiller, chapt 26 file, with yearly data
chap26_file = str(FILE_ROOT) + "../../../code/data/shiller/chapt26.xlsx"
colnames = pd.read_excel(chap26_file, sheet_name='Data', skiprows=2, nrows=1).columns
df_chap26 = pd.read_excel(chap26_file, sheet_name="Data", skiprows=7, names=colnames)

df_chap26 = df_chap26.iloc[:-7].copy()

# Rename the first column to 'Year'
df_chap26.rename(columns={df_chap26.columns[0]: 'Year'}, inplace=True)
df_chap26['Year'] = df_chap26['Year'].astype(int)
```

```
df_chap26["T-Bill Return"] = (1. + df_chap26["R"]/100. ).shift(1).fillna(1).cumprod()
df_chap26["T-Bill Real Return"] = df_chap26["RealR"].shift(1).fillna(1).cumprod()
```

```
display(df_chap26.head())
display(df_chap26.tail())
```

|   | Year | P    | D    | E    | R    | RLONG | CPI       | RealR    | C \     |
|---|------|------|------|------|------|-------|-----------|----------|---------|
| 0 | 1871 | 4.44 | 0.26 | 0.40 | 6.35 | 5.32  | 12.464061 | 1.047504 | Tab)    |
| 1 | 1872 | 4.86 | 0.30 | 0.43 | 7.81 | 5.36  | 12.654392 | 1.054320 | in 2005 |
| 2 | 1873 | 5.11 | 0.33 | 0.46 | 8.35 | 5.58  | 12.939807 | 1.133511 | dollars |
| 3 | 1874 | 4.66 | 0.33 | 0.46 | 6.86 | 5.47  | 12.368896 | 1.148076 | NaN     |
| 4 | 1875 | 4.54 | 0.30 | 0.36 | 4.96 | 5.07  | 11.512651 | 1.114055 | NaN     |

|   | Unnamed: 9 | ... | P*C | RealD    | Return   | ln(1+ret) | RealE    | P/E \     |
|---|------------|-----|-----|----------|----------|-----------|----------|-----------|
| 0 | 1871.0     | ... | NaN | 4.731385 | 0.135809 | 0.127345  | 7.279054 | NaN       |
| 1 | 1872.0     | ... | NaN | 5.338874 | 0.088615 | 0.084907  | 7.652386 | 12.150000 |
| 2 | 1873.0     | ... | NaN | 6.143830 | 0.021590 | 0.021360  | 8.564127 | 11.883721 |
| 3 | 1874.0     | ... | NaN | 6.600773 | 0.122790 | 0.115817  | 9.201078 | 10.130435 |
| 4 | 1875.0     | ... | NaN | 6.369199 | 0.112843 | 0.106918  | 7.643039 | 9.869565  |

|   | E10 | P/E10 | T-Bill Return | T-Bill Real Return |
|---|-----|-------|---------------|--------------------|
| 0 | NaN | NaN   | 1.000000      | 1.000000           |
| 1 | NaN | NaN   | 1.063500      | 1.047504           |
| 2 | NaN | NaN   | 1.146559      | 1.104405           |
| 3 | NaN | NaN   | 1.242297      | 1.251855           |
| 4 | NaN | NaN   | 1.327519      | 1.437225           |

[5 rows x 23 columns]

|     | Year | P       | D     | E      | R   | RLONG | CPI     | RealR    | C   | \ |
|-----|------|---------|-------|--------|-----|-------|---------|----------|-----|---|
| 141 | 2012 | 1300.58 | 31.25 | 86.51  | NaN | 1.97  | 226.665 | 0.984302 | NaN |   |
| 142 | 2013 | 1480.4  | 34.99 | 100.20 | NaN | 1.91  | 230.280 | NaN      | NaN |   |
| 143 | 2014 | 1822.36 | 39.44 | 102.31 | NaN | 2.86  | 233.916 | NaN      | NaN |   |
| 144 | 2015 | 2028.18 | 43.39 | NaN    | NaN | 1.88  | 233.707 | NaN      | NaN |   |
| 145 | 2016 | 1918.6  | NaN   | NaN    | NaN | NaN   | NaN     | NaN      | NaN |   |

|     | Unnamed: 9 | ... | P*C | RealD | Return   | ln(1+ret) | RealE | P/E       | \ |
|-----|------------|-----|-----|-------|----------|-----------|-------|-----------|---|
| 141 | 2012.0     | ... | NaN | 31.25 | 0.144043 | 0.134569  | NaN   | 14.957792 |   |
| 142 | 2013.0     | ... | NaN | NaN   | NaN      | NaN       | NaN   | NaN       |   |
| 143 | NaN        | ... | NaN | NaN   | NaN      | NaN       | NaN   | NaN       |   |
| 144 | NaN        | ... | NaN | NaN   | NaN      | NaN       | NaN   | NaN       |   |
| 145 | NaN        | ... | NaN | NaN   | NaN      | NaN       | NaN   | NaN       |   |

|     | E10       | P/E10     | T-Bill Return | T-Bill Real Return |
|-----|-----------|-----------|---------------|--------------------|
| 141 | 68.558322 | 20.265948 | 631.02331     | 34.699283          |
| 142 | 69.552621 | 21.593294 | 631.02331     | 34.154564          |
| 143 | NaN       | NaN       | 631.02331     | 34.154564          |
| 144 | NaN       | NaN       | 631.02331     | 34.154564          |
| 145 | NaN       | NaN       | 631.02331     | 34.154564          |

[5 rows x 23 columns]

**Monthly total bond return.** Given interest of 10-year bonds, with  $N = 10$  years

$$\begin{aligned}
 M_n &= \underbrace{\frac{I_n}{12 \cdot 100}}_{\text{month coupon}} + \underbrace{\frac{I_n}{I_{n+1}} + \left(1 + \frac{I_{n+1}}{12 \cdot 100}\right)^{-\frac{1}{12N-1}} \cdot \left(1 - \frac{I_n}{I_{n+1}}\right)}_{\text{price change}} = \\
 &= \underbrace{\frac{i_n}{12}}_{\text{month coupon}} + \underbrace{\frac{i_n}{i_{n+1}} + \left(1 + \frac{i_{n+1}}{12}\right)^{-\frac{1}{12N-1}} \cdot \left(1 - \frac{i_n}{i_{n+1}}\right)}_{\text{price change}},
 \end{aligned}$$

being  $I = 100 \cdot i$ , the rates as a percentage.

**Real total bond return.**

$$T_{n+1}^R = T_n^R \cdot M_n \cdot \frac{CPI_n}{CPI_{n+1}},$$

being  $\frac{CPI_n}{CPI_{n+1}}$  the “realization” factor.

```
#> Inspection
# Rate GS10      : Interest Rate of 10Y Bond
# Returns        : Monthly Total Bond Returns
# Returns.1      : Real Total Bond Returns
# Real Return    : 10Y Annualized Stock Real Return
# Real Return.1  : 10Y Annualized Bonds Real Return
# Real Return.2  : Real 10Y Excess Annualized Returns
print(df_shiller.columns)
# display(df_shiller.head())
# display(df_shiller.tail())
```

```
Index(['Date', 'P', 'D', 'E', 'CPI', 'Fraction', 'Rate GS10', 'Price',
      'Dividend', 'Price.1', 'Earnings', 'Earnings.1', 'CAPE', 'Unnamed: 13',
      'TR CAPE', 'Unnamed: 15', 'Yield', 'Returns', 'Returns.1',
```

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```
'Real Return', 'Real Return.1', 'Returns.2'],
dtype='object')
```

## 22.1 Return

```
#> Evaluate nominal return, not contained in Shiller data
# df_shiller["10Y Bond Nominal Total Return"] = 1.
df_shiller["10Y Bond Nominal Total Return"] = df_shiller['Returns'].shift(1).
    ↪ fillna(1).cumprod()
```

```
display(df_shiller.head())
display(df_shiller.tail())
```

|   | Date       | P    | D    | E   | CPI       | Fraction    | Rate     | GS10 | Price      | \ |
|---|------------|------|------|-----|-----------|-------------|----------|------|------------|---|
| 0 | 1871-01-01 | 4.44 | 0.26 | 0.4 | 12.464061 | 1871.041667 |          | 5.32 | 114.692610 |   |
| 1 | 1871-02-01 | 4.5  | 0.26 | 0.4 | 12.844641 | 1871.125000 | 5.323333 |      | 112.798304 |   |
| 2 | 1871-03-01 | 4.61 | 0.26 | 0.4 | 13.034972 | 1871.208333 | 5.326667 |      | 113.868307 |   |
| 3 | 1871-04-01 | 4.74 | 0.26 | 0.4 | 12.559226 | 1871.291667 |          | 5.33 | 121.514327 |   |
| 4 | 1871-05-01 | 4.86 | 0.26 | 0.4 | 12.273812 | 1871.375000 | 5.333333 |      | 127.487866 |   |

|   | Dividend | Price.1    | ... | Unnamed: 13 | TR  | CAPE | Unnamed: 15 | Yield | \ |
|---|----------|------------|-----|-------------|-----|------|-------------|-------|---|
| 0 | 6.716234 | 114.692610 | ... | NaN         | NaN | NaN  | NaN         | NaN   |   |
| 1 | 6.517235 | 113.341406 | ... | NaN         | NaN | NaN  | NaN         | NaN   |   |
| 2 | 6.422074 | 114.954311 | ... | NaN         | NaN | NaN  | NaN         | NaN   |   |
| 3 | 6.665343 | 123.233997 | ... | NaN         | NaN | NaN  | NaN         | NaN   |   |
| 4 | 6.820339 | 129.868479 | ... | NaN         | NaN | NaN  | NaN         | NaN   |   |

|   | Returns  | Returns.1 | Real Return | Real Return.1 | Returns.2 | \ |
|---|----------|-----------|-------------|---------------|-----------|---|
| 0 | 1.004177 | 1.000000  | 0.130609    | 0.092504      | 0.038106  |   |
| 1 | 1.004180 | 0.974424  | 0.130858    | 0.094635      | 0.036224  |   |
| 2 | 1.004183 | 0.964209  | 0.130951    | 0.096186      | 0.034765  |   |
| 3 | 1.004185 | 1.004919  | 0.122056    | 0.090972      | 0.031084  |   |
| 4 | 1.004188 | 1.032591  | 0.122638    | 0.089488      | 0.033150  |   |

|   | 10Y Bond Nominal Total Return |
|---|-------------------------------|
| 0 | 1.000000                      |
| 1 | 1.004177                      |
| 2 | 1.008374                      |
| 3 | 1.012592                      |
| 4 | 1.016830                      |

[5 rows x 23 columns]

|      | Date       | P           | D         | E      | CPI      | Fraction    | \ |
|------|------------|-------------|-----------|--------|----------|-------------|---|
| 1850 | 2025-03-01 | 5683.983333 | 76.145301 | 216.69 | 319.799  | 2025.208333 |   |
| 1851 | 2025-04-01 | 5369.495714 | 76.546867 | NaN    | 320.795  | 2025.291667 |   |
| 1852 | 2025-05-01 | 5810.919524 | 76.948434 | NaN    | 321.465  | 2025.375000 |   |
| 1853 | 2025-06-01 | 6029.9515   | 77.350000 | NaN    | 321.8    | 2025.458333 |   |
| 1854 | 2025-07-01 | 6279.35     | NaN       | NaN    | 321.9675 | 2025.541667 |   |

|  | Rate | GS10 | Price | Dividend | Price.1 | ... | Unnamed: 13 | \ |
|--|------|------|-------|----------|---------|-----|-------------|---|
|--|------|------|-------|----------|---------|-----|-------------|---|

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|      |      |             |           |             |     |     |
|------|------|-------------|-----------|-------------|-----|-----|
| 1850 | 4.28 | 5722.525411 | 76.661629 | 3845325.219 | ... | NaN |
| 1851 | 4.28 | 5389.121125 | 76.826645 | 3625591.938 | ... | NaN |
| 1852 | 4.42 | 5820.002899 | 77.068716 | 3919793.241 | ... | NaN |
| 1853 | 4.38 | 6033.090148 | 77.390261 | 4067651.835 | ... | NaN |
| 1854 | 4.3  | 6279.350000 | NaN       | 4233686.042 | ... | NaN |

|      | TR        | CAPE | Unnamed: 15 | Yield    | Returns  | Returns.1 | Real Return | \   |
|------|-----------|------|-------------|----------|----------|-----------|-------------|-----|
| 1850 | 37.348615 |      | NaN         | 0.016749 | 1.003567 | 39.625790 |             | NaN |
| 1851 | 35.015341 |      | NaN         | 0.018767 | 0.992343 | 39.643654 |             | NaN |
| 1852 | 37.769849 |      | NaN         | 0.014834 | 1.006896 | 39.258095 |             | NaN |
| 1853 | 39.098992 |      | NaN         | 0.014026 | 1.010099 | 39.487674 |             | NaN |
| 1854 | 40.590130 |      | NaN         | 0.013847 | NaN      | 39.865727 |             | NaN |

|      | Real Return.1 | Returns.2 | 10Y Bond Nominal | Total Return |
|------|---------------|-----------|------------------|--------------|
| 1850 | NaN           | NaN       |                  | 1016.706201  |
| 1851 | NaN           | NaN       |                  | 1020.332454  |
| 1852 | NaN           | NaN       |                  | 1012.519392  |
| 1853 | NaN           | NaN       |                  | 1019.501865  |
| 1854 | NaN           | NaN       |                  | 1029.798273  |

[5 rows x 23 columns]

```
#> Nominal and real value
fig = make_subplots(rows=2, cols=1, shared_xaxes=True, # vertical_spacing=0.1,
                    subplot_titles=("10Y T-Bond - Nominal Total Return", "10Y T-Bond - Real Total
Return"),
)

#> 10Y T-Bonds
fig.add_trace(go.Scatter(x=df_shiller["Date"], y=df_shiller["10Y Bond Nominal Total
Return"], name="10Y", line=dict(color="steelblue")),
              row=1, col=1
)
fig.add_trace(go.Scatter(x=df_shiller["Date"], y=df_shiller["Returns.1"], name="10Y",
line=dict(color="steelblue"), showlegend=False),
              row=2, col=1
)

#> 1Y T-Bills
fig.add_trace(go.Scatter(x=df_chap26["Year"], y=df_chap26["T-Bill Return"], name="1Y",
line=dict(color="orange")),
              row=1, col=1
)
fig.add_trace(go.Scatter(x=df_chap26["Year"], y=df_chap26["T-Bill Real Return"], name=
"1Y", line=dict(color="orange"), showlegend=False),
              row=2, col=1
)

fig.update_yaxes(type="log", row=1, col=1)
fig.update_yaxes(type="log", row=2, col=1)

fig.update_layout(showlegend=True,)

fig.show()
```



## USA EQUITY MARKET

In 1926 the Standard Statistics Company developed a 90-stock index. On March 4, 1957, Standard and Poor's - the name of the company after the merging of the SSC and the Poor's Publishing - expanded the index to the current number of 500 companies, renamed the S&P500 Stock Composite Index.

On August 31, 1976, The Vanguard Group launched the first mutual fund to retail investors that tracked the index.

### Table of contents.

- **Return** over different time frames, both in nominal and real value
- **Sectors**

```
#> Mount GDrive if required - for Colab editing
from pathlib import Path

# Detect Colab and mount Drive if needed
try:
    from google.colab import drive
    IN_COLAB = True
    pio.renderers.default = "colab"
except ImportError:
    IN_COLAB = False
    pio.renderers.default = "iframe"

if IN_COLAB:
    drive.mount('/content/drive')
    %cd /content/drive/MyDrive/basics-books/repos/financial-edu/bbooks-financial-edu/
    ↪ch/history

FILE_ROOT = Path.cwd()
# DATA_DIR = REPO_ROOT / 'data'

print("Repo root:", FILE_ROOT)
# print("Data dir:", DATA_DIR)
```

```
Repo root: /home/davide/Documents/basics-books/books/bbooks-financial-edu/ch/
↪history
```

## 23.1 Return

- Time history of the total return index
- Return over different time spans: probability function and statistics, composition of returns
- Volatility: drag, volatility clusters,...

### 23.1.1 Total Return - Shiller data

Shiller data, from 1871 to 2025

```
count      1854.000000
mean         0.006515
std          0.040827
min         -0.261879
25%         -0.013347
50%          0.009404
75%          0.029377
max          0.524294
Name: return_month, dtype: float64
Skewness: 0.5564109443798456
Kurtosis: 17.94821481233623
```

## 23.2 Sectors

**Message:** market composition changes - rise and fall of dominant sectors; diversification can disappear as market concentration in one or few sectors increases; survivorship and return bias.

**Methods:**

- Evolution of largest companies of a time range over longer time ranges
- SPDR sector ETFs, available since 1998/12

**References:**

- Wikipedia, top 10 companies 2000-2025, [https://en.wikipedia.org/wiki/List\\_of\\_public\\_corporations\\_by\\_market\\_capitalization](https://en.wikipedia.org/wiki/List_of_public_corporations_by_market_capitalization)
- Forbes, top 50 companies: 1917, 1967, 2017, <https://www.forbes.com/sites/jeffkauflin/2017/09/19/americas-top-50-companies-1917-2017/>
- finhacker, top 20 US companies 1989-2026, <https://www.finhacker.cz/en/top-20-sp-500-companies-by-market-cap/>
- B.Taylor, Finaeon, 200 years of market concentration, <https://finaeon.com/200-years-of-market-concentration/>
- K.Zhu, Visual Capitalist, Charted: S&P 500 Market Concentration Over 145 Years, <https://www.visualcapitalist.com/charted-sp-500-market-concentration-over-145-years/>
- JPM Asset Management, picture in media folder



## 23.3 References

- Kenneth Franch Data Library: industry portfolios by market cap, long history, clean and ideal for education, [https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data\\_library.html](https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html)
- Robert Shiller data
  - <http://www.econ.yale.edu/~shiller/data.htm>
  - <https://shillerdata.com/>
- Wikipedia for top companies: scratch largest companies by mkt cap every 5-10 year and then show the evolution of the market cap



...

```
#> Libraries
import numpy as np

import pandas as pd
# from IPython.display import display # import not required, maybe

import plotly.express as px
import plotly.graph_objects as go
import plotly.io as pio
from plotly.subplots import make_subplots

import yfinance as yf
```

```
#> Mount GDrive if required - for Colab editing
from pathlib import Path

# Detect Colab and mount Drive if needed
try:
    from google.colab import drive
    IN_COLAB = True
    pio.renderers.default = "colab"
except ImportError:
    IN_COLAB = False
    pio.renderers.default = "iframe"

if IN_COLAB:
    drive.mount('/content/drive')
    %cd /content/drive/MyDrive/basics-books/repos/financial-edu/bbooks-financial-edu/
    ↪ch/history

FILE_ROOT = Path.cwd()

print("Repo root:", FILE_ROOT)
```

```
Repo root: /home/davide/Documents/basics-books/books/bbooks-financial-edu/ch/
↪history
```

```
#> Import Shiller data (.xls)
shiller_file = str(FILE_ROOT) + "../../../code/data/shiller/ie_data.xlsx"
```

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(continued from previous page)

```
df_shiller = pd.read_excel(shiller_file, sheet_name="Data", skiprows=7) #, engine=
↳ "xlrd")
```

```
#> Import gold data: monthly prices (.csv)
gold_file = str(FILE_ROOT) + "../..code/data/gold/monthly.csv"
df_gold = pd.read_csv(gold_file,)
```

```
#> Clean data
#> Shiller
#> Remove non-data last row
df_shiller = df_shiller.iloc[:-1].copy()

#> Convert Data to proper datetime type
df_shiller["Date"] = df_shiller["Date"].map(lambda x: f"{x:.2f}")
df_shiller["Date"] = pd.to_datetime(df_shiller["Date"], format="%Y.%m")
```

```
#> Gold
df_gold["Date"] = pd.to_datetime(df_gold["Date"], format="%Y-%m")
```

```
#> Inspection
display(df_shiller.head())
display(df_shiller.tail())
display(df_gold.head())
display(df_gold.tail())
```

|   | Date       | P    | D    | E   | CPI       | Fraction    | Rate     | GS10 | Price      | \ |
|---|------------|------|------|-----|-----------|-------------|----------|------|------------|---|
| 0 | 1871-01-01 | 4.44 | 0.26 | 0.4 | 12.464061 | 1871.041667 |          | 5.32 | 114.692610 |   |
| 1 | 1871-02-01 | 4.5  | 0.26 | 0.4 | 12.844641 | 1871.125000 | 5.323333 |      | 112.798304 |   |
| 2 | 1871-03-01 | 4.61 | 0.26 | 0.4 | 13.034972 | 1871.208333 | 5.326667 |      | 113.868307 |   |
| 3 | 1871-04-01 | 4.74 | 0.26 | 0.4 | 12.559226 | 1871.291667 |          | 5.33 | 121.514327 |   |
| 4 | 1871-05-01 | 4.86 | 0.26 | 0.4 | 12.273812 | 1871.375000 | 5.333333 |      | 127.487866 |   |

|   | Dividend | Price.1    | ... | CAPE | Unnamed: 13 | TR | CAPE | Unnamed: 15 | Yield | \ |
|---|----------|------------|-----|------|-------------|----|------|-------------|-------|---|
| 0 | 6.716234 | 114.692610 | ... | NaN  | NaN         |    | NaN  |             | NaN   |   |
| 1 | 6.517235 | 113.341406 | ... | NaN  | NaN         |    | NaN  |             | NaN   |   |
| 2 | 6.422074 | 114.954311 | ... | NaN  | NaN         |    | NaN  |             | NaN   |   |
| 3 | 6.665343 | 123.233997 | ... | NaN  | NaN         |    | NaN  |             | NaN   |   |
| 4 | 6.820339 | 129.868479 | ... | NaN  | NaN         |    | NaN  |             | NaN   |   |

|   | Returns  | Returns.1 | Real Return | Real Return.1 | Returns.2 |
|---|----------|-----------|-------------|---------------|-----------|
| 0 | 1.004177 | 1.000000  | 0.130609    | 0.092504      | 0.038106  |
| 1 | 1.004180 | 0.974424  | 0.130858    | 0.094635      | 0.036224  |
| 2 | 1.004183 | 0.964209  | 0.130951    | 0.096186      | 0.034765  |
| 3 | 1.004185 | 1.004919  | 0.122056    | 0.090972      | 0.031084  |
| 4 | 1.004188 | 1.032591  | 0.122638    | 0.089488      | 0.033150  |

[5 rows x 22 columns]

|      | Date       | P           | D         | E      | CPI     | Fraction    | \ |
|------|------------|-------------|-----------|--------|---------|-------------|---|
| 1850 | 2025-03-01 | 5683.983333 | 76.145301 | 216.69 | 319.799 | 2025.208333 |   |
| 1851 | 2025-04-01 | 5369.495714 | 76.546867 | NaN    | 320.795 | 2025.291667 |   |
| 1852 | 2025-05-01 | 5810.919524 | 76.948434 | NaN    | 321.465 | 2025.375000 |   |

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```

1853 2025-06-01      6029.9515  77.350000      NaN      321.8  2025.458333
1854 2025-07-01      6279.35      NaN      NaN      321.9675  2025.541667

      Rate GS10      Price      Dividend      Price.1  ...      CAPE  \
1850      4.28  5722.525411  76.661629  3845325.219  ...  34.784750
1851      4.28  5389.121125  76.826645  3625591.938  ...  32.621270
1852      4.42  5820.002899  77.068716  3919793.241  ...  35.172023
1853      4.38  6033.090148  77.390261  4067651.835  ...  36.394510
1854      4.3   6279.350000      NaN  4233686.042  ...  37.806545

      Unnamed: 13      TR CAPE      Unnamed: 15      Yield      Returns      Returns.1  \
1850      NaN      37.348615      NaN      0.016749  1.003567  39.625790
1851      NaN      35.015341      NaN      0.018767  0.992343  39.643654
1852      NaN      37.769849      NaN      0.014834  1.006896  39.258095
1853      NaN      39.098992      NaN      0.014026  1.010099  39.487674
1854      NaN      40.590130      NaN      0.013847      NaN  39.865727

      Real Return      Real Return.1      Returns.2
1850      NaN      NaN      NaN
1851      NaN      NaN      NaN
1852      NaN      NaN      NaN
1853      NaN      NaN      NaN
1854      NaN      NaN      NaN

[5 rows x 22 columns]

```

```

      Date      Price
0 1833-01-01  18.93
1 1833-02-01  18.93
2 1833-03-01  18.93
3 1833-04-01  18.93
4 1833-05-01  18.93

```

```

      Date      Price
2312 2025-09-01  3836.23
2313 2025-10-01  4002.77
2314 2025-11-01  4217.36
2315 2025-12-01  4322.26
2316 2026-01-01  5024.17

```

```

#> Merge Shiller and gold data
df_gold.rename(columns={"Price": "Gold"}, inplace=True)
df = df_gold.merge(df_shiller, on="Date", how="left")

```

```

#> Filter by date
mask = df['Date'].between('1871-01', '2025-07')
df = df[mask]

#> Add a column with gold real value (today value)
base_cpi = df['CPI'].iloc[-1]
print(base_cpi)
df['Gold Real'] = (df['Gold'] / df['CPI']) * base_cpi

```

```
321.9675
```

```
display(df.head())
display(df.tail(10))
```

```

      Date   Gold   P   D   E      CPI      Fraction Rate GS10  \
456 1871-01-01 18.93 4.44 0.26 0.4 12.464061 1871.041667 5.32
457 1871-02-01 18.93 4.5 0.26 0.4 12.844641 1871.125000 5.323333
458 1871-03-01 18.93 4.61 0.26 0.4 13.034972 1871.208333 5.326667
459 1871-04-01 18.93 4.74 0.26 0.4 12.559226 1871.291667 5.33
460 1871-05-01 18.93 4.86 0.26 0.4 12.273812 1871.375000 5.333333

```

```

      Price Dividend ... Unnamed: 13 TR CAPE Unnamed: 15 Yield  \
456 114.692610 6.716234 ... NaN NaN NaN NaN
457 112.798304 6.517235 ... NaN NaN NaN NaN
458 113.868307 6.422074 ... NaN NaN NaN NaN
459 121.514327 6.665343 ... NaN NaN NaN NaN
460 127.487866 6.820339 ... NaN NaN NaN NaN

```

```

      Returns Returns.1 Real Return Real Return.1 Returns.2 Gold Real
456 1.004177 1.000000 0.130609 0.092504 0.038106 488.993491
457 1.004180 0.974424 0.130858 0.094635 0.036224 474.504863
458 1.004183 0.964209 0.130951 0.096186 0.034765 467.576365
459 1.004185 1.004919 0.122056 0.090972 0.031084 485.28823
460 1.004188 1.032591 0.122638 0.089488 0.033150 496.573109

```

```
[5 rows x 24 columns]
```

```

      Date   Gold   P   D   E      CPI  \
2301 2024-10-01 2690.08 5792.319565 73.877418 203.570000 315.664
2302 2024-11-01 2651.13 5929.916 74.354836 206.870000 315.493
2303 2024-12-01 2648.01 6010.908571 74.832255 210.170000 315.605
2304 2025-01-01 2709.69 5979.5155 75.269937 212.343333 317.671
2305 2025-02-01 2894.73 6038.69 75.707619 214.516667 319.082
2306 2025-03-01 2983.25 5683.983333 76.145301 216.690000 319.799
2307 2025-04-01 3217.64 5369.495714 76.546867 NaN 320.795
2308 2025-05-01 3309.49 5810.919524 76.948434 NaN 321.465
2309 2025-06-01 3352.66 6029.9515 77.350000 NaN 321.8
2310 2025-07-01 3340.15 6279.35 NaN NaN 321.9675

```

```

      Fraction Rate GS10      Price Dividend ... Unnamed: 13  \
2301 2024.791667 4.1 5907.986497 75.352678 ... NaN
2302 2024.875000 4.36 6051.608846 75.880735 ... NaN
2303 2024.958333 4.39 6132.086645 76.340850 ... NaN
2304 2025.041667 4.63 6060.388442 76.287963 ... NaN
2305 2025.125000 4.45 6093.298659 76.392253 ... NaN
2306 2025.208333 4.28 5722.525411 76.661629 ... NaN
2307 2025.291667 4.28 5389.121125 76.826645 ... NaN
2308 2025.375000 4.42 5820.002899 77.068716 ... NaN
2309 2025.458333 4.38 6033.090148 77.390261 ... NaN
2310 2025.541667 4.3 6279.350000 NaN ... NaN

```

```

      TR CAPE Unnamed: 15      Yield Returns Returns.1 Real Return  \
2301 39.375226 NaN 0.015221 0.982514 40.006731 NaN
2302 40.176632 NaN 0.012564 1.001225 39.328482 NaN

```

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|      |           |     |          |          |           |     |
|------|-----------|-----|----------|----------|-----------|-----|
| 2303 | 40.544845 | NaN | 0.012630 | 0.984603 | 39.362679 | NaN |
| 2304 | 39.899969 | NaN | 0.011804 | 1.018269 | 38.504542 | NaN |
| 2305 | 39.943400 | NaN | 0.013572 | 1.017426 | 39.034610 | NaN |
| 2306 | 37.348615 | NaN | 0.016749 | 1.003567 | 39.625790 | NaN |
| 2307 | 35.015341 | NaN | 0.018767 | 0.992343 | 39.643654 | NaN |
| 2308 | 37.769849 | NaN | 0.014834 | 1.006896 | 39.258095 | NaN |
| 2309 | 39.098992 | NaN | 0.014026 | 1.010099 | 39.487674 | NaN |
| 2310 | 40.590130 | NaN | 0.013847 | NaN      | 39.865727 | NaN |

|      | Real Return.1 | Returns.2 | Gold Real   |
|------|---------------|-----------|-------------|
| 2301 | NaN           | NaN       | 2743.798255 |
| 2302 | NaN           | NaN       | 2705.536092 |
| 2303 | NaN           | NaN       | 2701.393069 |
| 2304 | NaN           | NaN       | 2746.338555 |
| 2305 | NaN           | NaN       | 2920.90742  |
| 2306 | NaN           | NaN       | 3003.478886 |
| 2307 | NaN           | NaN       | 3229.400417 |
| 2308 | NaN           | NaN       | 3314.66325  |
| 2309 | NaN           | NaN       | 3354.405092 |
| 2310 | NaN           | NaN       | 3340.15     |

[10 rows x 24 columns]

## 24.1 Price history

Piecewise gold standard resisted until 1971, when the US abandoned the Bretton Woods system and ended the direct convertibility between gold and the USD.

Before this “Nixon shock”, the price of gold was anchored to the USD for long periods of time though this anchor was occasionally reset. The most dramatic shifts occurred during WWI to account for war-time currency devaluation, and again in 1933-34. In the latter, Executive Order 6102 and the Gold Reserve Act effectively nationalized private gold and revalued it from 20.67 to 35.00 per ounce—a deliberate act of monetary policy to combat the Great Depression by devaluing the dollar.

### References

- H. McCulloch, World War I, Gold, and the Great Depression, Cato Institute, <https://www.cato.org/blog/world-war-i-gold-great-depression>
- National Mining Association, Historical Gold Prices - 1833 to Present

```
#> Nominal and real value
fig = make_subplots(rows=2, cols=1, shared_xaxes=True, # vertical_spacing=0.1,
                    subplot_titles=("Gold price - Nominal", "Gold price - Real",))
)

#> S&P500 index
fig.add_trace(go.Scatter(x=df["Date"], y=df["Gold"], name="Gold price - Nominal",
                        line=dict(color="black")),
              row=1, col=1
)
fig.add_trace(go.Scatter(x=df["Date"], y=df["Gold Real"]/df["Gold Real"].iloc[0],
                        name="Gold price - Real, 1871 = 1", line=dict(color="black")),
              row=2, col=1
)
```

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```

)

fig.update_yaxes(type="log", row=1, col=1)
fig.update_yaxes(type="log", row=2, col=1)

fig.update_layout(showlegend=False,)

fig.show()

# fig = go.Figure()
# fig.add_trace(go.Scatter(x=df_gold['Date'], y=df_gold['Price']))

# fig.update_yaxes(type="log")
# fig.show()

```

## 24.2 Quantitative Characteristics

Some statistics...

### 24.2.1 Return and Volatility

Since 1971, from the Nixon shock and the end of the convertibility between USD and gold.

```

#> Get the df since 1971
month_window = 1
year_window = 12

df_1971 = df.copy()
mask = df['Date'].between('1971-01', '2025-07')
df_1971 = df_1971[mask]

```

```

#> Evaluate nominal returns
df_1971['Gold Return 1m'] = df_1971['Gold'].pct_change(month_window)
df_1971['Gold Return 1m Annual'] = ( 1. + df_1971['Gold Return 1m'] )**(12/month_
↪window) - 1
df_1971['Gold Return 1y'] = df_1971['Gold'].pct_change( year_window)

df_1971['Gold Vol 1y'] = df_1971['Gold Return 1y'].rolling(window= year_window).std()↪
↪* np.sqrt(12)

#> Evaluate real returns
df_1971['Gold Real Return 1m'] = df_1971['Gold Real'].pct_change(month_window)
df_1971['Gold Real Return 1m Annual'] = ( 1. + df_1971['Gold Real Return 1m'] )**(12/
↪month_window) - 1
df_1971['Gold Real Return 1y'] = df_1971['Gold Real'].pct_change( year_window)

df_1971['Gold Real Vol 1y'] = df_1971['Gold Real Return 1y'].rolling(window= year_
↪window).std() * np.sqrt(12)

```

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```
# display(df.tail(10))
```

```
#> Nominal returns
fig = make_subplots(rows=2, cols=1, shared_xaxes=True, # vertical_spacing=0.1,
                    subplot_titles=("Monthly returns", "1-year rolling statistics",)
)

#> S&P500 index
colors_1m = np.where(df_1971['Gold Return 1m'] >= 0, 'green', 'red')
fig.add_trace(
    go.Bar(
        x=df_1971["Date"], y=df_1971["Gold Return 1m"],
        name="1m return (annualized)",
        marker_color=colors_1m,
        showlegend=False
    ),
    row=1, col=1
)

colors_1y = np.where(df_1971['Gold Return 1y'] >= 0, 'green', 'red')
fig.add_trace(
    go.Bar(
        x=df_1971["Date"], y=df_1971["Gold Return 1y"],
        name="1y return",
        marker_color=colors_1y,
        showlegend=False
    ),
    row=2, col=1
)
fig.add_trace(go.Scatter(x=df_1971["Date"], y=df_1971["Gold Vol 1y"], name="1y vol",),
               row=2, col=1
)

fig.update_layout(title="Nominal Return")

fig.show()
```

```
#> Nominal returns
fig = make_subplots(rows=2, cols=1, shared_xaxes=True, # vertical_spacing=0.1,
                    subplot_titles=("Monthly returns", "1-year rolling statistics",)
)

#> S&P500 index
colors_1m = np.where(df_1971['Gold Real Return 1m'] >= 0, 'green', 'red')
fig.add_trace(
    go.Bar(
        x=df_1971["Date"], y=df_1971["Gold Real Return 1m"],
        name="1m return (annualized)",
        marker_color=colors_1m,
        showlegend=False
    ),
    row=1, col=1
)
```

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```

)

colors_1y = np.where(df_1971['Gold Real Return 1y'] >= 0, 'green', 'red')
fig.add_trace(
    go.Bar(
        x=df_1971["Date"], y=df_1971["Gold Real Return 1y"],
        name="1y return",
        marker_color=colors_1y,
        showlegend=False
    ),
    row=2, col=1
)
fig.add_trace(go.Scatter(x=df_1971["Date"], y=df_1971["Gold Real Vol 1y"], name="1y_
    vol"),
    row=2, col=1
)

fig.update_layout(title="Real Return")

fig.show()

```

## 24.2.2 Drawdown history

```

from textwrap import fill
#> Calculate drawdowns with rolling max
rolling_max = df_1971['Gold'].cummax()
df_1971['Drawdown'] = (df_1971['Gold'] / rolling_max) - 1

rolling_max = df_1971['Gold Real'].cummax()
df_1971['Drawdown Real'] = (df_1971['Gold Real'] / rolling_max) - 1

#> Plot nominal and real drawdowns
fig_dd = go.Figure()

colors = {
    'gold': '255, 215, 0',
    'black': '0, 0, 0',
    'firebrick': '178, 34, 34',
    'royalblue': '65, 105, 225'
}

fig_dd.add_trace(go.Scatter(
    x=df_1971['Date'],
    y=df_1971['Drawdown Real'],
    fill='tozeroy',
    fillcolor=f'rgba({colors["black"]}, 0.2)',
    mode='lines',
    line_color='black',
    name='Real'
),
)

```

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```
fig_dd.add_trace(go.Scatter(
    x=df_1971['Date'],
    y=df_1971['Drawdown'],
    fill='tozero',
    fillcolor=f'rgba({colors["firebrick"]}, 0.2)',
    mode='lines',
    line=dict(color='firebrick', dash='dot'),
    name='Nominal'
),
)

# 4. Styling
fig_dd.update_layout(
    title='Gold Price Drawdown (%)',
    yaxis_title='Percentage Decline',
    yaxis_tickformat='.1%', # Formats as -5.0%, -10.0%, etc.
    template='plotly_white',
    # hovermode='x unified'
)

fig_dd.show()
```

### 24.2.3 Correlation with other assets

- low correlation with equities and bonds. Always? *Example:* evaluate the rolling 36-month correlation between gold and other assets.

```
df_1971["SP500 Total Return 1m"] = df_1971["Price.1"].pct_change(month_window)

#> Calculate 12-Month Rolling Correlation
window = 12 # Starting from monthly data
df_1971['Corr(Gold, SP500)'] = df_1971['Gold Return 1m'].rolling(window).corr(df_1971[
    ↪ 'SP500 Total Return 1m'])
```

```
#> Define equity stress (S&P500 drawdown > 10%)
sp500_max = df_1971['Price.1'].cummax()
df_1971['SP500 Total Drawdown'] = (df_1971['Price.1'] / sp500_max) - 1
stress_mask = df_1971['SP500 Total Drawdown'] < -0.1

starts = df_1971["Date"][stress_mask & ~stress_mask.shift(1).fillna(False)]
ends = df_1971["Date"][stress_mask & ~stress_mask.shift(-1).fillna(False)]
```

```
# 3. Plotting with Plotly
fig_corr = go.Figure()

# Gold vs S&P 500 (Solid line)
fig_corr.add_trace(go.Scatter(
    x=df_1971['Date'], y=df_1971['Corr(Gold, SP500)'],
    mode='lines', name='Gold vs S&P 500',
    line=dict(color='rgba(65, 105, 225, 1)', width=2)
))
```

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```

# # Gold vs DXY (Dashed line for accessibility)
# fig_corr.add_trace(go.Scatter(
#     x=df.index, y=df['Corr_Gold_DXY'],
#     mode='lines', name='Gold vs US Dollar (DXY)',
#     line=dict(color='rgba(178, 34, 34, 1)', width=2, dash='dash')
# ))

# # Add a zero-line for reference
# fig_corr.add_hline(y=0, line_dash="dot", line_color="black", opacity=0.5)

# fig_corr.update_layout(
#     title='12-Month Rolling Correlation of Gold',
#     yaxis=dict(title='Correlation Coefficient', range=[-1, 1]),
#     template='plotly_white',
#     hovermode='x unified'
# )

# 4. Add the Background Shading for Equity Drawdowns
# We can loop through the index to find 'stress' clusters
for (start, end) in zip(starts, ends):
    fig_corr.add_vrect(
        x0=start, x1=end,
        fillcolor="rgba(128, 128, 128, 0.2)", # Light gray
        layer="below", line_width=0,
        # annotation_text="Equity Stress"
    )

fig_corr.update_layout(
    title='12-month Cross Correlation between gold and other assets',
    showlegend=True,
)

fig_corr.show()

```

### 24.2.4 Gold as a macro indicator

- relationship with real rates (yield - inflation)
- safe haven in major crisis
- currency hedge (correlation between USD and gold)

## 24.3 The role in a portfolio

- **Low correlation**, see section above
- **“Safe Heaven”**, see section above



## **Part VII**

### **Extra**





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CHAPTER  
**TWENTYFIVE**

---

**EXTRA AND RANDOM**



## EURISTHICS AND HISTORICAL CORRELATIONS

### 26.1 Expected returns in the stoch market

#### 26.1.1 Shiller P/E and S&P500 10-year annualised forward returns

- from Invesco, Applied philosophy: The Shiller P/E and the S&P500 returns

#### 26.1.2 Bogle Expected Return Formula

- Comment by *Ben Carlson*, on his website



## **Part VIII**

# **People**



## RESOURCES, PEOPLE AND FIRMS

### 27.1 Tools

These tools have not been tested and verified here. I decline any responsibility about their use.

#### 27.1.1 Simple tools for investors

[Simple tools for investors] (<https://www.simpletoolsforinvestors.eu/index.shtml>): **bond** monitor, calculators, tools (minus-eater, laddering,...)

#### 27.1.2 Interactive Asset Allocation - by Research Affiliates

Expected return vs. volatility plot of different asset classes on different time horizons by *Research Affiliates*.

#### 27.1.3 Gregory Gundersen Blog

Gregory Gundersen blog

### 27.2 People

#### 27.2.1 Arnott, Robert

Founder of *Research Affiliates*.

#### 27.2.2 Faber, Meb(ane)

Co-founder and Chief Investment Officer at *Cambria Investment Management*.

### 27.2.3 Wigglesworth, Robin

Financial Times' global finance correspondent, author of *Trillions* a book on the past, present and future of passive investing.

### 27.2.4 Zweig, Jason

Columnist for the Wall Street Journal since 2008. *A Safe Heaven for Intelligent Investor*.

### 27.2.5 Ritholtz, Barry

### 27.2.6 Maggiulli, Nick

### 27.2.7 Carlson, Ben

Director of Institutional Asset Management at *Ritholtz Wealth Management*. Author of the website *A Wealth of Common Sense*

### 27.2.8 Green, Micheal

Forseen “**Volmageddon**” of the 5 February 2018, the collapse of inverse ETFs or ETP: VIX, SVXY and VMIN, as a consequence of the VIX daily surge from 17 to 37 (approximately +115%, that erased daily inverse products)

*Even a moderate (~4%) equity sell-off can unleash devastating volatility swings if the short-vol market is crowded. Rebalancing demands can worsen volatility when liquidity dries up near market close.*

- *The Bull*, 192. Micheal Green: perché l'investimento passivo distorce il mercato (e come comportarci)

### 27.2.9 Yardeni, Edward

## 27.3 Firms

### 27.3.1 Reserach Affiliates

Founded by *Robert Arnott*

### 27.3.2 Cambria Investment Management

Co-founded by *Mebane Faber*



### 27.3.3 Ritholtz Wealth Management

*Barry Ritholtz, Nick Maggiulli, Ben Carlson*



## THE BULL

“The Bull, il tuo podcast di finanza personale”, Riccardo Spada. [Youtube channel](#)

- Jason Zweig, American financial journalist, columnist for the WSJ since 2008. [Jason Zweig - A Safe Haven for Intelligent Investor](#)
  - **224** what it takes to become an intelligent investor
- Nick Maggiulli, COO for Ritholtz Wealth Management LCC, financial blogger at [Of Dollars And Data - Act Smarter. Live Richer](#). Author of “Just Keep Buying” about the power of compounding.
  - **221** Just keep buying
- Davide Serra, founder and CEO of Algebris Investment
  - **216** The change we’re living and consequences for investors
- William Bernstein, american financial theorist and neurologist
  - **214** The 4 pillars of investing and how to manage risk and uncertainty
- Barry Ritholtz
  - **206** How **not** to invest
- Robin Wigglesworth
  - **203** the case for passive investing: navigative crises through simplicity
- Ben Carlson, Director of Institutional Asset Management at [Ritholtz Wealth Management](#). Author of the webiste [A Wealth of Common Sense](#)
  - **200** Investing is mostly a matter of common sense (and patience)
- Robert Arnott
  - **196** Fundamental investing
- Micheal Green
  - **192** Why passive investing is distorting the markets
  - [Haddad, Valentin](#), associate professor at the UCLA: how competitive is stock market? why is asset demand inelastic?
- Edward Yardeni,
  - **183** Why should we be optimistic?
- Meb Faber,
  - **180** Investing with Common Sense
- Eugene Fama, efficient market, **factor** investing, and factor ETFs

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### **bond-risk-inflation-zero-coupon-example**

bond-risk-inflation-zero-coupon-example (*ch/assets/bonds-risks*), [62](#)

### **coin-kelly-criterion**

coin-kelly-criterion  
(*code/notebooks/rebalancing\_intro*), [47](#)

### **fair-coin**

fair-coin (*code/notebooks/rebalancing\_intro*), [47](#)

### **shannon-demon-coin**

shannon-demon-coin  
(*code/notebooks/rebalancing\_intro*), [45](#)